BIO-MATH HANDBOOK

Biology math questions explained with examples

For CSIR UGC NET LS/JRF, GATE, JAM, ICMR, IISC, BHU, JNU, ICAR, SET, BITS PILANI entrance Preparation

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Introduction:
This book is an effort to collect Math problems from different topics of life sciences and to discuss about methods to solve those mathematical questions from biological science.

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Chapter 1

Biochemistry Problems

Introduction:

Biochemistry is a very imporatnat chapter to ask mathematical questions as it is filled with chemical formula and equations. Every year more that 2-3 questions are asked from this chapter in CSIR UGC NET and GATE exams. This chapter is also very importanat for IIT-JAM entrance test. Most of the questions asked from this chapter in the mentioned competitive exams are of mathematical type (Appx. 70%). So, you need to practice all these problems to grasp the actual concepts of Biochemistry.

Chapter content:

1. Biological macromolecules Problems
2. Acid Base problems
3. Half-Life and radioactive decay problems
4. Enzyme kinetics problems
5. Gibb's free energy and thermodynamics problems
6. Beer-Lambert Law problems

Scan the following QR code to Know how to solve Biochemistry problems

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1. Biological macromolecules Problems

Practice problems and solutions

Chapter 1: Acid-Base Equilibrium and Spectrophotometry

1.1 Acid-Base Equilibrium:
What is the pH of the following solutions?
a) 0.35 M hydrochloric acid
b) 0.35 M acetic acid (pKa = 4.76)
c) 0.035 M acetic acid.

1.2 Acid-Base Equilibrium:
A weak acid, HA, has a total concentration of 0.20M and is ionized (dissociated) to 2%;
a) Calculate the Ka for this acid.
b) Calculate the pH for this acidic solution.

1.3 Acid-Base Equilibrium:
Calculate the pH of the following mixtures:
a) 1M acetic acid and 0.5M sodium acetate
b) 0.3M phosphoric acid and 0.8M KH2PO4 (pKa=2.14)

1.4 Acid-Base Equilibrium:
You need to prepare a buffer solution at pH = 7.00 with KH2PO4 and Na2HPO4 (pKa=7.21). If you use a 0.1M solution of KH2PO4, what would be the concentration of Na2HPO4 needed?

1.5 Acid-Base Equilibrium:
You need to prepare a buffer solution at pH = 7.00 with KH2PO4 and Na2HPO4. What would be the respective concentration of these substances if you wished to obtain a final phosphate concentration ([HPO4-2] + [H2PO4 -1]) of 0.3M?

1.6 Spectrophotometry:
What is the concentration of the amino acid tyrosine (ε=1 420 L mol-1 cm-1) if you obtain an absorbance of 0.71 with a 1 cm cuvette? With a 0.1 cm cuvette?

1.7 Spectrophotometry:
What would be the absorbance reading of a 37-mM solution of tyrosine?

1.8 Spectrophotometry:
You wish to determine the concentration of haemoglobin in a blood sample by spectrophotometry. You first create a standard curve of the absorbance at 412 nm of several solutions of known haemoglobin concentrations. The data for the standard curve is shown below. What is the concentration (in µg/mL) of haemoglobin in your sample if the absorbance obtained at 412 nm was 0.303?

Absorbance (412nm)	Concentration of standard solution (µg/ml)
0.069	1
0.113	2
0.201	4
0.377	8
0.730	16

Chapter 2: Amino acids

* 2.1. Molecular mass of an amino acid.
1.812 g of a crystallized α-amino acid (pKa1: 2.4; pKa2; 9.7) has a pH of 10.4 when dissolved in 100 mL of 0.1M NaOH. Calculate the molecular mass of this amino acid.

2.2. Titration curve
Calculate the pI of histidine and draw its titration curve. Indicate the position of all pKas and the pI as well as the percentages of each ionic form at the start and finish of the titration and at all pKas. The list of pKas for all 20 amino acids can be found at the end of the "Problems" section of this problem set.

2.3. Net charges of amino acids
What is the net charge (+, 0, -) of the amino acids glycine, serine, aspartic acid, glutamine and arginine at:
a) pH 2.01 b) pH 3.96 c) pH 5.68 d) pH 10.76

2.4. Ionic exchange chromatography
A mixture of lysine, glycine, alanine, isoleucine and glutamic acid are separated by ionic exchange chromatography. What is the order of elution of these amino acids if you use gradient
buffer system from pH 10 to pH 2:
a) with a cation exchange resin?
b) with an anion exchange resin?
Which column would give the best separation?

2.5. Amino acids
What amino acids can be converted into another amino acid with gentle hydrolysis, resulting in release of ammonia?

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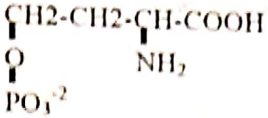
2.7. *Amino acids*

Phosphoserine is found after enzymatic hydrolysis of casein, a milk protein. However, it does not belong to the 20 amino acids coded during protein synthesis. Give a plausible explanation.



Phosphoserine

***2.8. Ionic exchange chromatography**

Glycine, alanine, valine and leucine can be successfully separated by ionic exchange chromatography even though their pKas are almost identical. Explain the behaviour of these amino acids.

2.9. *Peptides.*

A peptide is hydrolyzed and its amino acid content analyzed. Hydrolysis destroys the amino acid tryptophan; therefore, the content of tryptophan can be estimated with spectrophotometry.

Establish the empirical formula of the peptide with the following information.

Amino acids	*mmol*
Ala	2.74
Glu	1.41
Leu	0.69
Lys	2.81
Arg	0.72
Trp	0.65

2.10. *Peptides.*

Draw the structure of the following peptide GWYQR. Indicate the ionic form of the peptide at the following pH:

a) pH 2.0 b) pH 7.0 c) pH 10.5

Chapter 3. General properties and purification of proteins

3.1. Protein Purification

Why do we often use ammonium sulphate precipitation in initial purification steps of proteins?

3.2. Protein Purification

DEAE cellulose columns are rarely used at pH greater than 8.5. Why?

3.3. Protein Purification

6-phosphogluconate dehydrogenase has a pI of 6. Explain why the buffer used for a chromatography on DEAE-cellulose must have a pH greater than 6 but less than 9 in order to ensure the enzyme is efficiently bound to the column.

3.4. Protein Purification.

Would the enzyme, 6-phosphogluconate dehydrogenase bind to a CM-cellulose resin if the same conditions as the previous problem were used? Why?

3.5. Protein Purification.

What pH would the buffer need to be in order to permit the dehydrogenase in the previous problem to bind to the CM-cellulose resin?

3.6. Protein Purification.

We load a DEAE-cellulose column adjusted to a pH of 6.5 with the following mixture of proteins: ovalbumin (pI = 4.6), urease (pI = 5.0), and myoglobin (pI = 7.0). The proteins are eluted first with a buffer of weak ionic strength at a pH of 6.5, and then the same buffer containing increasing amounts of sodium chloride is used to elute the proteins. What order are the proteins eluted?

3.7. Protein Purification.

An enzyme (MW 24 kDa, pI 5.5) is contaminated with two other proteins, one with a similar molecular mass and a pI of 7.0 while the other has a molecular mass of 100 kDa and a pI of 5.4.

Suggest a procedure to purify the contaminated enzyme.

3.8. Protein Purification.

A procedure used to purify 6-gluconate dehydrogenase from E. coli is presented below. a) Calculate (1) the specific activity, (2) the percent yield based on the initial quantity of the enzyme and (3) the degree of purification for each step (i.e. fold increase in purification).

b) Indicate which step purifies the protein the most.

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c) Assuming the protein is pure after gel permeation chromatography (on Bio-Gel A), what percent of the initial extract contained 6-gluconate dehydrogenase?

Purification step	*Volume (mL)*	*Total protein (mg)*	*Enzymatic activity (µg/min)*
1- Cellular extract	2 800	70 000	2 700
2- Ammonium sulfate	3 000	25 400	2 300
3- Heat denaturation	3 000	16 500	1 980
4- DEAE chromatography	80.00	390.00	1 680
5- CM-cellulose chromatography	50.00	47.00	1 350
6- Bio-Gel A chromatography	7.00	35.00	1 120

3.9 Protein Purification.
Why is SDS omitted when proteins need to undergo isoelectric focusing?

3.10. Protein Purification.
A series of proteins with known molecular mass and an enzyme of unknown molecular mass are separated by chromatography on a Sephadex G-200 column. The elution volume (Ve) for each protein is indicated in the table below. Estimate the molecular mass of the unknown protein.

Protein	*Mr*	V_e *(mL)*
Blue dextran	1 000 kDa	85.00
lysozyme	14 kDa	200.00
Chymotrypsinogen	25 kDa	190.00
ovalbumin	45 kDa	170.00
Serum albumin	65 kDa	150.00
aldolase	150 kDa	125.00
urease	500 kDa	90.00
ferritin	700 kDa	92.00
ovomucoid	28 kDa	160.00
unknown	?	130.00

3.11. Protein Purification.
Referring to the previous problem, give a plausible explanation for the bizarre behavior ferritin's elution from the sephadex column.

3.12. Protein Purification.
A student isolates a protein from anaerobic bacteria and analyses the protein by polyacrylamide gel electrophoresis containing SDS (PAGE-SDS). Following protein staining, a single band appears, which excites the student's supervisor. To be certain, the supervisor suggests that the student run a second electrophoresis under *native* conditions (i.e. non-denaturing, or without SDS). This gel shows two bands after staining. Assuming no errors were committed during these experiments, explain the observations.

3.13. Protein Purification.
A student from CHMI 2227 analyses bovine serum albumin (BSA) with a polyacrylamide gel electrophoresis (PAGE-SDS). During the experiment, the student forgets to add β- mercaptoethanol to the sample. When comparing his sample to those of his classmates he realizes that the molecular mass of his BSA sample determined by PAGE-SDS is 57 kDa, while all the other students (those that added β-mercaptoethanol) found a molecular mass of 68 kDa.
Explain this difference.

3.14. Polypeptide sequencing
Consider the following peptide:
A-L-K-M-P-E-Y-I-S-T-D-Q-S-N-W-H-H-R
Indicate the fragments generated after the following digestions:
a) trypsin b) pepsin c) protease V8 d) cyanogen bromide

3.15 Polypeptide sequencing
Deduce the polypeptide sequence that generated the following results:
a) acid hydrolysis: (Ala2 , Arg , Lys2 , Met, Phe, Ser2);
b) Carboxypeptidase A digestion: Ala;
c) Trypsin digestion: (Ala, Arg)
(Lys, Phe, Ser)
(Lys)
(Ala, Met, Ser)

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d) cyanogen bromide treatment: (Ala, Arg, Lys2 , Met, Phe, Ser)
(Ala, Ser)

e) thermolysine digestion: (Ala)
(Ala, Arg, Ser)
(Lys2 , Met, Phe, Ser)

3.16. Polypeptide sequencing

A polypeptide is reduced by β-mercaptoethanol to yield two peptide fragments with the following sequences:
fragment 1: A-C-F-P-K-R-W-C-R-R-V-C
fragment 2: C-Y-C-F-C
The non-reduced polypeptide is digested with thermolysine and yields the following fragments:
(A,C,C,V)
(R,K,F,P)
(R,R,C,C,W,Y)
(C,C,F)
Indicate the positions of disulfide bridges in the polypeptide.

3.17. Polypeptide sequencing

An analysis of the polypeptide Shawi isolated from the bacteria *Chretientus negativii*, yields the following results:
a) acid hydrolysis: (Ala4 , Val, Lys2 , Arg, Gly, Asp, Met, Pro, Trp)
b) carboxypeptidase digestion: Lys
c) dinitrofluorobenzene treatment: Val
d) cyanogen bromide treatment: generates two polypeptides:
peptide A: (Gly, Arg, Trp, Asp, Lys, Ala); Treatment of this peptide with DNBF and carboxypeptidase yields :
DNFB: Gly Carboxypeptidase: Lys
peptide B: (Ala3 , Lys, Val, Met, Pro); Treatment of this peptide with DNFB and carboxypeptidase yields:
DNFB: Val Carboxypeptidase: Met
e) trypsine digestion: yields three peptides
peptide C: (Lys, Trp, Ala); Treatment of this peptide with DNFB and carboxypeptidase yields :
DNFB: Trp
peptide D: (Ala3 , Val, Lys, Pro)
peptide E: (Met, Asp, Gly, Arg); Treatment of this peptide with DNFB and carboxypeptidase yields :
DNFB: Met
Finally, treating peptide D with thermolysine yields the following:
Val
Ala
Ala
(Ala, Lys, Pro)
What is the primary structure of this peptide?

Chapter 4. Three dimensional structures of proteins

4.1. 3-D Structures of proteins

What amino acids among the following would you expect to find a) inside, and b) at the surface of a typical globular protein in an aqueous solution of pH 7?
Glu Arg Val
Phe Ileu Asn
Lys Ser Thr

4.2. 3-D Structures of proteins

According to the structure of urea, deduce how this compound can promote denaturation of proteins.

4.3. 3-D Structures of proteins

Phenylalanine, a hydrophobic amino acid, is frequently found at the surface of natives and functional proteins. Give the most probable role of phenylalanine in this situation.

****4.4. 3-D Structures of proteins***

Aspartic acid, a charged amino acid, is frequently found inside of native and functional proteins.
Give the most probable role of phenylalanine in this situation.

4.5. 3-D Structures of proteins

The following table describes the amino acid compositions of three proteins.

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	Number of residues per molecule		
Amino acids	*protein 1*	*protein 2*	*protein 3*
Polar residues			
Arg	12.00	4.00	7.00
Asn	9.00	6.00	5.00
Asp	14.00	5.00	9.00
Cys	7.00	2.00	6.00
Gln	8.00	7.00	6.00
Glu	11.00	4.00	6.00
His	4.00	2.00	4.00
Lys	22.00	6.00	15.00
Ser	20.00	5.00	11.00
Thr	15.00	3.00	11.00
Trp	2.00	3.00	3.00
Tyr	7.00	7.00	6.00
Non-polar residues			
Ala	14.00	28.00	25.00
Gly	9.00	9.00	8.00
Ileu	5.00	16.00	9.00
Leu	3.00	19.00	7.00
Met	7.00	11.00	9.00
Phe	9.00	13.00	11.00
Pro	8.00	13.00	10.00
Val	16.00	29.00	21.00

protein C is a globular protein with four identical sub-units, deduce the corresponding amino acid composition of these proteins.

4.6. 3-D Structures of proteins

Indicate which secondary structure or structures (α -helix, β -pleated, random coil) will the following peptide adopt in an aqueous solution at pH 7

Ileu-Glu-Asn-Glu-Gln-Asn-Met-Ala-His-Phe-Trp-Tyr

4.7. 3-D Structures of proteins

Indicate which secondary structure or structures (α -helix, β -pleated, random coil) will the following peptide adopt in an aqueous solution at pH 7

Gly-Ala-Gly-Ala-Gly-Ser-Gly-Ala-Gly-Ser-Gly-Ala

4.8. 3-D Structures of proteins

Indicate which secondary structure or structures (α -helix, β -pleated, random coil) will the following peptide adopt in an aqueous solution at pH 7

Lys-Gly-Arg-Arg-Lys-Gly-Arg-Gly-Arg-Pro

4.9. 3-D Structures of proteins

Indicate which secondary structure or structures (α -helix, β -pleated, random coil) will the following peptide adopt in an aqueous solution at pH 7

1 10

Gly-Pro-Glu-Ser-Ala-Tyr-Lys-Thr-Leu-Phe-Asp-Val-Pro-Asp-Asp-Glu-Asp-Gly-Gly

20 26

Ser-Ala-Gly-Ser-Ser-Gly-Ala

4.10. 3-D Structures of proteins

The following table describes the amino acid composition of three proteins. Determine what structure these proteins will adopt: α-helical, β-pleated or a triple helix of collagen.

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protein	A	B	C	protein	A	B	C
Ala	29.40	5.00	10.70	Leu	0.50	6.90	2.40
Arg	0.50	7.20	5.00	Lys	0.30	2.30	3.40
Asp	1.30	6.00	4.50	Met	-	0.50	0.80
Cys	-	11.20	-	Phe	0.50	2.50	1.20
Glu	1.00	12.10	7.10	Pro	0.30	7.50	12.20
Gly	44.60	8.10	33.00	Ser	12.20	10.20	4.30
His	0.20	0.70	0.40	Trp	0.20	1.20	-
Hypro	-	-	9.40	Tyr	5.20	4.20	0.40
Ileu	0.70	2.80	0.90	Val	2.20	5.10	2.30

Chapter 5. Structure and properties of nucleic acids.

5.1. Nucleic acid structure.

Consider the following polynucleotide:

AUUACGUGGUGCACUCGGGAACAUCCCGAGUGCACCACGUAAUGGA

Draw the two most stable **intramolecular** secondary structures this polymer can adopt

**5.2. Nucleic acid structure.*

A solution of double stranded DNA is heated and then cooled to room temperature for two minutes. Predict, qualitatively, the variation in absorbance at 260 nm in the following conditions:

a) the solution is heated to a temperature slightly above Tm before being cooled;

b) the solution is heated to a temperature way above Tm before being cooled;

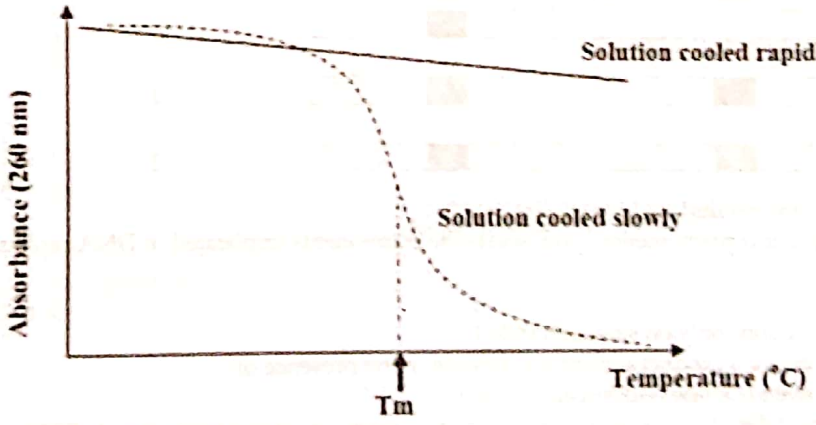
c) suggest the structure of two polynucleotides (synthetic or natural) which will result in an absorbance profile following a cooling which is the perfect inverse of the pattern obtained in (b).

5.3. Nucleic acid structure.

Explain why, RNA, and not DNA, is hydrolyzed under basic pH conditions.

5.4. Nucleic acid structure.

The following results were obtained during a denaturation/renaturation experiment of a simple nucleic acid (polyA : polyU). How would you interpret these results?



5.5. Nucleic acid structure.

IMP (inosine monophosphate) is present in chez *E. coli* as an intermediate of biosynthesis of purines and it is possible to incorporate IMP to DNA if the ITP (inosine triphosphate) is present in the reaction medium. However, in nature, IMP is never present in DNA. Propose an explanation.

5.6. Nucleic acid structure

What are the products of the digestion of the oligoribonucleotide 5'pACGAUGCUAUC3' by each of the following enzymes:

a) pancreatic ribonuclease;

b) T2 ribonuclease;

c) T1 ribonuclease;

5.7. Nucleic acid structure

Lets proceed to the analysis of an RNA molecule. Its global base composition is 2A, 2C, 1U, 1G.

Its treatment with the serpent venom phosphodiesterase yields pC.

Its hydrolysis by pancreatic ribonuclease yields 1C, a dinucleotide containing A and C, and a trinucleotide containing A, G, and U.

The action of RNase T2 yields pAp, a dinucleotide containing U and C and a trinucleotide containing A, G and C.

What is the primary structure of this RNA?

5.8. Nucleic acid structure

Let's proceed to the analysis of an RNA molecule whose global base composition is 2A, 4C, 2G, 1U.

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Pancreatic ribonuclease treatment yields 2Cp, two dinucleotides, one containing G and C and the other containing A and U, and a trinucleotide containing A, C and G.
A mixture of RNase T1 and RNase T2 yields C, Ap, pGp and two trinucleotides, one containing A and C and the second containing CG and U. The serpent venom phosphodiesterase yields pC.
What is the formula of this RNA?

5.9. Nucleic acid structure

What is the global charge of the trinucleotide ApGpUpC at neutral pH?

6.10. Nucleic acid structure

Why does a circular double stranded DNA renature more rapidly than a linear double stranded DNA?

6.11. Nucleic acid structure

Why does DNA denature in pure water, that is where the ionic strength is close to zero?

6.12. Nucleic acid structure

The size of the E. coli chromosome is 4000 kpb. What length of DNA does it contain?

6.13. Nucleic acid synthesis.

During an experiment similar to that performed by Meselson and Stahl, you grow bacteria for 3 generations (instead of 2 as in the classic experiment) in a mixture containing only 14N.
Following DNA isolation and analysis by analytical centrifugation, what proportion of heavy DNA, hybrid DNA and light DNA will you obtain?

6.14. Nucleic acid synthesis.

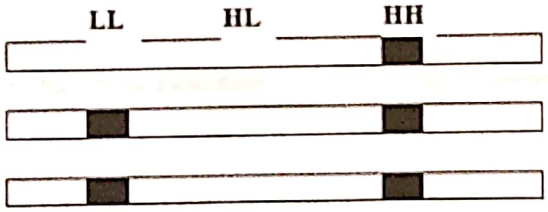
An isolated strand (+) of DNA (base composition: 10% of A, 20% of G, 30% of C and 40% of T) is replicated by E. coli DNA polymerase into a complimentary strand (-). The double stranded
DNA is then used as a model for the E. coli RNA polymerase which transcribes the (-) strand.
Indicate the base composition of the formed strand (in % of A, C, G of T/U).

****6.15. Nucleic acid synthesis.***

The time required to completely synthesise the E.coli genome is 40 minutes. However, it takes only 20 minutes for these bacteria to produce one generation. Can you explain this paradox?

*****6.16. Nucleic acid synthesis.***

You are the first scientist to successfully analyze a micro-organism found on Mars. Because this bacterium contains double-stranded DNA as genetic material, you decide to analyze using Meselson-Stahl techniques. You obtain the following results:



a) how would you interpret these results?
b) in order to better understand this phenomenon, you isolate the components implicated in DNA replication in this organism. You identify:
- a RNA polymerase activity;
- a DNA polymerase which functions only on single-stranded;
- a new enzyme which can generate a product sensitive to DNAse in the presence of NADH and a product insensitive to DNase and resistant to heat.

According to this information, deduce the mechanism by which this micro-organism replicates its DNA.

Solutions with explanations

Acid-Base Equilibrium and Spectrophotometry

1.1 Acid-base equilibrium:

a) Since HCl is a strong acid, it will completely dissociate when in solution:
HCl ^ H+ + Cl-
Stoichiometry tells us that, since the initial HCl concentration is 0.35M, the final concentration of H+ in the solution will also be 0.35M. This gives us:
pH = - log[H+] = -log 0.35 = 0.46

b) Acetic acid will also dissociate in solution:
CH3-COOH 4 * CH3-COO- + H+
However, since it is a weak acid, it will not completely dissociate, and we have to take into account the association constant (Ka) in our calculations. This constant is described as follows:
pKa = - log Ka
$Ka = 1/10^{pKa} = 1.74 \times 10^{-5}M$

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We can now easily determine the H+ concentration:
Ka = [H+] [CH3COO-]
[CH3COOH]
1.74×10^{-5}M = [H+] [CH3COO-]
0.35 M
1.74×10^{-5}M x 0.35M = [H+] [CH3COO-] = $[H+]^2$
$[H+] = (6.09 \times 10^{-6} M^2)^{1/2} = 2.47 \times 10^{-3}$M
Finally: pH = - log [H+] = - log $2,47 \times 10^{-3}$ = 2.61
c) Following the same steps as in (b), we get a pH of 3.11.

1.2 Acid-base equilibrium:

a) We have a weak acid. The acid-base equilibrium is:
HA + * H+ + A
We can determine Ka as follows:
Ka = rH+l [A-]
[HA]
The question stipulates that this acid is only 2% ionised (or 0.02, that's the same thing). This allows us to obtain the respective concentrations of species HA, H+ and A- :
[H+] = [A-] = 0.20M x 0.02 = 0.004M
[HA] = 0.2M - [H+] = 0.196 M
Therefore :
Ka = [H+] [A-]
[HA]
Ka = 0.004M x 0.004M 0.196 M
$Ka = 8.16 \times 10^{-5}$ M
b) The pH of this solution is: pH = - log [H+] = - log 0.004M = 2.39

1.3 Acid-base equilibrium:

a) This mixture is a buffer solution made of acetic acid and its conjugated base, sodium acetate:
CH3COOH → H+ + CH3COO- Na+
This pH of this type of solution can be determined with the Henderson-Hasselbach equation:
pH = pKa + log [Conjugated base] [Acid]
pH = 4.76 + log 0.5M = 4.46 1 M
b) We have the following acid-base equilibrium:
H3PO4 ←→H+ + H2PO4- K+
Using the same procedure as in (a), we get:
pH = pKa + log M7PO4-] [h3po4]
pH = 2.14 + log 0.8 M 0.3 M
pH = 2.57

1.4 Acid-base equilibrium:

We have the following equilibrium:
H2PO4- → H+ + HPO4-2
And the pKa for this equilibrium is 7.21.
The Henderson-Hasselbach equation gives us:
pH = pKa + log rHPO4-2l [H2PO4-]
7.00 = 7.21 + log rxl 0.1 M
-0.21 = log x - log 0.1 M -0.21 + log 0.1 M = log x = -1.21
$x = 10^{\log x} = 0.062$ M

1.5 Acid-base equilibrium:

We have the following acid-base equilibrium:
H2PO4- → H+ + HPO4-2
According to the question, we have: $[H_2PO_4^-] + [HPO4"^2] = 0.3M$
Hence: [H2PO4"] = 0.3 M - [HPO4"2]
From the Henderson-Hasselbach equation, we have:
pH = pKa + log [HPO4-2] [H2PO4-]
7.00 = 7.21 + log [HPO4-2]
[H2PO4-]
7.00 = 7.21 + log [HPO4-2]
[H2PO4-]
-0.21 = log [HPO4-2] [H2PO4-]
$10^{-021} = [HPO4^{-2}]$
[H2PO4-]
$0.616 = [HPO_4^{-2}]$
[H2PO4-]
0.616 x [H2PO4-] = [HPO4-2]

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Which is identical to :
$0.616 \times (0.3M - [HPO_4^{-2}]) = [HPO_4^{-2}]$
$0.185 - 0.616 \times [HPO4^{-2}] = [HPO4^{-2}]$
$0.185 = 1.616 \times [HPO4^{-2}]$
$0.114\ M = [HPO4^{-2}]$
And the concentration in $H_2PO_4^-$ will be :
$[H_2PO_4^-] = 0.3M - [HPO_4^{-2}] = 0.186\ M$

1.6 Spectrophotometry

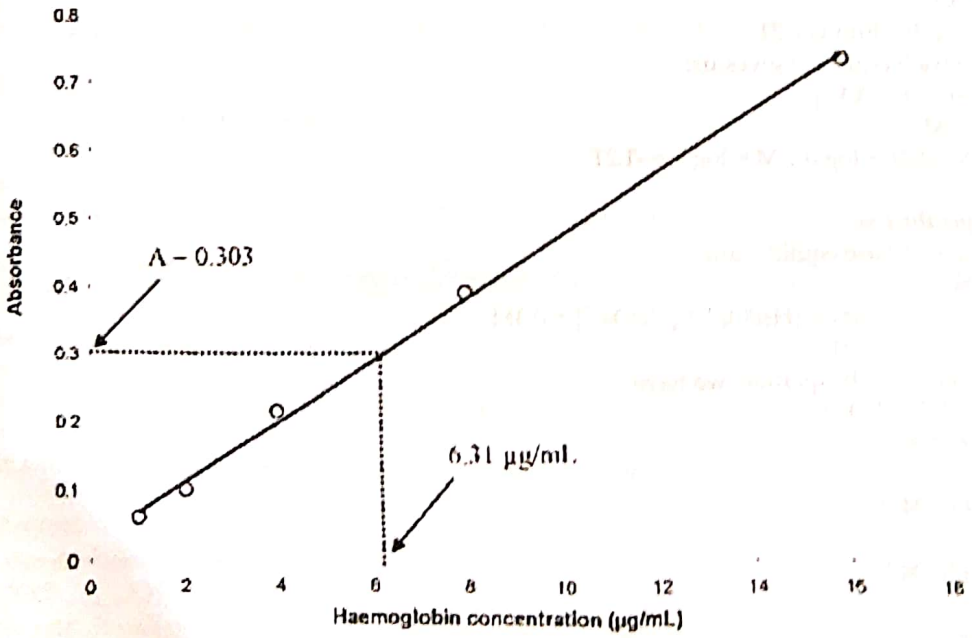
The relationship between the absorbance and the concentration of a solution is given by the Beer-Lambert equation:
A = scl
Where:A = absorbance
s = Molar extinction coefficient (units: litres x mol^{-1} x cm^{-1}) c = concentration (units: mol/l = M) l = light path (thickness of the cuvette, units: cm)
We get the following:
0.71 = 1.420 L mol^{-1} cm^{-1} x c x 1 cm c = 0.71 mol cm 1420 L x 1 cm
c = 5 x 10^{-4} M
If we use a cuvette where c=0.1 cm, we get:
0.71 = 1.420 L mol^{-1} cm^{-1} x c x 0.1 cm c = 0.71 mol cm 1420 L x 0.1 cm
c = 0.005 M

1.7 Spectrophotometry

With the Beer'Lambert equation, we have:
A = scl
Therefore:
A = 1420 L mol^{-1} cm^{-1} x (37 x 10^{-3}M) x 1cm A = 52.54

1.8 Spectrophotometry

We first have to graph the standard cuve of the absorbance as a function of the concentration of the haemoglobin standards. This graph is shown on the following page.
Since the unknown has an absorbance of 0.303, we can use the standard curve to determine the corresponding haemoglobin concentration, in this case 6.31pg/mL.
A more accurate value can be obtained by using the familiar equation:
y = mx + b
where y = value on the y axis x = value on the x axis m = slope
b = intersect on the y axis
The values for m and b are easily obtained from the graph or by linear regression of the data (the latter being, by far, the best method).
Therefore, we get:
y = (0.0441 mLpg^{-1})x + 0.0246
x = (y-b)/m
x = (0.303-0.0246)/0.0441mLpg^{-1} = 6.31 pg/mL.



Chapter 2: Amino acids

**** 2.1. Molecular mass of an amino acid.***

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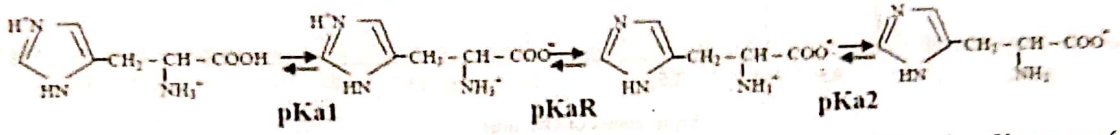
By adding NaOH, we shift the acid-base equilibrium of the amino acid towards the base:
NH_3+-CH(R)-COO- → NH_2-CH(R)-COO-
With the Henderson-Hasselbach equation, it is possible to obtain the proportion of the amino acid in the base and acid form after the addition of NaOH:
pH = pKa + Log [base] / [acid]
10.4 = 9.7 + Log [base] / [acid]
0.7 = Log [base] / [acid]
5.011 = [base] / [acid]
Since 0.01 mol of NaOH (100 mL of a 0.1M solution) was required to obtain a pH value of 10.4, this indicates that 0.01 mol of this amino acid has been converted to the basic form. Therefore:
5.011 = [base] / [acid]
5,011 = 0,01 mol /[acid]
[acid] = 0,002 mol
The sum of the amount of amino acid in the base and acid forms will give us the total amount of amino acid in the solution, which is 0,012 mol. All we have to do next is determine the molecular mass:
1,812 g = 0,012 mol
Therefore, the molecular mass of the amion acid is: g / mol = 1,812 g /0,012 mol = 151 g / mol

2.2. *Titration of amino acids.*

The titration of histidine involves the following equilibria:



To draw the titration curve, we need the values for the pKas and the inflexion points. For histidine, the pKas are as follows:
pKa1: 1.82 pKaR: 6.00 pKa2: 9.17
By definition, the pKa is the pH where half the the amino acid lost a proton, while the other half is still protonated.
The inflection points are obtained by the average value of two consecutive pKas:
Inflection poin # 1: (1.82 + 6.00) / 2 = 3.91
Inflection point # 2: (6.00 + 9.17) / 2 = 7.59
The pI is defined as the pH where the net charge of the amino acid is 0. It is determined by calculating the average of the pKas preceding and following the inflexion point where the amion acid carries no net charge. For histidine, the pI will be 7,59 (inflection point #2).
We can now easily draw the titration curve (see next page).

2.3. *Net charge of amino acids.*

To determine the net charge of amino acids at various pH values, we first need to determine the pI for each amino acid With this information, we can deduce the charge of the amino acids: by definition, the amino acid carries no net charges with the pH equals the pI. When the pH is lower than the pI, the amino acid will carry a net positive charge. When the pH is superior to the pI, then the amino acid carries a net negative charge.
Therefore, we obtain the following:

pH	*Glycine (pI: 5.97)*	*Serine (pI: 5.68)*	*Aspartic Acid (pI: 2.77)*	*Glutamine (pI: 5.65)*	*Arginine (pI: 10.76)*
2.01	+	+	+	+	+
3.96	+	+	-	+	+
5.68	+	0	-	0	+
10.76	-	-	-	-	0

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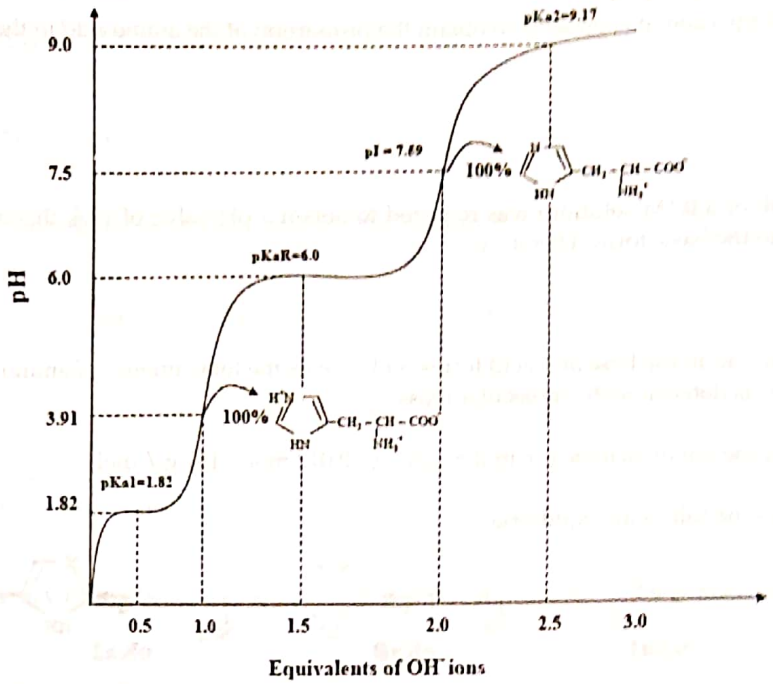
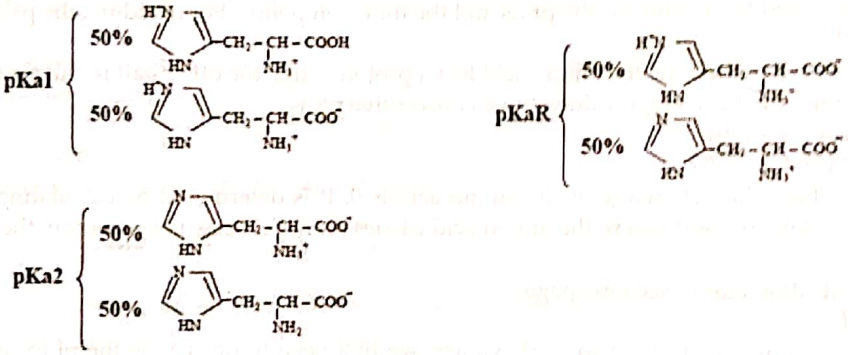
Problem 2.2.





2.4. *Ion exchange chromatography.*

We start the chromatography at pH 10. At this particular pH, all the amino acids in the mixture are negatively charged (the pH greater than the pI of each amino acid).

a) using a ***cation*** exchange chromatography, none of the amino acids will stick to the resin and they will all be found in the eluent.

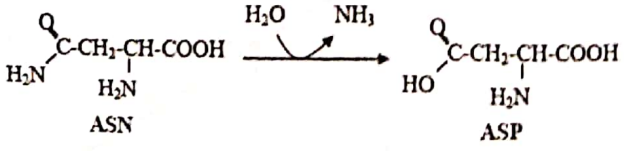
b) using an anion exchange chromatography, all the amino acids will bind the resin. As we decrease the pH, the amino acids will progressively elute when the pH of the buffer becomes lower that their pI. We will collect the amino acids in this order:

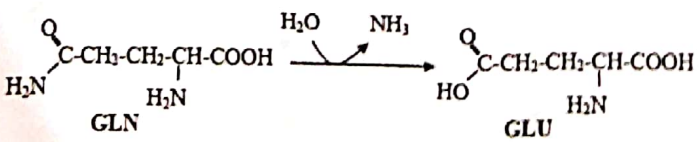
1- lysine
2- isoleucine, alanine, glycine (all three will elute pretty much at the same time since their pI is similar)
3- glutamic acid.

Obviously, the anion exchange resin is the best choice for this experiment.

2.5. *Amino acids*

Only two amino acids can be converted to other amino acids and at the same time generate ammonia: asparagine and glutamine:





2.6. *Amino acids

This observation can only be explained if we assume that serine is first inserted into casein, and that a phosphate group is then added one protein synthesis is over, producing phosphoserine. The experimental data confirm this hypothesis.

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*2.7. *Ion exchange chromatography.*

Since these four amino acids can be separated by ion exchange chromatography, but that their pI is virtually identical, another physico-chemical property must be responsible for this behaviour. A close look at the structure of glycine, alanine, valine and leucine reveals a progressive increase in the hydrophobic character of their side chain. We can deduce that these amino acids can establish hydrophobic interactions with the ion exchange resin, allowing their separation.

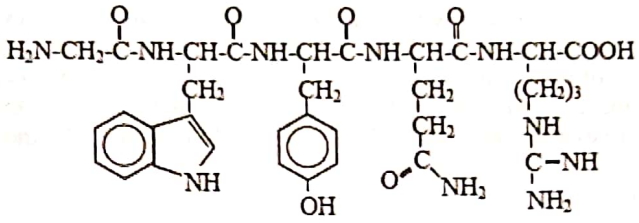
2.8. Peptides.

Since the yield in leucine, arginine and tryptophan is similar, we can conclude that they are present in equal propotions in the polypeptide. Twice this amount was obtained as glutamate. Finally, alanine and lysine were recovered in proportions corresponding to four times the amount of arginine/leucine/tryptophan. Therefore, the empirical formula of this peptide would be:

$(Arg, Leu, Trp, Glu^{\wedge} Ala^{*} Lys4)_n$ This experiment does not allow us to determine the amino acid sequence of this peptide.

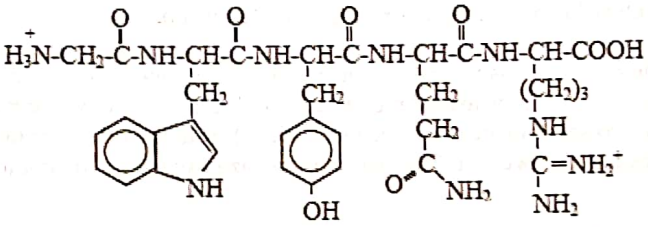
2.9. Peptides.

The structure of the peptide GWYQR (Glycyl-Tryptophanyl-Tyrosyl-Glutaminyl-Arginine) is:

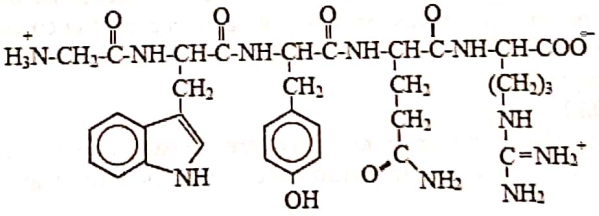


To determine the form of this peptide at each pH, all we have to do is note the pKa values of all the charged groups (including the side chains), and to deduce the ionization status of each of these groups (pH<pKa: protonated; pH>pKa: non-protonated):

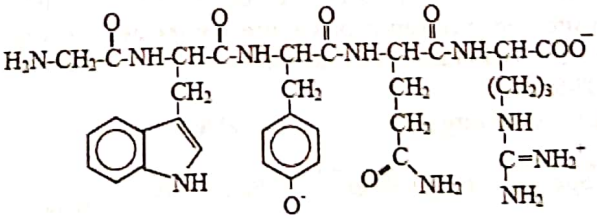
At pH 2: Every group is protonated:



At pH 7: The carboxyl group of arginine is ionized (pH>pKa). However, the side chain of arginine and the amino group of glycine remain protonated (pH<pKa):



At pH 10.5: Here, the amino group of glycine and the side chain of tyrosine are deprotonated (pH>pKa). However, the amino group of the side chain of arginine remains protonated (pH<pKa):



Chapter 3. General properties and purification of proteins

3.1. Protein purification.

Ammonium sulphate precipitation enables the concentration of our favorite protein by the nonspecific precipitation of a large proportion of the proteins of the extract. We can therefore easily obtain a partial purification of our favorite protein, which can then be purified further using other methods.

3.2. Protein purification.

The diethylamino group ($-CH_2-CH_2-NH^{+}-CH_2-CH_3$) of DEAE-cellulose carries a positive charge which is responsible for the ion-binding properties of this resin. Effectively, negatively charged amino acids/proteins will interact with the diethylamino group (via electrostatic interactions), while positively charged amino acids/proteins will be eluted. Since the diethylamino group has a pKa close to 8.5, it will be deprotonated at pH values above 8.5 and will use all ability to bind negatively charged molecules.

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3.3. *Protein purification.*
At a pI above 6, 6-phosphogluconate dehydrogenase has a net negative charge (pH>pI): it will bind the resin. At a pH value above 9, the diethylamino group of the resin is deprotonated, preventing any separation of the enzyme as a function of its charge.

3.4. *Protein purification.*
No because CM-cellulose (CM = carboxymethyl = $-CH_2-COOH$) is a cation-exchange resin: at a pH above 6, 6-phosphogluconate dehydrogenase is negatively charged (see problem 3.3) and will not bind the resin.

3.5. *Protein purification.*
In order to separate 6-phosphogluconate dehydrogenase using a CM-cellulose column, we must ensure that the protein has a net positive charge. The pH of the buffer will have to be below the protein's pI, therefore below a value of 6.

3.6. *Protein purification.*
The biochemist has 2 choices when it comes to eluting proteins bound to ion exchange resins: use a pH gradient, or a salt gradient (usually NaCl).
In the situation where a salt gradient is chosen, one would usually start with a buffer of low NaCl concentration (i.e. low ionic strength), and then progressively introduce a buffer with a greater and greater salt concentration. The Na^+ or Cl^- ions will elute proteins from the column by neutralizing the negative or positive charges on the proteins which interact with the resin. The end result is the elution of proteins as a function of their charge density: to be eluted from a cation exchange resin, those proteins with less positive charges will require less Cl^- ions (thus, a lower NaCl concentration) to neutralize them than those proteins with more positive charges. The same logic can be used to explain the ability of Na^+ ions to elute proteins bound to anion exchange resins (e.g. CM-cellulose).
Regarding question 3.6, we can conclude that:
- myoglobin will elute first, since it will not bind to the resin (pH<pI: positive net charge);
- the pI of urease (5,4) is closer to the pH of the buffer that is the pI of ovalbumin (4,6): urease will therefore carry a a lower number of negative charges than ovalbumin. We can therefore predict that urease will elute at a NaCl concentration which will be lower than the one required to elute ovalbumin.

The order of protein elution will therefore be: myoglobine, urease, and ovalbumin.

3.7. *Protein purification.*
A fast and simple way to separate these proteins is to first perform an ion exchange chromatography: it will then be possible to get rid to the protein whose pI is different that our favorite enzyme. As a second step, we can perform a molecular sieve (or size exclusion, that's the same thing) chromatography to separate our enzyme (Mr 24 kDa) from the other contaminating protein (Mr 100 kDa).
Note: Similar results would be obtained if one were to first perform the size exclusion chromatography and then the ion exchange chromatography.

3.8. *Protein purification.*
a) ***Specific activity*** is defined as the enzymatic activity per mg protein and is an indication of the relative concentration of the enzyme in the solution (the greater the amount of enzyme in the solution ***relative to all other proteins,*** the greater the specific activity will be). All we have to do is to divide the enzymatic activity obtained at each step by the corresponding protein concentration. For example, for the heat treatment step, we get:
specific activity = enzymatic activity / mg protein
specific activity = 1 980 U/16 500 mg = 0,12 U/mg
The ***percent yield*** is defined as the amount of enzyme (or protein) recovered at each step with reference to the amount present at the start of the purification procedure. This value is obtained by dividing the enzymatic activity at step X by the initial enzymatic activity. Again, for the heat treatment step, we get
Percent yield = Enzyme activity step X / Initial enzyme activity
Percent yield = (1 980 U/2 700U) x 100 = 73,3 %
The ***degree of purification*** is obtained by dividing the specific activity after step X by the initial specific activity. We therefore obtain the fold increase in enzyme purification after the purification procedure. For the heat treatment step, we get:
Degree of purification = Specific activity step X/ Initial specific activity
Degree of purification = 0,12/0,039 = 3,07-fold.
The results for each step of the purification procedure are shown in the table below:

Purification step	*Specific activity (U/mg)*	*Percent yield*	*Degree of purification (fold increase)*
Cell extract	0,039	100%	---
Ammonium sulphate	0,09	85,2%	2,30
Heat treatment	0,12	73,3%	3,07

Purification step	*Specific activity (U/mg)*	*Percent yield*	*Degree of purification (fold increase)*
DEAE chromato.	4,31	62,2%	110,50

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CM-cellulose chromato.	28,72	50%	736,40
Bio-Gel A	32,00	41,5%	820,50

b) To determine which purification step was the most effective, all one has to do is divide the value for the degree of purification of step X by the value for the preceding step. Thus, the DEAE chromatography step was the most efficient, with a 36 fold increase in enzyme purity.

c) Considering that the protein is pure after molecular sieve chromatography, 35 mg of 6- gluconate dehydrogenase were obtained. This corresponds to 41,5 % of the amount of enzyme initially present in the extract (refer to percent yield). Thus, in the initial extract we had:

35 mg / 41,5% = 75,9 mg

And this amount of enzyme was initially present in an extract containinfg a total of 70 000 mg proteins. Thus, in the initial extract we had:

(75,9 mg / 70 000 mg) x 100 = 0,108 % of 6-gluconate dehydrogenase

3.9. ***Protein purification.***

The aim of isoelectric focusing is to separate proteins according to their pI, thus according to their charge at different pH values. Adding SDS to the protein sample would give all proteins the same charge density and would prevent their separation by this type of electrophoresis.

3.10. ***Protein purification.***

To determine the molecular mass of our unknown protein, we must first draw the graph of the elution profile of our sandards as a function of the log of their molecular mass (see graph on following page).

Protein	Log Mr	Vel (ml)
dextran blue	6.000	85.00
Lysozyme	4.146	200.00
Chymotrypsinogen	4.398	190.00
Ovalbumin	4.653	170.00
serum albumin	4.813	150.00
aldolase	5.176	125.00

Protein	Log Mr	Vei (ml)
urease	5.699	90.00
Ferritin	5.845	92.00
ovomucoide	4.447	160.00

From this graph, we obtain a molecular mass of 139 kDa for our unknown protein.

Note: ferritin and ovomucoid were excluded of the standard curve because they obviously behaved differently to the other proteins when subjected to molecular sieve chromatography.

Graph, problem 3.10

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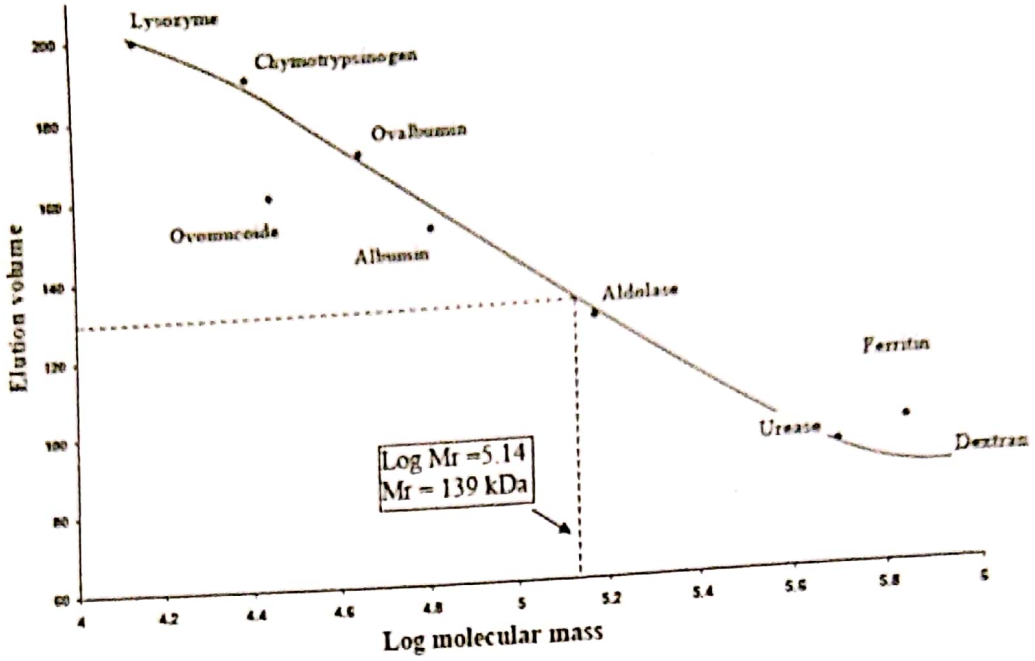

*3.11. *Protein purification.*
Ferritin has an iron core, giving it a greater density than other proteins of similar size and influencing its behaviour when subjected to molecular sieve chromatography.

3.12. Protein purification.
In the presence of SDS, all the proteins have an identical negative charge density: this is what allows us to use PAGE-SDS to determine the molecular mass of proteins. Therefore, two proteins of different pI but identical molecular mass will co-migrate as a single band on PAGE- SDS. However, in the absence of SDS (thus under ***native*** or ***non-denaturing*** conditions), protein migration towards the positive or negative electrodes will be driven by its net charge, in other words by the protein's pI. In this case, two proteins of identical molecular mass but different pI will give two distinct bands upon gel staining.

3.13. *Protein purification*
In the absence of P-mercaptoethanol, the protein's disulfide bonds remain intact. Thus, the protein will have a more compact shape and will migrate more rapidly during PAGE-SDS than the same protein whose disulfide bonds have been reduced.

3.14. *Peptide sequencing.*
a) Trypsin hydrolyzes the peptide bond on the carboxyl-side of the basic amino acids lysine and arginine. Therefore, every peptide fragment generated by trypsin will have Arg or Lys at their C- terminus (with the exception, of course, of the fragment corresponding to the C-terminal end of the peptide). Therefore, the peptide given in this example will give us the following fragments:
A-L-K M-P-E-Y-I-S-T-D-Q-S-N-W-H-H-R
b) Pepsin hydrolyzes the peptide bond on the N-terminal side of the aromatic amino acids Phe, Trp, and Tyr. Therefore, the fragments obtained after pepsin digestion will all contain Tyr, Phe or Trp at their N-terminus (with the notable exception of the fragment corresponding to the N- terminus of the initial peptide). Using the peptide shown here, we obtain the following fragments:
A-L-K-M-P-E Y-I-S-T-D-Q-S-N W-H-H-R
c) Protease V8 hydrolyzes the peptide bond on the C-terminal side of the acidic amino acids Asp and Glu. Therefore, every peptide fragment generated by protease V8 will have Asp or Glu at their C-terminus (with the exception, of course, of the fragment corresponding to the C-terminal end of the peptide). Therefore, the peptide given in this example will give us the following fragments:
A-L-K-M-P-E Y-I-S-T-D Q-S-N-W-H-H-R
d) Cyanogen bromide hydrolyzes the peptide bond on the C-terminal side of Met. Therefore, every peptide fragment generated by cyanogen bromide will have Met at their C-terminus (with the exception, of course, of the fragment corresponding to the C-terminal end of the peptide). Therefore, the peptide given in this example will give us the following fragments:
A-L-K-M P-E-Y-I-S-T-D-Q-S-N-W-H-H-R

3.15. Peptide sequencing
Digesting with carboxypeptidase A tells us that the C-terminal residue of the peptide is Ala.
Digesting with trypsin allows us to partially order two of the four fragments (remember: trypsing generates fragments whose C-terminal end is Arg or Lys):
Ala-Arg (Phe, Ser)-Lys
Furthermore, since digesting with trypsin generates a free Lys residue, this indicates that this Lys is on the C-terminal side of either Arg or Lys.
Trypsin digestion also indicates that the tripeptide (Ala, Met, Ser) is the C-terminus of the peptide (it doesn't end with Arg or Lys).
Furthermore, CNBr digestion allows us to determine the position of Met in this tripeptide:
Met-(Ala, Ser)

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Since Ala is the C-terminal residu of this peptide (see digestion with carboxypeptidase A), the sequence of the last 3 amino acids of the polypeptide will be:
Met-Ser-Ala
Thermolysin cuts the peptide bond on the N-terminal side of hydrophobic amino acids: we can therefore deduce the position of the hydrophobic amino acids of the two fragments:
Ala-(Arg, Ser) and Phe-(Lys, Lys,)-Met-Ser
With this information, and considering the trypsin digestion pattern, we can conclude that the sequence of this polypeptide is:
Ala-Arg-Ser-Phe-Ly s-Lys-Met-Ser-Ala
3.15. Peptide sequencing
Digesting with carboxypeptidase A tells us that the C-terminal residue of the peptide is Ala.
Digesting with trypsin allows us to partially order two of the four fragments (remember: trypsing generates fragments whose C-terminal end is Arg or Lys):
Ala-Arg (Phe, Ser)-Lys
Furthermore, since digesting with trypsin generates a free Lys residue, this indicates that this Lys is on the C-terminal side of either Arg or Lys.
Trypsin digestion also indicates that the tripeptide (Ala, Met, Ser) is the C-terminus of the peptide (it doesn't end with Arg or Lys).
Furthermore, CNBr digestion allows us to determine the position of Met in this tripeptide:
Met-(Ala, Ser)
Since Ala is the C-terminal residu of this peptide (see digestion with carboxypeptidase A), the sequence of the last 3 amino acids of the polypeptide will be:
Met-Ser-Ala
Thermolysin cuts the peptide bond on the N-terminal side of hydrophobic amino acids: we can therefore deduce the position of the hydrophobic amino acids of the two fragments:
Ala-(Arg, Ser) and Phe-(Lys, Lys,)-Met-Ser
With this information, and considering the trypsin digestion pattern, we can conclude that the sequence of this polypeptide is:
Ala-Arg-Ser-Phe-Ly s-Lys-Met-Ser-Ala
Since the disulfide bonds of the peptide were intact when the peptide was digested with thermolysin, some of the peptide fragments shown above will be linked together with disulphide bonds involving Cys residues. From the fragments obtained, we can deduce the position of the disulphide bonds as follows:
A-C-F-P-K-R-W-C-R-R-V-C
S - S
C-Y-C-F-C - S – S J
Note: Don't forget that both inter-chain and intra-chain S-S- bonds can be present in the molecule.
3.17. Peptide sequencing
Digestion with carboxypeptidase tells us that the C-terminus is Lys.
DNFB treatment indicates that the N-terminus is Val.
Trypsin digestion allows us to partially order the peptides, as follows:
peptide C: Try-Ala-Lys peptide D: Val-(Ala, Ala, Ala, Pro)-Lys (remember: Val = N- terminus)
peptide E: Met-(Asp, Gly)-Arg
We can order the rest of the residues with the results from CNBr treatment:
Met-Gly-Asp-Arg
Finally treating peptide D with thermolysine allows us to order the three Ala and the Pro:
Val-Ala-Ala-Ala-Pro-Lys
Therefore, the sequence of the peptide is:
Val-Ala-Ala-Ala-Lys-Pro-Met-Gly-Asp-Arg-Try-Ala-Lys

Chapter 4. Three dimensional structures of proteins 4.1. 3-D structure of proteins
Generally speaking, hydrophobic amino acids are found buried inside proteins (away from water), while polar and charged amino acids are most often found on the surface of proteins. We will then get the following distribution for the amino acids:
Buried inside: Val, Phe, Ileu
On the surface: Glu, Arg, Asn, Lys, Ser, Thr
**4.2. 3-D structure of proteins*
The structure of urea suggests that this molecule denatures proteins by breaking the hydrogen interactions which stabilize the 3-D structure of the macromolecules (i.e. via the interaction of the amino and ketone groups of urea with the amino and ketone groups of the peptide bonds and side chains).
4.3. 3-D structure of proteins
Even though it is located at the surface of proteins, Phe must avoid contact with water. This can be accomplished if two or more protein subunits interact via hydrophobic regions (which could include Phe), keeping Phe in an hydroiphobic environment.
**4.4. 3-D structure of proteins*
The presence of Asp inside proteins is possible if its polar and charged groups are involved in intermolecular interactions. This is possible if Asp is part of asecondary structure like the a- helix.
4.5. 3-D structure of proteins

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Protein 1 has a high amount of hydrophilic amino acids, and very little hydrophobic residues (65% hydrophilic/35% hydrophobic). This suggests that a lot of these amino acids will be interacting with the solvent, which is the case for rod-shaped proteins (i.e. proteine A).

Protein 3 has the same amount of hydrophobic vs hydrophilic amino acids, while protein 2 has many more hydrophobic than hydrophilic residues (proteine 2 = 30%/70%). This suggests that protein 3 would be globular, with several hydrophobic amino acids buried inside and lots of hydrophilic amino acids on the surface. Protein 3 would therefore be protein B.

As for protein 2, its high content in hydrophobic amino acids little content in hydrophilic amino acids suggest that it could be protein C: the association of several subunits would regions of high content of hydrophobic amino acids, protecting them from the aquous environment.

4.6. *3-D structure of proteins*

The primary structure of this peptide doesn't have any features expected from a P-pleated sheet, and no Gly and Pro (which are known to disrupt secondary structures) are present. We can therefore deduce that this peptide would adopt an a-helical structure.

4.7. *3-D structure of proteins*

The primary structure of this peptide is typical for those arranged as P-pleated sheets.

4.8. *3-D structure of proteins*

The presence of several positively charged residues (Arg and Lys) in addition to Gly indicates that this peptide will most likely be a random coil.

4.9. *3-D structure of proteins*

With the criterias use in the 3 preceding problems, we can deduce that:

Amino acids 2-12: a-helix;

Amino acids 13-17: random coil;

Amino acids 18-26: P-pleated sheet

4.10. *3-D structure of proteins*

The high amount of Pro and Hypro indicates that protein C will be a collagen-like triple-helix.

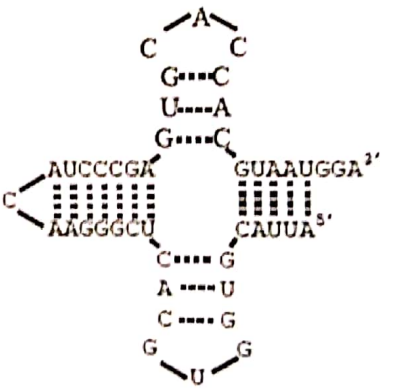
Protien A is rich in in amino acids with small side-chains (Gly, Ser, Ala): it will adopt a P- pleated sheet type of structure.

Protein B has a lot of amino acids that would be expected to be found in a-helices. However, the integrity of this helix would be severely perturbed by the presence of Gly and Pro and by the presence of several consecutive acidic or basic amino acids.

Chapter 5. Structure and properties of nucleic acids.

5.1. *Structure of nucleic acids.*

The polynucleotide shown here is an RNA molecule (notice the presence of uracil). RNA can for double-stranded structures if complementary sequences are present in the molecule. As for DNA, the more stable secondary double-stranded RNA structures will be those with the most base pairs (therefore, the most H-bonds). For the RNA molecule described here, we get these two secondary structures:



*5.2. *Structure of nucleic acids.*

a) Since Tm has not yet been reached, the two polynucleotide chains are not completely separated and are still aligned according to their complementary sequence. We would obtain the following melting curve:

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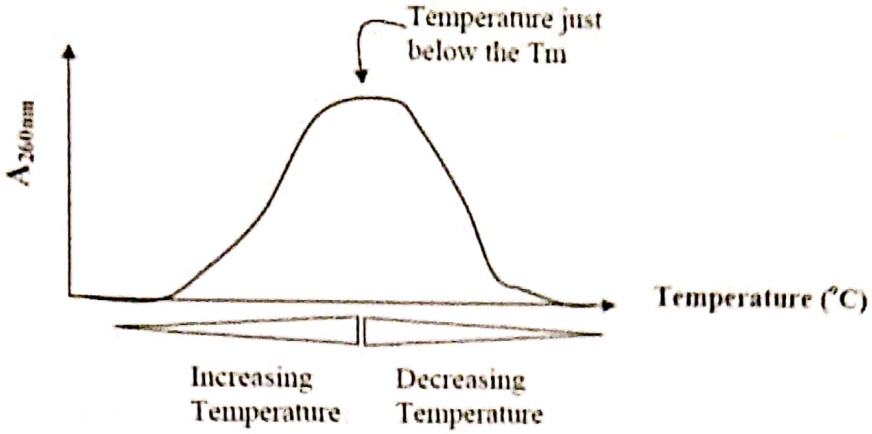
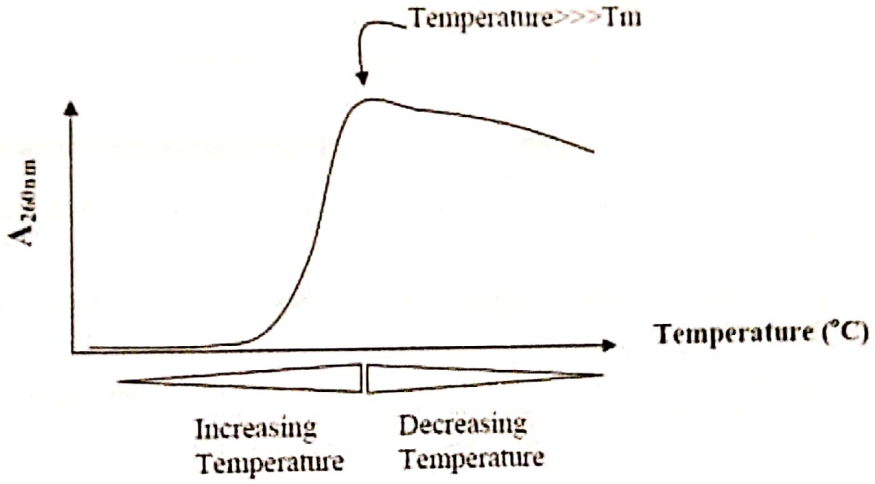


b) Since the temperature is increased way above the Tm, both polynucleotide chains have been separated. Upon cooling the temperature, both chains will have to realign to form complementary base pairs: since this is a random process, it can take a while. Since the time allotted after cooling is short (2 min), the DNA molecule will not have time to completely rehybridize, the absorbance will decrease only very slowly and we will get a curve like this one:

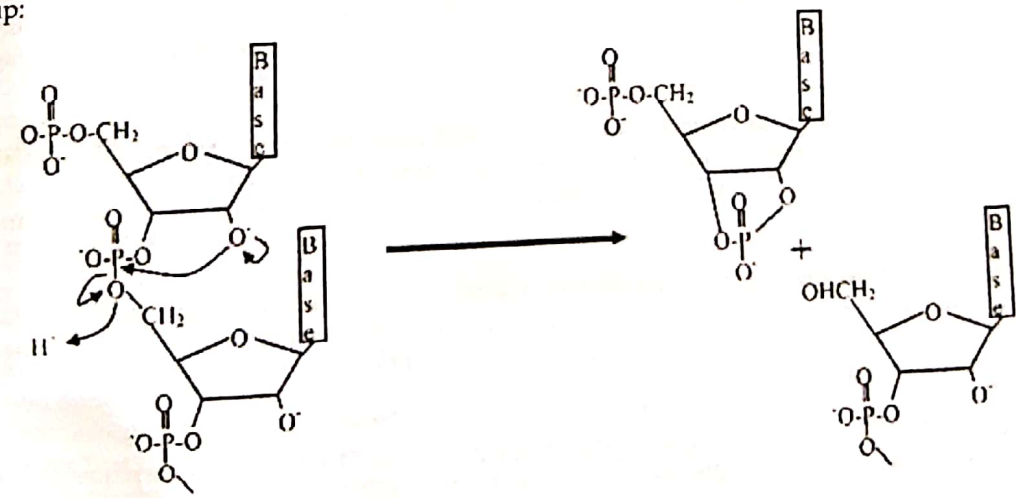


c)

c) A perfectly reversible melting curve can be obtained if the alignment between complementary base pairs can readily be achieved. This is the case when polynucleotide of low complexity are used, like poly(G):poly(C), or poly (AT):poly(AT).

5.3. Structure of nucleic acids.

Nucleic acid hydrolysis involves the break of a phosphodiester bond linking each nucleotide in the 5' -> 3' orientation. The only difference between RNA and DNA is the presence of a 2' hydroxyl (2' OH) group on the ribose sugar of RNA. Therefore, in the presence of a base, this group can be easily ionized, generating an O^- ion that can easily break the phosphodiester bond by attacking the phosphoryl group:



5.4. Structure of nucleic acids.

These data can be interpreted in the following fashion:

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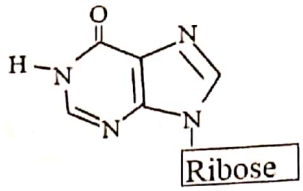
a) if we *rapidly* cool down the nucleic acid solution, the absorbance at 280nm will not vary by much, indicating that the polynucleotides remained single-stranded (in other words, the DNA molecule stays *denatured*).
b) if we slowly cool-down the nucleic acid solution, we observe a progressive but steady decline in the absorbance at 280 nm, indicating that our polynucleotide has become double-stranded (in other words: the two strands *rehybridized*).

5.5. Structure of nucleic acids.
The structure of IMP is as follows:



IMP can form H bonds with *both* adenosine and cytosine (compare the structure of IMP with AMP and CMP). Therefore, incorporating IMP into DNA would be highly detrimental to the organism by creating an ambiguity in base pairing.

5.6. Structure of nucleic acids.
a) pancreatic ribonuclease cut RNA on the 3′ end of *pyrimidine* nucleotides, generating a 3′ phosphate end. In the polynucleotide given in this problem, pancreatic ribonuclease will yield the following digestion products:
^{5}pACp$^{3'}$ $^{5'}$GAUp$^{3'}$ ^{5}GCp$^{3'}$ $^{5'}$Up$^{3'}$ ^{5}AUp$^{3'}$ ^{5}C$^{3'}$
b) ribonuclease T2 cuts on the 3′ end of adenosine, and generates a 3′ phosphate end:
^{5}pAp$^{3'}$ ^{5}CGAp$^{3'}$ ^{5}UGCUA$^{3'}$ $^{5'}$UC$^{3'}$
c) ribonuclease T1 cuts on the 3' end of guanosines and generates a 3′ phosphate end:
^{5}pACGp$^{3'}$ $^{5'}$AUGp$^{3'}$ $^{5'}$CUAUC$^{3'}$

5.7. Structure of nucleic acids.
Snake venom phosphodiesterase only cuts the nucleotide at the 3′ end of RNA molecules, liberating a nucleotide with a 5′ phosphate. This tells us that Cp is located at the 3′ end of our polynucleotide.
Pancreatic ribonuclease cuts on the 3′ end of pyrimidines, generating small fragments ending with a pyrimidine with a 3′ phosphate end. From the data provided, we can deduce the following nucleotide sequences:
$^{5'}$ACp$^{3'}$ ^{5}(A, G) Up$^{3'}$
Finally, ribonuclease T2 cuts after adenosine, generating fragments ending with adenosine 3′ phosphate. We can use this information to order the sequence of the product:
^{5}(CG)Ap$^{3'}$
Ain addition, the presence of pAp after digestion with ribonuclease T2 tells us that A is at the 5′ end of the polynucleotide.
Putting together all these clues, we get the following complete sequence:
^{5}pACGAUCp$^{3'}$

5.8. Structure of nucleic acids.
Snake venom phosphodiesterase cuts the nucleotide located at the 3′ end of the fragment. The fact that we get pC tells us that this is the nucleotide at the 3′ end of the nucleic acid.
Pancreatic ribonuclease cuts RNA molecules on the 3′ side of pyrimidines, generating fragments ending with pyrimidine 3′ phosphate. With the data at hand, we can deduce part of the nucleotide sequence, as follows:
^{5}GCp$^{3'}$ $^{5'}$AUp$^{3'}$ ^{5}(A, G) Cp$^{3'}$
Also, the fact that we get 2 Cp indicates that these nucleotides immediately follow U or C.
RNAse T1 cuts on the 3′ side of guanosine, generating fragments ending with guanosine 3′phosphate. RNAse T2 cuts on the 3′ side of adenosine, generating fragments ending with adenosine 3′phosphate. The combined use of these two enzymes will give us a mixture of fragments generated by the action of either enyme, or both. We can deduce the sequence of the fragments obtained, as follows:
$^{5'}$CCAp$^{3'}$, ^{5}(C,U)Gp$^{3'}$
Note: we can deduce the presence of the two cytidines in the first fragment, since it is mentionned that this fragment is a trinucleotide, and that one C is lacking when we count the nucleotides obtained after digestion with RNAses T1 + T2.
From the data obtained after digestion with pancreatic ribonuclease, we can tell that adenosine precedes uridine. Therefore, we get the following fragment:
^{5}AUCGp$^{3'}$
Also, again from the results obtained after the digestion with RNAse T1 + T2, the fact that we obtained pGp tells us that this is the 5′ nucleotide. Furthermore, the result obtained with pancreatic ribonuclease tells us that G is followed by C. The fact that C and Ap were obtained also tells us that these two nucleotides immediately follow A or G.
This allows us to order the nucleotides as follows:
$^{5'}$pGC....AUCG....C$^{3'}$
All we have to do now is to place A and C. Since digesting the RNA with RNAse T1 + T2 gave us the sequence ^{5}CCAp$^{3'}$, this suggests that C is located between pGC and AUCG. Finallt, the last nucleotide, A, would logically be placed before the last C of the nucleic acid.
Taking into account all these pieces of information, we get the following sequence:
^{5}pGCCAUCGAC$^{3'}$

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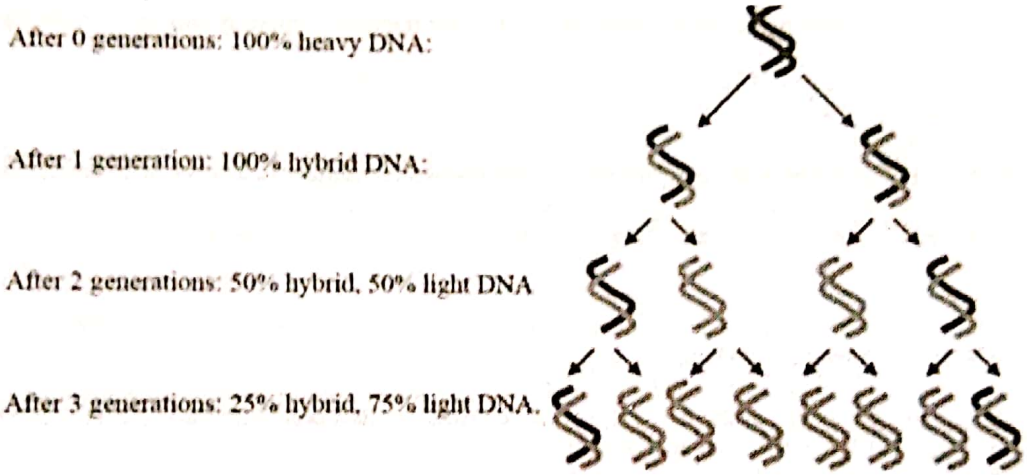
5.9. Structure of nucleic acids.
At neutral pH, the nitrogenated bases are not ionized, and the charges will only come from the phosphate groups. Since, in the example given here, three phosphates for the phosphodiester bonds, we get 3 x (1-) = 3 negative charges.

6.10. Structure of nucleic acids.
When a circular DNA molecule is denatured, the result is two tangled-up single-stranded circles. When a linear DNA molecule is denatures, both stanns diffuse away from each other. Upon renaturation, it will be much easier for the circular, tangled-up single strands to form complementary base pairs (because they are closer), this compared to a linear molecule of identical sequence (which have to rely on random collisions in order to meet).

6.11. Structure of nucleic acids.
The fusion temperature depends on the ionic strength (i.e. the salt concentration) of the solution. If we decrease the ionic strength, the fusion temperature will also decrease. In the extreme case where the ionic strength is equal to zero, the the negatively charged phosphate groups will not be neutrolized by counter-ions: this will be sufficient to decrease the fusion temperature to below 20°C.

6.12. Structure of nucleic acids.
B-type DNA of 10 base pairs has a length of 3,4 nm (34 A). The E. coli chromosome, being 4 000 kbp long (i.e. 4 000 000 bp), will therefore have a length of $1{,}36 \times 10^6$ nm (approximately 1,4 mm).

6.13. Nucleic acid synthesis.
In the Meselson and Stahl experiment, we first incubate bacteria in culture media containing only the ^{15}N isotope of nitrogen. We then transfer those bacteria into a culture media containing only ^{14}N. We obtain the following results:
After 0 generations: 100% heavy DNA:



6.14. Nucleic acid synthesis.
The starting strand (+) is made of: 10% A, 20%G, 30%C and 40%T. After its replication, the complementary (-) strand will have the following composition: 40%A, 30%G, 20%C and 10%T (since A base pair only with T, and C only with G). Finally, after transcription the RNA produced will have a nucleotide composition that will be complementary to the (-) strand, which is 10%A, 20%G, 30%C and 40% U.

6.15. Nucleic acid synthesis.
This observation can only be explained if the E. coli genome is replicated 1.5 times when the bacteria are seperating.

***6.16. Nucleic acid synthesis.*
a) since no hybrid molecule is obtained, we must conclude that replication occurs through a ***convervative*** mechanism, during which the two template (old) strands re-anneal together after each round of replication.
b) The presence of RNA polymerase suggests that the bacterial DNA is first transcribed into an RNA molecule. Afterwards, this RNA molecule (resistant to the action of *DNAse)* is converted into a DNA molecule *(DNAse sensitive)* through the reduction of the 2'carbon via the new enzymatic activity and NADH. Finally, this DNA is used as a template by DNA polymerase to form a double-stranded DNA molecule identical to the initial parental duplex.

2. Half Life Problems

Half Life Theory

What follows is not a calculus-based discussion concerning half-life calculations. Later on, you may learn the calculus-based approach. And that will be a good thing.
Doing half-life problems is focused on using several equations. The order in which you use them depends on the data given and what is being asked. Here is the first equation:
$(1/2)^{\text{number of half-lives}}$ equals the decimal amount remaining
Let us use several different half-lives to illustrate this equation.
$(1/2)^0 = 1$

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In this example, zero half-lives have elapsed. The 1 represents the decimal fraction remaining. In other words, at the very start, before any decay has taken place, 100% of the material is on hand. Keep in mind that the decimal amount times 100 becomes the percentage.
$(1/2)^1 = 0.5$
This second example shows one half-life elapsed. At this point 0.5 of the original amount remains. 0.5 expressed as a percentage is 50%, however, please be aware that it is the decimal amount that will be used in the calculations.
Just below are the amounts remaining after 2, 3, and 4 half-lives.
$(1/2)^2 = 0.25$
$(1/2)^3 = 0.125$
$(1/2)^4 = 0.0625$
I would like to stress that the decimal amount is the amount that remains after a given number of half-lives. Many times, a problem will specify how much has decayed and you must use that information to determine how much remains. It's the amount that remains that goes into the calculation.
Sometimes, you will see the equation expressed thusly:
$1/2^n$ <--- the exponent is applied only to the 2, as in one over 2^n
This is often done in order to highlight the amount remaining progression in a fractional way. Thusly:
$1/2^0 = 1$ <--- remember, that 1 represents 100%
$1/2^1 = 1/2$
$1/2^2 = 1/4$
$1/2^3 = 1/8$
$1/2^4 = 1/16$
Learn to recognize which whole-number half-lives are associated with their fractional amounts and the decimal amount. Thusly:
one half-life: $1/2^1 = 1/2 = 0.5$
two half-lives: $1/2^2 = 1/4 = 0.25$
three half-lives: $1/2^3 = 1/8 = 0.125$
four half-lives: $1/2^4 = 1/16 = 0.0625$
Decay problems at the start of your study of these problems will often be whole number half-lives, as in one half-life, two half-lves, three half-lives and so on. However, as you advance, you will see values like 2.45 and 0.5882 for the number of half-lives elapsed. In that case, you need to turn to your calculator and do a calculation using the y^x (or x^y) key. Thusly:
$(1/2)^{2.45} = 0.18301$
$(1/2)^{0.5882} = 0.66517$
I generally raise 0.5 to the proper power. If you want, you can raise 2 to the power and then use the 1/x key.
The second equation to be aware of concerns the number of half-lives that have occurred. It can be expressed as
total time elapsed ÷ length of half-life = number of half-lives elapsed
Often, the problem will tell you the length of the half-life (say, 5730 years, the half-life of carbon-14) and then ask you about how much remains after, for example, 17190 years. In that case, you do this:
17190 yr / 5730 yr = 3 half-lives
Or, 12,300 years:
12300 / 5730 = 2.1466 half-lives
Usually, you then go to the first equation discussed above in order to determine the decimal amount remaining. After you've memorized the hole-number half-lives as I recommended above, you'll recognize three half-lives as being associated with 0.125 remaining. For 2.1466 half-lives, you turn to the calculator:
$(1/2)^{2.1466} = 0.225844$
The third equation is this:
starting amount times decimal amount remaining = remaining amount
To use symbols, we have this:
(H_o) (decimal amount remaining) = H
By the way, H is not a standard symbol, it's just the letter I decided to use. The subscripted 'o' is standard usage and it indicates the starting (or original, hence the letter 'o') amount.
A bit of a warning: you will sometimes see equation two (the one to calculate the number of half-lives inserted as a fraction into the exponent portion. This is often done when one of the two values in equation two is the desired final answer. As in this:
$H_o \times (1/2)^{\text{total time / half-life time}} = H$
If you see it set up like that, don't try and solve it directly. Just use one or more of the above equations instead. For example, problem #1 just below can be set up like this:
88.0 times $(1/2)^{x / 12.3} = 11.0$
I've also solved Example #3 both a several-step approach and using the single equation just above.

How to solve problems?

1. If 100.0 g of carbon-14 decays until only 25.0 g of carbon is left after 11 460 y, what is the half-life of carbon-14?

Solution
Step 1: List the given and unknown values.
Given: initial mass of sample = 100.0 g, final mass of sample = 25.0 g, total time of decay = 11 460 y

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Unknown: number of half-lives =? half-lives, half-life =? y
Step 2: Write down the equation relating half-life, the number of half-lives, and the decay time, and rearrange it to solve for half-life.

$$\text{total time of decay} = \text{number of half-lives} \times \frac{\text{number of years}}{\text{half-life}}$$

$$\frac{\text{number of years}}{\text{half-life}} = \frac{\text{total time of decay}}{\text{number of half-lives}}$$

Step 3: Calculate how many half-lives have passed during the decay of the 100.0 g sample.

$$\text{fraction of sample remaining} = \frac{\text{final mass of sample}}{\text{initial mass of sample}} = \frac{25.0\text{ g}}{100.0\text{ g}} = \frac{1}{4}$$

$$\text{after one half-life} = \frac{1}{2}\text{; after two half-lives} = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}\text{ of sample}$$

Two half-lives have passed.

Step 4: Calculate the half-life.

$$\frac{\text{number of years}}{\text{half-life}} = \frac{11\,460\text{ y}}{2\text{ half-lives}} = \frac{5730\text{ y}}{\text{half-life}}$$

$$\text{half-life of carbon-14} = 5730\text{ y}$$

2. Thallium-208 has a half-life of 3.053 min. How long will it take for 120.0 g to decay to 7.50 g?

Solution Step 1: List the given and unknown values.
Given: half-life = 3.053 min initial mass of sample = 120.0 g final mass of sample = 7.50 g Unknown: number of half-lives =? half-lives total time of decay =?
Step 2: Write down the equation relating half-life, the number of half-lives, and the decay time, and rearrange it to solve for the total time of decay.

$$\text{total time of decay} = \text{number of half-lives} \times \frac{\text{number of min}}{\text{half-life}}$$

Step 3: Calculate how many half-lives have passed during the decay of the 120.0 g sample

$$\text{fraction of sample remaining} = \frac{7.50\text{ g}}{120.0\text{ g}} = 0.0625 = \frac{1}{16}$$

$$\text{after one half-life} = \frac{1}{2}\text{; after two half-lives} = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}\text{;}$$

$$\text{after three half-lives} = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}\text{; after four half-lives} =$$

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}\text{ of sample. Four half-lives have passed.}$$

Step 4: Calculate the total time required for the radioactive decay

$$\text{total time of decay} = 4\text{ half-lives} \times \frac{3.053\text{ min}}{\text{half-life}}$$

$$\text{total time of decay} = 12.21\text{ min}$$

3. Gold-198 has a half-life of 2.7 days. How much of a 96-g sample of gold- 198 will be left after 8.1 days?

Step 1: List the given and unknown values.
Given: half-life = 2.7 days total time of decay = 8.1 days initial mass of sample = 96 g Unknown: number of half-lives =? half-lives final mass of sample =? g
Step 2: Write down the equation relating half-life, the number of half-lives, and the decay time, and rearrange it to solve for the number of half-lives.

$$\text{total time of decay} = \text{number of half-lives} \times \frac{\text{number of days}}{\text{half-life}}$$

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$$\text{number of half-lives} = \frac{\text{total time of decay}}{\frac{\text{number of days}}{\text{half-life}}}$$

Step 3: Calculate how many half-lives have passed during the decay of the 96-g sample.

$$\text{number of half-lives} = \frac{8.1 \text{ days}}{\frac{2.7 \text{ days}}{\text{half-life}}} = 3.0 \text{ half-lives}$$

Step 4: Calculate how much of the sample will remain after 3.0 half-lives.

final mass of sample = initial mass of sample × fraction of sample remaining

$$\text{fraction of sample remaining after three half-lives} = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$$

$$\text{final mass of sample} = 96 \text{ g} \times \frac{1}{8} = 12 \text{ g}$$

Sample problems and explanations

Example #1: How many years will it take for 88.0 grams of tritium to decay to an 11.0-gram sample? (The half-life of tritium is 12.3 years.)

Solution:

1) Determine the decimal amount remaining:

11.0 g / 88.0 g = 0.125

2) Determine the number of half-lives:

$(1/2)^n = 0.125$

n log 0.5 = log 0.125

n = log 0.125 / log 0.5

n = 3

3) Determine years elapsed:

12.3 yr times 3 = 36.9 yr

Example #2: How many years will it take for 84.0 grams of tritium to decay to a 23.5-gram sample? (The half-life of tritium is 12.3 years.)

Solution:

1) Determine the decimal amount remaining:

23.5 g / 84.0 g = 0.279762

Note that I rounded off, but not too much.

2) Determine the number of half-lives:

$(1/2)^n = 0.279762$

n log 0.5 = log 0.279762

n = 1.837728

3) Determine years elapsed:

12.3 yr times 1.837728 = 22.6 yr

Comment: in these calculations, I carry several extra digits (called guard digits) from each step to the next. I only round off to the proper number of significant figures at the end of the calculation.

Example #3: A 208 g sample of sodium-24 decays to 13.0 g of sodium-24 in 60.0 hr. What is the half-life of this radioactive isotope?

Solution #1:

1) Determine the decimal amount remaining:

13.0 g / 208 g = 0.0625

2) Determine the number of half-lives:

recognize 0.0625 as being associated with 4 half-lives

or

$(1/2)^n = 0.0625$

n log 0.5 = log 0.0625

n = 4

3) Determine the length of the half-life:

60.0 hr / 4 = 15.0 hr

Solution #2:

1) Set it up in one equation:

208 times $(1/2)^{(60/n)} = 13$

2) Divide both sides by 208:

$0.0625 = (1/2)^{(60/n)}$

Note: the 0.0625 comes from 13 divided by 208

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3) Take the log of both sides:
$\log(0.0625) = \log(1/2)^{(60/n)}$
$\log(0.0625) = (60/n) \log(1/2)$
$-1.2041 = (60/n) (-0.3010)$
4) Divide by -0.3010:
$4 = 60/n$
$4n = 60$
$n = 15$ hr

Example #4: Calculate the half-life of a sample of X which decays from 27.5 g to 0.598 g in a period of 574 years.
Solution:
0.598 / 27.5 = 0.02174545
$(1/2)^n = 0.02174545$
$n = 5.52314$
574 yr / 5.52314 = 104 yr (to three sig figs)

Example #5: The radioactive element carbon-14 has a half-life of 5730. years. The percentage of carbon-14 present in the remains of plants and animals can be used to determine age. How old is an object that has lost 60% of its carbon-14?
Solution:
1) Determine how many half-lives have elapsed:
$(1/2)^n = 0.40$
$n \log 0.5 = \log 0.40$
$n = 1.32193$
2) Calculate the age of the object:
5730 yr times 1.32193 = 7575 yr
Note that the problem said 60% was lost. Since the solution technique uses the amount remaining, I used 40%. Be aware that this is a common thing: to give you the amount lost, which you have to convert into the amount remaining.
Also, note that I used the decimal equivalent of 40%. Don't use a percentage in the calculation, use the decimal amount, 0.4 in this example.

Example #6: An element has a half-life of 25 years. You currently have 20 grams. How many grams did you have 100 years ago?
Solution:
100 yr / 25 yr = 4 <— this is the number of half-lives
$(1/2)^4 = 0.0625$ <— this is the decimal amount remaining after 4 half-lives
20 g / 0.0625 = 320 g <— the answer
Here's the 320-g decaying over four half-lives:
320 —> 160
160 —> 80
80 —> 40
40 —> 20

Example #7: P-32 is radioactive isotope with a half-life of 14.3 days. If you currently have 73.3 g of P-32, how much P-32 was present 4 days ago?
Solution:
1) How many half-lives have elapsed in 4 days?
4 days / 14.3 days = 0.27972
2) The decimal amount that remains after 0.27972 half-lives have elapsed:
$(1/2)^{0.27972} = 0.82375$
3) Determine amount from 4 days prior:
73.3 is to 0.82375 as x is to 1
$x = 88.9833$ g
rounded to three sig figs, 89.0 g

Example #8: If a sample contains 144 atoms of Au-179 (half-life = 7.5 s), approximately how many such atoms were present 30 seconds earlier?
Solution:
30 s / 7.5 s = 4 <— four half-lives elapsed during the 30 s
$(1/2)^4 = 0.0625$ <— decimal amount remaining after four half-lives
x is to 1 as 144 is to 0.0625
144 / 0.0625 = 2304

Example #9: Find the half-life of an element which decays at a rate of 3.402% per day.
Solution:
1) After one day, this much remains:
100% minus 3.402% = 96.598%
We will use it as 0.96598
2) How many half-lives have elapsed?
$(1/2)^n = 0.96598$
$n \log 0.5 = \log 0.96598$

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n = 0.049935
3) Find the half-life:
1 day is to 0.049935 as x is to 1
x = 20 days
Comment: you could set up a spreadsheet and do it by brute force, subtracting 3.402% of the material on hand each day, with the half-life being the number of days needed to arrive at 50%.
Day 19 to 20 would be this:
x - 0.03402x = 50%
where x is equal to 51.76%
Example #10: The half-life of In-111 is 0.007685 years; how long (in hours) would it take for the amount of In-11 to decrease to 43.24% of its initial amount?
Solution #1:
$(1/2)^n = 0.4324$
n log 0.5 = log 0.4324
n = log 0.4324 / log 0.5
n = 1.20956 <--- the number of half-lives
1.20956 times 0.007685 yr = 0.00930 yr
0.00930 yr times (365 day / yr) times (24 hr / day) = 81.468 hr
To three sig figs, this is 81.5 hours
Solution #2:
$\ln A/A_o = - (0.693/t_{1/2}) t$
$\ln (0.4324) = - (0.693/0.007685) t$
$t = 9.30 \times 10^{-3}$ yr
Convert years to hours as above in first solution.
Alternatively, you could convert the half-life into hours and use that:
0.007685 yr x 365 day/yr x 24 hr/day = 67.32 hr
$\ln (0.4324) = -(0.693/t_{1/2}) t$
Note: in the original answer on Yahoo Answers, the value of 81.4 hrs. was obtained for the Solution #2 procedure. If you use ln 2 in place of 0.693 (and do not round it off), you'll probably obtain 81.5 hrs. I didn't try it out.
Problem #11: The half-life of iodine-131 is 8.040 days. What percentage of an iodine-131 sample will remain after 40.20 days?
Solution:
40.20 d / 8.040 d = 5
$(1/2)^5 = 0.03125$
percent remaining = 3.125%
Problem #12: The half-life of thorium-227 is 18.72 days How many days are required for three-fourths of a given amount to decay?
Solution:
3/4 = 0.75 <--- amount decayed
1 - 0.75 = 0.25 <--- amount remaining
$(1/2)^n = 0.25$
n = 2
18.72 d times 2 = 37.44 d
Problem #13: If you start with 5.32×10^9 atoms of Cs-137, how much time will pass before the amount remaining is 5.20×10^6 atoms? The half-life of Cs-137 is 30.17 years.
Solution:
$5.20 \times 10^6 / 5.32 \times 10^9 = 0.0009774436$ (the decimal amount remaining)
$(1/2)^n = 0.0009774436$
n log 0.5 = log 0.0009774436
n = 9.99869892 half-lives
30.17 yr times 10 = 301.7 yr
Problem #14: The half-life of the radioactive isotope phosphorus-32 is 14.3 days. How long until a sample loses 99% of it radioactivity?
Solution:
99% loss means 1% remaining
1% = 0.01
$(1/2)^n = 0.01$
n log 0.5 = log 0.01
n = 6.643856
14.3-day times 6.643856 = 95.0 day
Problem #15: The half-life of Palladium-100 is 4 days. After 12 days a sample of Pd-100 has been reduced to a mass of 4.00 mg. (a) Determine the starting mass. (b) What is the mass 8 weeks after the start?
Solution:
12 day / 4 day = 3
$(1/2)^3 = 0.125$

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4.00 mg / 0.125 = 32.0 mg
8 weeks = 56 days
56 d / 4 = 14 half-lives
$(1/2)^{14} = 0.000061035$
32.0 mg times 0.000061035 = 0.00195 mg (rounded to three figs)

Problem #16: Rn-222 has a half-life of 3.82 days. How long before only 1/16 of the original sample remains?

Solution:
recognize 1/16 as a fraction associated with 4 half-lives (from $1/2^4 = 1/16$)
3.82 days x 4 = 15.3 days

Problem #17: U-238 has a half-life of 4.46×10^9 years. Estimates of the age of the universe range from 9×10^9 years to 23×10^9 years (Cauldrons in the Cosmos: Nuclear Astrophysics, C.E. Rolfs and W.S. Rodney, Univ. of Chicago, 1988, p. 477). What fraction of this isotope present at the start of the universe remains today? Calculate for both minimum and maximum values, as well as a median value of 16×10^9 years.

Solution:
1) Calculation for the median value:
$(16 \times 10^9) / (4.46 \times 10^9) = 3.587$ half-lives
2) What fraction remains?
$(1/2)^{3.587} = 0.0832$
8.32% remains

Problem #18: A sample of Se-83 registers 10^{12} disintegrations per second when first tested. What rate would you predict for this sample 3.5 hours later, if the half-life is 22.3 minutes?

Solution:
210 min / 22.3 min = 9.42 half-lives (210 min is 3.5 hours)
$(1/2)^{9.42} = 0.00146$ (the decimal fraction remaining)
$10^{12} \times 0.00146 = 1.46 \times 10^9$ disintegrations per second remaining

Problem #19: Iodine-131 has a half-life of 8.040 days. If we start with a 40.0-gram sample, how much will remain after 24.0 days?

Solution:
24.0 days / 8.040 days = 2.985 half-lives
$(1/2)^{2.985} = 0.1263$ (the decimal fraction remaining)
40.0 g x 0.1263 = 5.05 g

Problem #20: If you start with 2.97×10^{22} atoms of molybdenum-99 (half-life = 65.94 hours), how many atoms will remain after one week?

Solution:
one week = 168 hours
168 / 65.94 = 2.548
$(1/2)^{2.548} = 0.171$ (the decimal fraction remaining)
$(2.97 \times 10^{22}) \times 0.171 = 5.08 \times 10^{21}$

Problem #21: The isotope H-3 has a half-life of 12.26 years. Find the fraction remaining after 49 years.

Solution:
49 / 12.26 = 3.9967
$(1/2)^{3.9967} = 0.0626$

Problem #22: How long will it take for a 64.0 g sample of Rn-222 (half-life = 3.8235 days) to decay to 8.00 g?

Solution:
8.00 / 64.0 = 0.125 (the decimal fraction remaining)
$(1/2)^n = 0.125$
by experience, n = 3 (remember that 0.125 is 1/8)
3.8235 x 3 = 11.4705 days

Problem #23: A scientist needs 10.0 micrograms of Ca-47 (half-life = 4.50 days) to do an experiment on an animal. If the delivery time is 50.0 hours, how many micrograms of $^{47}CaCO_3$ must the scientist order?

Solution:
4.50 days x 24 hrs/day = 108 hrs
50/108 = 0.463 half-lives
$(1/2)^{0.463} = 0.725$ (the decimal portion of Ca-47 remaining after 50 hrs)
10.0 mg / 0.725 = 13.8 mg

Problem #24: What precentage of the parent isotope remains after 0.5 half-lives have passed?

Solution:
$(1/2)^n$ = decimal amount remaining
where n = the number of half-lives
$(1/2)^{0.5} = 0.707$
The question asks for a percentage, so 70.7%

Problem #25: Manganese-56 has a half-life of 2.6 h. What is the mass of manganese-56 in a 1.0 g sample of the isotope at the end of 10.4 h?

Solution:

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10.4 / 2.6 = 4
4 half-lives = 0.0625 remaining
0.0625 g

Practice problems: Apply your knowledge

Apply Your Knowledge:

Problem #1: The half-life of Zn-71 is 2.4 minutes. If one had 100.0 g at the beginning, how many grams would be left after 7.2 minutes has elapsed?

Problem #2: Pd-100 has a half-life of 3.6 days. If one had 6.02×10^{23} atoms at the start, how many atoms would be present after 20.0 days?

Problem #3: Os-182 has a half-life of 21.5 hours. How many grams of a 10.0-gram sample would have decayed after exactly three half-lives?

Problem #4: After 24.0 days, 2.00 milligrams of an original 128.0 milligram sample remain. What is the half-life of the sample?

Problem #5: U-238 has a half-life of 4.46×10^9 years. How much U-238 should be present in a sample 2.5×10^9-year-old, if 2.00 grams was present initially?

Problem #6: How long will it take for a 40.0-gram sample of I-131 (half-life = 8.040 days) to decay to 1/100 its original mass?

Problem #7: Fermium-253 has a half-life of 0.334 seconds. A radioactive sample is considered to be completely decayed after 10 half-lives. How much time will elapse for this sample to be considered gone?

Problem #8: At time zero, there are 10.0 grams of W-187. If the half-life is 23.9 hours, how much will be present at the end of one day? Two days? Seven days?

Problem #9: 100.0 grams of an isotope with a half-life of 36.0 hours is present at time zero. How much time will have elapsed when 5.00 grams remains?

Problem #10: How much time will be required for a sample of H-3 to lose 75% of its radioactivity? The half-life of tritium is 12.26 years.

Answers with Explanations:

1. Solution:
7.2 / 2.4 = 3 half-lives
$(1/2)^3 = 0.125$ (the amount remaining after 3 half-lives)
100.0 g x 0.125 = 12.5 g remaining

2. Solution:
20.0 / 3.6 = 5.56 half-lives
$(1/2)^{5.56} = 0.0213$ (the decimal fraction remaining after 5.56 half-lives)
$(6.02 \times 10^{23})(0.0213) = 1.28 \times 10^{22}$ atoms remain

3. Solution:
$(1/2)^3 = 0.125$ (the amount remaining after 3 half-lives)
10.0 g x 0.125 = 1.25 g remain
10.0 g - 1.25 g = 8.75 g have decayed
Note that the length of the half-life played no role in this calculation. In addition, note that the question asked for the amount that decayed, not the amount that remaining.

4. Solution:
2.00 mg / 128.0 mg = 0.015625
How many half-lives must have elapsed to get to 0.015625 remaining?
$(1/2)^n = 0.015625$
n log 0.5 = log 0.015625
n = log 0.5 / log 0.015625
n = 6
24 days / 6 half-lives = 4.00 days (the length of the half-life)

5. Solution:
$(2.5 \times 10^9) / (4.46 \times 10^9) = 0.560$ (the number of half-lives that have elapsed)
$(1/2)^{0.560} = 0.678$ (the decimal fraction of U-238 remaining)
2.00 g x 0.678 = 1.36 g remain

6. Solution:
$(1/2)^n = 0.01$
n log 0.5 = log 0.01
n = 6.64
6.64 x 8.040 days = 53.4 days

7. Solution:
0.334 x 10 = 3.34 seconds]

8. Solution:
24.0 hr / 23.9 hr/half-life = 1.0042 half-lives
One day = one half-life; $(1/2)^{1.0042} = 0.4985465$ remaining = 4.98 g
Two days = two half-lives; $(1/2)^{2.0084} = 0.2485486$ remaining = 2.48 g

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Seven days = 7 half-lives; $(1/2)^{7.0294}$ = 0.0076549 remaining = 0.0765 g

9. Solution:

5.00 / 100.0 = 0.05 (decimal fraction remaining)

$(1/2)^n = 0.05$

n log 0.5 = log 0.05

n = 4.32 half-lives

36.0 hours x 4.32 = 155.6 hours

10. Solution:

If you lose 75%, then 25% remains. Use 0.25 rather than 25%.

$(1/2)^n = 0.25$

n = 2 (remember $(1/2)^2 = 1/4$ and 1/4 = 0.25)

12.26 x 2 = 24.52 years

Comment: the more general explanation follows:

$(1/2)^n = 0.25$

n log 0.5 = log 0.25

n = log 0.25 / log 0.5

n = 2

3. Enzyme kinetics problems

Sample problems and explanations

1. **For a given enzyme catalyzed reaction, the Michaelis constant is 0.6mM and the substrate concentration is 1.0mM. What is the fractional saturation of the enzyme under these conditions?**

Possible Answers:

62.5%

50%

60.0%

37.5%

40.0%

Explanation:

The fractional saturation of an enzyme is defined as the amount of enzyme that is bound to substrate divided by the total amount of enzyme. To calculate the fractional saturation, we'll need to use the Michaelis-Menton equation:

$$V_o = \frac{V_{max}[S]}{K_M + [S]}$$

In addition, we'll need to define the rate and maximum rate in terms of enzyme concentrations:

$$V_o = k_{cat}[ES]$$

$$V_{max} = k_{cat}[E]_T$$

From the above equations, we can calculate the fractional saturation of the enzyme:

$$Fractional\ Saturation = \frac{[ES]}{[E]_T} = \frac{V_o}{V_{max}} = \frac{[S]}{K_M + [S]}$$

$$\frac{[S]}{K_M + [S]} = \frac{1.0\,mM}{0.6\,mM + 1.0\,mM} = 0.625 = 62.5\%$$

2. **Given an enzyme with KM of 0.5mM. at what substrate concentration will the velocity of the enzyme reach 1/4 of the Vmax?**

Vmax=200mmol/s

Possible Answers:

0.5mM

0.12mM

0.17mM

0.20mM

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To solve this, we need the solve for $[S]$ in the Michaelis-Menten equation:

$$V_o = \frac{V_{max}[S]}{K_m + [S]}$$

We know the following information:

$$K_M = 0.5mM$$

$$V_{max} = 200\frac{mmol}{s}$$

$$V_o = \frac{1}{4}V_{max} = 50\frac{mmol}{s}$$

Plug in these numbers and solve for substrate concentration.

$$50 = \frac{200[S]}{.5 + [S]}$$

$$\frac{1}{4} = \frac{[S]}{.5 + [S]}$$

$$.5 + [S] = 4([S])$$

$$.5 = 3[S]$$

$$\frac{1}{6}mM = [S] \approx 0.17mM$$

3. Suppose that an enzyme mixture contains an enzyme with a michaelis constant of $5.0*10^{-6}$M. If the substrate concentration in this mixture is $4.5*10^{-6}$M, what is the fractional saturation of this enzyme mixture?

Possible Answers:
26.31%
68.13%
23.68%
52.63%
47.37%

Explanation:
This question is asking us to determine the fractional saturation of a solution containing enzyme and substrate. Let's start by considering what we have, and what we are trying to solve for.
We know that the substrate concentration is 4.5*10−6M and we also know that this enzyme has a Michaelis constant of 5.0×10^{-6}M. Also, let's consider what fractional saturation is. Fractional saturation refers to the proportion of enzyme molecules in a solution that are bound to substrate. This value can range from 0% (all enzymes are not bound to substrate) to 100% (all enzymes are bound to, or saturated with, substrate). So, we are looking to calculate the number of enzyme-substrate complexes divided by the total amount of enzyme in solution, which we can express as $[ES]/[E]_T$. We can also relate these terms by the following equations.

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$$V_0 = k_{cat}\,[ES]$$

$$V_{max} = k_{cat}[E]_T$$

Dividing these equation by each other, we obtain:

$$\frac{V_0}{V_{max}} = \frac{[ES]}{[E]_T}$$

Furthermore, we can relate these terms by considering the Michaelis-Menten equation:

$$V_0 = \frac{V_{max}[S]}{K_M + [S]}$$

Or, written another way:

$$\frac{V_0}{V_{max}} = \frac{[S]}{K_M + [S]}$$

And combining everything we have done so far, we have:

$$\frac{V_0}{V_{max}} = \frac{[ES]}{[E]_T} = \frac{[S]}{K_M + [S]}$$

And plugging in values, we can solve:

$$Fractional\ Saturation = \frac{[ES]}{[E]_T} = \frac{4.5 * 10^{-6}\ M}{5.0 * 10^{-6}\ M + 4.5 * 10^{-6}\ M} = 47.37\ \%$$

4. **An enzyme with a K_M value of 5 mM has a reaction rate of $200\ \frac{\text{mmol}}{\text{min}}$ at a substrate concentration of 0.5 mmol. What is the maximum reaction rate that this enzyme can achieve when it is saturated with substrate?**

Possible Answers:

$550\ \frac{\text{mmol}}{\text{min}}$

$2000\ \frac{\text{mmol}}{\text{min}}$

$1100\ \frac{\text{mmol}}{\text{min}}$

$1000\ \frac{\text{mmol}}{\text{min}}$

$2200\ \frac{\text{mmol}}{\text{min}}$

Explanation:

For this question, we're provided with the michaelis constant for an enzyme, as well as the reaction rate for that enzyme at a particular substrate concentration. We're asked to determine the maximum possible reaction rate that this enzyme can achieve when it is saturated with substrate.

To solve this problem, we'll need to use the michaelis-menten equation, which is expressed as follows.

$$V_o = \frac{V_{max}[S]}{K_M + [S]}$$

Then, we can rearrange the equation above in order to isolate the V_{max} term.

$$V_{max} = \frac{V_o(K_M + [S])}{[S]}$$

Now, we can plug in the values given to us in the question stem in order to solve for our answer.

$$V_{max} = \frac{(200\ \frac{mmol}{min})(5\ mmol + 0.5\ mmol)}{0.5\ mmol}$$

$$V_{max} = 2200\ \frac{mmol}{min}$$

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5. **What is the maximum reaction rate if adding $3\ \mathrm{M}$ of substrate produces a reaction rate of $15\ \frac{\mathrm{mol}}{\mathrm{s}}$?**

$(K_m = 3\ \mathrm{M})$

Possible Answers:

$7.5\ \frac{\mathrm{mol}}{\mathrm{s}}$

Cannot be determined from the given information

$30\ \frac{\mathrm{mol}}{\mathrm{s}}$

$15\ \frac{\mathrm{mol}}{\mathrm{s}}$

Explanation:

The trick to this question is to notice that the K_m and the substrate concentration are the same. Recall that K_m is the substrate concentration required to reach half the maximum reaction rate. Since we are told that the substrate concentration and K_m are the same, we can conclude that that reaction rate of $15\frac{mol}{s}$ is half the maximum reaction rate; therefore, the maximum reaction rate is $30\frac{mol}{s}$.

Questions on Turnover Number

5. The rate of an enzyme catalyzed reaction is 35 µmol/min at [S] = 10^{-4} M, ($K_M = 2 \times 10^{-5}$).
Calculate the velocity at [S] = 2×10^{-6} M.

Answer:
First calculate V_M using the Michaelis-Menton eqn:

$$v = \frac{V_M[S]}{K_M + [S]}, \text{ so: } 35 = \frac{V_M(10^{-4})}{2 \times 10^{-5} + 10^{-4}} = \frac{V_M(10^{-4})}{1.2 \times 10^{-4}}$$

$V_M = 1.2(35) = 42$ mmol/min;

then calculate v:

$$v = \frac{42\,(2 \times 10^{-6})}{2 \times 10^{-5} + 2 \times 10^{-6}} = \frac{84 \times 10^{-6}}{22 \times 10^{-6}} = 3.8 \text{ mmol/min}$$

6. An enzyme (1.84 µgm, MW = 36800) catalyzes a reaction in presence of excess substrate at a rate of 4.2 µmol substrate/min. What is the TON ?

Answer:

$$\mu \text{ mol E} = \frac{1.84\ \mu\text{gm}}{36800\ \mu\text{gm}/\ \mu\text{mol}} = 5 \times 10^{-5}\ \mu\text{mol E}$$

a. µmol/min

$$\text{TON} = \frac{}{5 \times 10^{-5}\ \mu\text{mol}} = 84000\ \text{min}^{-1}$$

What is the value of this TON ($84000\ \text{min}^{-1}$) in units of sec^{-1} ?

$$\text{TON E} = \frac{84000\ \text{min}^{-1}}{} \times \frac{1\ \text{sec}^{-1}}{60\ \text{min}^{-1}} = 1400\ \text{sec}^{-1}$$

7. Question on turnover number.
1.5 ug of enzyme (Mr 30,000 gives a Vmax of 3 umol product produced per minute.

1. What is the turnover number for this enzyme?
2. What is the turnover number if the enzyme is trimeric with 3 identical subunits?
3. What is the turnover number if the enzyme is dimeric with 2 nonidentical subunits?
4. What is the turnover number if 3 ug enzyme gave a Vmax of 3 umol/min.?

Answer with explanation:

1. *1.5 ug = 5 x 10-5* umol enzyme = Et.

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Vmax = 3 umol/min
Vmax = Kcat x Et
and kcat = Vmax / Et = turnover number = 3 / 5 x 10-5 = 0.6 x 105 umol/min/umol
= 1000 umol/sec/umol
turnover number = 1000 sec-1
2. if trimeric, 1 mol of enzyme = 3 mol of catalytic site
therefore, turnover number per catlaytic site = 1000 / 3 = 333 sec-1
3. if dimeric with nonidentical subunits
Mr per catalytic site = 30,000
and turnover number/catalytic site = 1000 sec-1
4. 3 ug enzyme gives Vmax = 3 umol/min.
twice as much enzyme gives same Vmax
therefore: Vmax per mol of enzyme is half that of the original question
and turnover number = 500 sec-1

4. Gibb's free energy problems

Theory behind Gibb's free energy

Free Energy
J. Willard Gibbs (1839-1903) can be considered as one of the founding fathers of the field of chemical thermodynamics. He introduced a quantity known as Gibbs free energy, G, that represents the amount of energy available in a chemical system that can do useful work. If we are to consider a chemical change, we are interested in the change of free energy, or ΔG. Thus, ΔG is a measure of the amount of energy in a chemical change that is free to do work on another physical or chemical system.
The most notable aspect of the ΔG concept is that its sign allows us to predict the spontaneity of a chemical reaction that occurs at a constant temperature and pressure:

1. If ΔG is negative, the reaction is spontaneous.
2. If ΔG is positive, the reaction is nonspontaneous.
3. If ΔG is zero, the reacting system is at equilibrium, and there will be no change in the reaction on the macroscopic level.

Standard state values of ΔG, symbolized as ΔG°, are commonly found in tables of thermodynamic quantities. Recall that the thermodynamic standard state conditions are 25°C, 1 atm pressure for gases, and 1 M concentrations for solutions. Calculation of ΔG° for a reaction is given by

$$\Delta G^\circ = \Sigma n\Delta G_f^\circ{}_{\text{products}} - \Sigma n\Delta G_f^\circ{}_{\text{reactants}} \qquad \text{(Eq.1)}$$

where ΔGf° is the standard free energy of formation; that is, the free energy change that occurs when one mole of a compound is formed from elements in their standard states. Note the similarity of this equation to the equation used to calculate ΔH° for a reaction, which was introduced earlier in your study of chemistry.

Sample problems and solutions

EXAMPLE 1
Calculate the free energy change for the complete combustion of one mole of methane, CH4(g), the main component of natural gas. Is this reaction spontaneous?
SOLUTION
We begin by writing the equation that represents this reaction. Recall that "complete combustion," or burning, is a reaction with oxygen from the atmosphere, forming carbon dioxide and water:
$CH_4(g) + 2\,O_2(g) \rightarrow CO_2(g) + 2\,H_2O(l)$
Now we apply Equation 23.1 and then consult a table of thermodynamic values:

$$\Delta G^\circ = (1\text{ mol})[\Delta G_f^\circ \text{ for } CO_2(g)] + (2\text{ mol})[\Delta G_f^\circ \text{ for } H_2O(\ell)] -$$
$$(1\text{ mol})[\Delta G_f^\circ \text{ for } CH_4(g)] - (2\text{ mol})[\Delta G_f^\circ \text{ for } O_2(g)] =$$
$$(1\text{ mol})(-394.4\text{ kJ/mol}) + (2\text{ mol})(-237.0\text{ kJ/mol}) - (1\text{ mol})(-50.8\text{ kJ/mol}) -$$
$$(2\text{ mol})(0) = -817.6\text{ kJ}$$

(Thermodynamic values are experimentally determined, so the values in your textbook may differ slightly from these.)
The negative value of ΔG° indicates that the reaction is spontaneous. This matches our experiences in everyday life, where we have seen that natural gas burns spontaneously.
Free Energy, Enthalpy, and Entropy
By definition, Gibbs free energy is
G = H - TS (Eq.3)
We are interested in the change of free energy associated with chemical reactions rather than absolute quantities, so we have
ΔG = ΔH - A(TS) (Eq.4)
and if we consider constant temperature processes,
ΔG = ΔH - TΔS (Eq.5)

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and, finally, adding in standard state conditions, we have

$\Delta G° = \Delta H° - T\Delta S°$ (Eq.6)

The Gibbs-Helmholtz equation, tells us that the standard free energy change depends on both the change in enthalpy and the change in entropy. We will explore this idea further during the workshop activities; for now, let's see how $\Delta G°$ can be calculated from $\Delta H°$ and $\Delta S°$ values.

EXAMPLE 2

Calculate $\Delta G°$ for the reaction in Example 2, the decomposition of calcium carbonate, from $\Delta H°$ and $\Delta S°$ values.

SOLUTION

We have already calculated $\Delta S°$ for the reaction $CaCO3(s) — CaO(s) + CO2(g)$ as 160.6 J/K. We can find $\Delta H°$ for the reaction in a similar manner:

$\Delta H°$ = (1 mol)[$\Delta H°$ for CaO(s)] + (1 mol)[$\Delta H°$ for CO2(g)] - (1 mol)[$\Delta H°$ for CaCO3(s)] =

(1 mol) (-635.3 kJ/mol) + (1 mol) (-393.5 kJ/mol) - (1 mol) (-1207.0 kJ/mol) = 178.2 kJ

Now we use Equation 23.6 to find the value of $\Delta G°$:

$\Delta G° = \Delta H° - T\Delta S° = 178.2 \text{ kJ} - 298.15 \text{ K} \times \frac{160.6 \text{ J}}{\text{K}} \times \frac{1 \text{ kJ}}{1000 \text{ J}} = 130.3 \text{ kJ}$

Notice that we used 298.15 K, or 25°C as the value of T. This is thermodynamic standard temperature. Also note how the $\Delta S°$ value was in joules per kelvin, while the $\Delta H°$ value was in kJ. The J ** kJ conversion must be accounted for in the calculation.

Nonstandard Conditions and Free Energy

Up to this point, we have only considered thermodynamic changes under standard conditions. ΔG at any conditions can be determined by

$\Delta G = \Delta G° + RT \ln Q$ (Eq.7)

where Q is the reaction quotient. This is the same quantity that was introduced while we were studying chemical equilibria.

EXAMPLE 3

Consider the reaction of nitrogen monoxide and chlorine to form nitrosyl chloride:

2 NO(g) + Cl2(g) - 2 NOCl(g)

a) Calculate $\Delta G°$ for the reaction.

b) Calculate ΔG when pno = 0.30 atm, pci_2 = 0.10 atm, and pNOCl = 0.45 atm.

SOLUTION

a) $\Delta G°$ is found by applying Equation 23.1:

$\Delta G°$ = (2 mol)[$\Delta Gf°$ for NOCl(g)] - (2 mol)[$\Delta Gf°$ for NO(g)] - (1 mol)[$\Delta Gf°$ for Ch(g)] =

(2 mol) (66.2 kJ/mol) - (2 mol) (86.6 kJ/mol) - (1 mol) (0) = -40.8 kJ

b) ΔG at nonstandard conditions (the pressures are not 1 atm in this case) is found by applying Equation 23.7: $\Delta G = \Delta G° + RT \ln Q$. Let's begin by calculating Q:

$$Q = \frac{(p_{NOCl})^2}{(p_{NO})^2 (p_{Cl_2})} = \frac{(0.45)^2}{(0.30)^2 (0.10)} = 23$$

Now we can find ΔG:

$$\Delta G = \Delta G° + RT \ln Q = -40.8 \text{ kJ} + \frac{8.314 \text{ J}}{\text{mol K}} \times 298.15 \text{ K} \times \ln 23 \times \frac{1 \text{ kJ}}{1000 \text{ J}} =$$

–33.0 kJ

The Equilibrium Constant and Free Energy

Starting from Equation Eq.7,

$\Delta G = \Delta G° + RT \ln Q$ (Eq.7)

and using the facts that at equilibrium, Q = K and ΔG = 0,

$0 = \Delta G° + RT \ln K$ (Eq.8)

and finally rearranging to isolate $\Delta G°$,

$\Delta G° = -RT \ln K$ (Eq.9)

we obtain the relationship between the change in free energy for a reaction and the equilibrium constant.

EXAMPLE 4

K_{sp} for the reaction $BaS04(s) \rightarrow Ba^{2+}(aq) + So4^{2-}(aq)$ is 1.1×10^{-10}. Use thermodynamic data to determine $\Delta G°$ for this reaction, and then calculate K from Equation 9. How do the K values compare?

SOLUTION

$\Delta G°$ is found in the usual manner:

$\Delta G°$ = (1 mol)[$\Delta Gf°$ for $Ba^{2+}(aq)$] + (1 mol)[$\Delta Gf°$ for $SO4^{2-}(aq)$] - (1 mol)[$\Delta Gf°$ for BaSO4(s)] =

(1 mol) (-560.8 kJ/mol) + (1 mol) (-744.5 kJ/mol) - (1 mol) (-1362.3 kJ/mol) = 57.0 kJ

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Now we can use Equation 23.9 to find K:

$$\Delta G^\circ = -RT \ln K \quad \ln K = \frac{-\Delta G^\circ}{RT} \qquad K = e^{(-\Delta G^\circ/RT)}$$

Let's get the coefficient of e first:

$$\frac{-\Delta G^\circ}{RT} = -57.0\ \text{kJ} \times \frac{K}{8.314\ \text{J}} \times \frac{1}{298.15\ \text{K}} \times \frac{1000\ \text{J}}{\text{kJ}} = -23.0$$

↑ assume molar quantities

$$K = e^{-23.0} = 1.0 \times 10^{-10}$$

The K calculated from ΔGf° values agrees with the tabulated K_{sp} value to ±1 in the doubtful digit.

Practice problems and solutions

1. The initial state of one mole of a monatomic ideal gas is P_x = 5 atm and T, = 300 K. Calculate the entropy change of the gas for.

(a) an isothermal decrease in pressure to 1 atm.

(b) a constant volume decrease in the pressure to 1 atm.

2. Calcium boils at 1440°C. The standard free energy of vaporization of liquid calcium is given by

ΔG° (cal/mol) = 41,030 + 5.83T log (7) - 42.237,

where log = $\log_{10}$ (personal communication, 1959, R. Schuhmann, Jr., Department of Metallurgical Engineering, Purdue University, West Lafayette, Indiana).

(a) Find the linear Gibbs free energy equation of the form which approximates the above equation as closely as possible near 1500 K.

(b) Calculate the boiling point from the linear equation and compare with the actual value.

3. Phosphorous-bearing vanadium deposits of the western U.S. are a source of vanadium used in the steel industry as an alloy component. X-ray examination of a representative sample of raw material from these deposits reveals that the major impurity present in the ore is iron in the form of Fe2P. A suggested method for removing the iron is to convert Fe2P to Fe3P and pure iron (ferrite) by vacuum distillation of the phosphorous and to remove the iron magnetically. Assuming the best vacuum obtainable in a large-scale operation is 10^{-2} mm Hg, estimate the operating temperature of a vacuum furnace required to accomplish this distillation.

4. For the reaction SiC(s) -> Si*(s)* + *C(s)*, ΔG^0_t = 12,770 - 1.667cal/mol from 298-1680 K. Using this data, determine:

(a) ΔS^0 at 1000 K.

(b) ΔH^0 at 1500K

5. The thermodynamics of SiO is important in many materials applications because of its high vapor pressure. Develop an expression for the Gibbs free energy of formation of SiO(l), $\Delta G^{0}{}_{f_{SiO(l)}}$, as a function of temperature between 298-1700 K using the data in Appendix A, Table A.4.

6. One mole of a metal, m, at 1 atm pressure is heated at constant volume from 300 to 500 K. Calculate the hydrostatic pressure that results from such a process. What is the work done? Assume C_p is constant over the temperature interval. a_m = 5x 10^{-5} Kr^1, and ^ = 8 xlO^{-6} atm^{-1}.

Ans: $P \sim 1251$ atm; $W = 0$ since V is constant.

Answers:

1.

(a) an isothermal decrease in pressure to 1 atm.

Ans: ΔS = 13.38 J/(mol K).

(b) a constant volume decrease in the pressure to 1 atm.

Ans: ΔS = -20.07 J/(molK).

2.

(a) Find the linear Gibbs free energy equation of the form

$\Delta Gj = a + bT$

which approximates the above equation as closely as possible near 1500 K.

Ans: ΔGj (cal/mol) = 37,235 - 21.2T.

(b) Calculate the boiling point from the linear equation and compare with the actual value.

Ans: T= 1756 K (=2.5% error).

3. *Ans:* T = 1411°C.

4. (a) ΔS^0 at 1000 K.

Ans: ΔS^0 = 1.66 cal/(mol-K).

(b) ΔH^0 at 1500K

Ans: $\Delta H^0{}_{1500}$= 12,770 cal/mol.

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5. Ans: ΔG^ (J/mol) = -91,848 + 20.72T log(T) - 150T.
6. *Ans: P* ~ 1251 atm; *W* = 0 since *V* is constant.

More practice problems:

1. Solid elemental sulfur can be produced, along with liquid water, by the reaction of hydrogen sulfide and sulfur dioxide gases. Calculate the standard free energy change for this reaction.
2. Nitrogen monoxide gas spontaneously decomposes into dinitrogen oxide and nitrogen dioxide gases. What is the standard entropy change for the decomposition of 3.0 mol of nitrogen monoxide?
3. A hypothetical reaction has AH° = -200.3 kJ and AS° = -77.0 J/K. Is this reaction spontaneous? Support your answer with the appropriate calculation.
4. Most of the direct energy needs of a cell are provided by the reaction of adenosine 5'-triphosphate (ATP) to form adenosine 5'-diphosphate (ADP) and hydrogen phosphate ion (Pi):
ATP → ADP + Pi
AG° = -30.0 kJ/mol for this reaction. What is AG when the concentrations in a cell are [ATP] = 3.2 x 10^{-3} M, [ADP] = 1.4 x 10^{-3} M, and [Pi] = 5.0 x 10^{-3} M?
5. Find the value of K_{sp} for iron(II) hydroxide from your textbook, then use Equation 9 to determine the value of AG° for the solution reaction of this slightly soluble solid. How does this value compare to the value determined by using AGf° values?
6. Consider the definition of Gibbs free energy, G = H - TS, and the equations that can be derived from this definition. Construct a "truth table" showing all possible combinations of enthalpy and entropy changes for a chemical reaction and the resulting ability to predict the spontaneity of the reaction. Use your textbook to find an example of a reaction that fits each condition in your truth table.
7. Consider the definition of Gibbs free energy, G = H - TS, and the equations that can be derived from this definition. Can a particular chemical reaction be nonspontaneous at one temperature but spontaneous at another temperature? If so, what criteria must be satisfied? If not, explain why.
8. Consider the equation that allows us to calculate free energy change under nonstandard conditions, AG = AG° + RT ln Q, and the equations that can be derived from this equation. Can the spontaneity of a reversible chemical reaction be determined solely from the equilibrium constant of that reaction? To answer this question, carefully consider the criteria for determining spontaneity and the relationship between the reaction quotient, Q, and the equilibrium constant, K. Explain your answer.
9. Determine the free energy change when 1.00 L of ethane, C2H6(g), at 25°C and 1.0 atm pressure, is completely oxidized.
10. Consider the decomposition of solid ammonium chloride to ammonia and hydrogen chloride gases. What do you predict for the sign of AS°? Consider a particulate-level explanation for your prediction. Calculate the value of AS° for the reaction, and compare it to your prediction. Now consider the decomposition of aqueous ammonium chloride to aqueous ammonia and hydrochloric acid. What do you predict for the sign of AS°? Why? Compare your prediction to the calculated value.
11. One of the reaction steps for the metabolism of glucose in animals is not spontaneous:
2-phosphoglycerate ^ phosphoenolpyruvate AG°' = 1.7 kJ/mol
(The prime indicates biochemistry standard state, which is the same as chemistry standard state except that biochemistry uses pH = 7.0 as a condition.) Will this reaction take place in a cell where [2-phosphoglycerate] = 2.3 x 10^{-4} M and [phosphoenolpyruvate] = 8.4 x 10^{-5} M?

5. Lambert-Beer Law problems:

Theory behind Beer-Lambert Law

Absorbance

Measuring the absorbance of a solution

If you have read the page about how an absorption spectrometer works, you will know that it passes a whole series of wavelengths of light through a solution of a substance (the sample cell) and also through an identical container (the reference cell) which only has solvent in it.

For each wavelength of light passing through the spectrometer, the intensity of the light passing through the reference cell is measured. This is usually referred to as I_o - that's I for Intensity.

The intensity of the light passing through the sample cell is also measured for that wavelength - given the symbol, I.

If I is less than I_o, then obviously the sample has absorbed some of the light. A simple bit of math is then done in the computer to convert this into something called the ***absorbance*** of the sample - given the symbol, A.

For reasons to do with the form of the Beer-Lambert Law (below), the relationship between A (the absorbance) and the two intensities is given by:

$$A = \log_{10} \frac{I_o}{I}$$

On most of the diagrams you will come across, the absorbance ranges from 0 to 1, but it can go higher than that.

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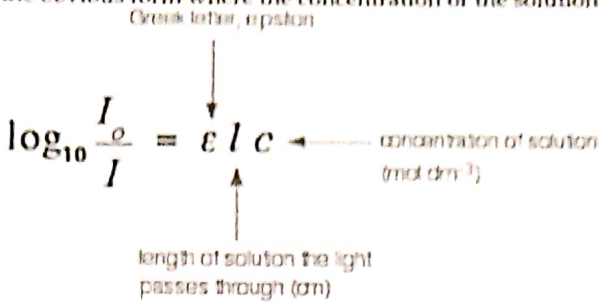
An absorbance of 0 at some wavelength means that no light of that particular wavelength has been absorbed. The intensities of the sample and reference beam are both the same, so the ratio I_o/I is 1. Log_{10} of 1 is zero.
An absorbance of 1 happens when 90% of the light at that wavelength has been absorbed - which means that the intensity is 10% of what it would otherwise be.
In that case, I_o/I is 100/10 (=10) and $\log_{10}$ of 10 is 1.

What the Law looks like

You will find that various different symbols are given for some of the terms in the equation - particularly for the concentration and the solution length. I'm going to use the obvious form where the concentration of the solution is "c" and the length is "l".



You should recognise the expression on the left of this equation as what we have just defined as the absorbance, A. You might also find the equation written in terms of A:

$$A = \varepsilon l c$$

That's obviously easier to remember than the first one, but you would still have to learn the equation for absorbance. It might be useful to learn it in the form:

$$A = \log_{10} \frac{I_o}{I} = \varepsilon l c$$

The Greek letter epsilon in these equations is called the molar absorptivity - or sometimes the molar absorption coefficient.

Molar absorptivity

If you rearrange the simplest of the equations above to give an expression for epsilon (the molar absorptivity), you get:

$$\varepsilon = \frac{A}{l c}$$

Remember that the absorbance of a solution will vary as the concentration or the size of the container varies. Molar absorptivity compensates for this by dividing by both the concentration and the length of the solution that the light passes through. Essentially, it works out a value for what the absorbance would be under a standard set of conditions - the light travelling 1 cm through a solution of 1 mol dm-3.

That means that you can then make comparisons between one compound and another without having to worry about the concentration or solution length.

Values for molar absorptivity can vary hugely. For example, ethanal has two absorption peaks in its UV-visible spectrum - both in the ultra-violet. One of these corresponds to an electron being promoted from a lone pair on the oxygen into a pi anti-bonding orbital; the other from a pi bonding orbital into a pi anti-bonding orbital.

Table1.1:

electron jump	wavelength of maximum absorption (nm)	molar absorptivity
lone pair to π anti-bonding orbital	290	15
π bonding to π anti-bonding orbital	180	10,000

The ethanal obviously absorbs much more strongly at 180 nm than it does at 290 nm. (Although, in fact, the 180-nm absorption peak is outside the range of most spectrometers.) You may come across diagrams of absorption spectra plotting absorptivity on the vertical axis rather than absorbance. However, if you look at the figures above and the scales that are going to be involved, you aren't really going to be able to spot the absorption at 290 nm. It will be a tiny little peak compared to the one at 180 nm. To get around this, you may also come across diagrams in which the vertical axis is plotted as $\log_{10}$(molar absorptivity).

If you take the logs of the two numbers in the table, 15 becomes 1.18, while 10,000 becomes 4. That makes it possible to plot both values easily, but produces strangely squashed-looking spectra!

Sample questions and answers

1. A solution of Tryptophan has an absorbance at 280 nm of 0.54 in a 0.5 cm length cuvette. Given the absorbance coefficient of trp is 6.4 × 103 LMol-1cm-1.

What is the concentration of solution? Solution:

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As ε = A / l c
l= 0.5 cm
A= 0.54
ε = 6.4 × 103 LMol-1cm-1
C=?
So c = A/ε l
= 0.54 / 6.4 × 103 × 0.5 Answer = 0.000168 M

2. A solution of thickness 2 cm transmits 40% incident light. Calculate the concentration of the solution, given that ε = 6000 dm3 /mol/cm.

Solution:
A = 2 - log10 %T = 2 - log10 40 = 2 – 1.6020 = 0.398
A = ε l c
l= 2cm
ε = 6000 dm3/mol/cm
A=0.398
c=?
So c = A/ ε l = 0.398/ 6000 × 2
Answer = 3.316 X 10 – 5 mol / dm3

3. A solution shows a transmittance of 20%, when taken in a cell of 2.5 cm thickness. Calculate its concentration, if the molar absorption coefficient is 12000 dm3/mol/cm.

Solution:
A = 2 - log10 %T = 2 - log10 20 = 2 – 1.301 = 0.698
A = ε l c
l= 2.5 cm
ε = 12000 dm3/mol/cm A=0.698
c=?
So c = A/ ε l
= 0.698/ 12000 × 2.5 Answer = 2.33 X 10 – 5 mol / dm3

4. Calculate the molar absorptivity of a 1 x 10-4 M solution, which has an absorbance of 0.20, when the path length is 2.5 cm.

Solution:
A = ε l c l= 2.5 cm A= 0.20
C= 1 x 10 – 4 M
ε =?
So ε = A / l c
= 0.20/ 2.5 ×1 x 10-4 Answer = 800 dm3/mol/cm.

5. The concentration of yeast t-RNA in an aqueous solution is 10 M. The absorbance is found to be 0.209 when this Solution is placed in a 1.00 cm cuvette and 258 nm radiations are passed through it.
a) Calculate the specific absorptivity, including units, of yeast t-RNA.
b) What will be the absorbance if the solution is 5 M?
c) What will be the absorbance if the path length of the original solution is increased to 5.00 cm?

Solution 5a
l = 1.00 cm c = 10.00 M A=0.209
So ε = A / l c
= 0.209 / 1.00 cm X 10 M Answer = 0.0209 dm3/mol/cm.
5b
ε = 0.0209 dm3/mol/cm. l = 1.00 cm
c = 5.00 M
A=?
So A = ε l c
A = 0.0209 dm3/mol/cm. X 1.00 cm X 5M
Answer = 0.1045 5c
ε = 0.0209 dm3/mol/cm. l = 5.00 cm
c = 10.00 M A=?
So A = ε l c
A = 0.0209 dm3/mol/cm X 5.00 cm X 10.00 M
Answer = 1.045

6. Calculate the molar absorptivity of a 0.5 x 10-3 M solution, which has an absorbance of 0.17, when the path length is 1.3 cm.

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Solution:
A = ε l c l= 1.3 cm A= 0.17
C= 0.5 x 10-3 M ε =?
So ε = A / l c
= 0.17/ 1.3 × 0.5 x 10-3 Answer = 261.53 dm3/mol/cm.

7. A CaCO3 solution shows a transmittance of 90%, when taken in a cell of 1.9 cm thickness. Calculate its concentration, if the molar absorption coefficient is 9000 dm3/mol/cm.
Solution:
A = 2 - log10 %T = 2 - log10 90 = 2 – 1.954 = 0.045
A = ε l c
l= 1.9 cm
ε = 9000 dm3/mol/cm A=0.045
c=?
So c = A/ ε l
= 0.045/ 9000 × 1.9
Answer = 2.631 × 10 -6 mol / dm3

8. Extinction coefficient of NADH at 340 nm is 6440 L/mol/cm. whereas NAD does not absorb at 340nm. What absorbance will be observed when light at 340nm passes through a 1cm cuvette containing 10uM NADH and 10 uM NAD.
Solution:
ε = 6440 L/mol/cm. l = 1.00 cm
c = 10.0 uM =10 X 10-6 M
A=?
So A = ε l c
A = 6440 L/mol/cm X 1.00 cm X 10 X 10-6 M
Answer = 0.0644
Note: this absorbance is only for NADH because NAD do not absorb at 340nm.

9. A 1.00 × 10–4 M solution of an analyte is placed in a sample cell with a path length of 1.00 cm. When measured at a wavelength of 350 nm, the solution's absorbance is 0.139. What is the analyte's molar absorptivity at this wavelength?
Solution
l = 1.00 cm
c = 1.00 × 10–4 M
A=0.139
ε =?
So ε = A / l c
= 0.139/ 1.0 × 1.00 x 10-4
Answer = 1390 cm−1 M−1

10. The absorbance of a Cu sulphate solution containing 0.500 mg Cu/mL was reported as 0.3500 at 440 nm.
a) Calculate the specific absorptivity, including units, of Cu sulphate on the assumption that a 1.00 cm cuvette was used.
b) What will be the absorbance if the solution is diluted to twice its original volume
Solution
a) l = 1.00 cm
c = 0.500 mg/ml A= 0.3500
ε =?
So ε = A / l c
= 0.3500 / 1.0 × 0.500
Answer = 0.7 cm-1(mg/mL)-1
b) c= 0.250 mg/ml
ε= 0.7 cm-1(mg/mL)-1
l = 1.00 cm
A=?
So A = ε l c
A= 0.7 cm-1(mg/mL)-1 X 1.00 cm X 0.250 mg/ml
Answer = 0.175

Practice problems and solutions

1. The absorbance is a dimensionless quantity, that is, a quantity without units. What will be the units of **e**, the specific absorptivity? Remember that 1 mL = 1 cm^3.

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From Beer-Lambert law:
A = elC
A is the absorbance e is the extinction coefficient l is the cell path length C is the concentration the units of e are
e = [A] / [cm X C] ^ e = [$cm^{-1}(C)^{-1}$]
where C is the concentration units that you are using.

2. How does the percent transmittance of a solution vary with (a) increasing concentration and (b) increasing path length?
   a. Percent Transmittance is inversely proportional to the concentration (more light goes through if the solution is very dilute).
   b. Percent Transmittance is inversely proportional to the Path length. Recall that A = -log[I/I_0], thus the relation between these quantities is logarithmic.

3. The absorbance of an iron thiocyanate solution containing 0.00500 mg Fe/mL was reported as 0.4900 at 540 nm.
a. Calculate the specific absorptivity, including units, of iron thiocyanate on the assumption that a 1.00 cm cuvette was used.
b. What will be the absorbance if (1) the solution is diluted to twice its original volume and (2) the solution is placed in a 5.00 cm cuvette?
4. Reffer to problem 3.
a. What percent of light is transmitted by the original iron thiocyanate solution?
b. What concentration of iron thiocyanate will absorb 50% of the entering light?
5. The concentration of yeast t-RNA in an aqueous solution is 10 pg/mL. The absorbance is found to be 0.209 when this solution is placed in a 1.00 cm cuvette and 258 nm radiation is passed through it.
a. Calculate the specific absorptivity, including units, of yeast t-RNA.
b. What will be the absorbance if the solution is diluted to 10 pg/mL?
c. What will be the absorbance if the path length of the original solution is increased to 5.00 cm?

Answers with explanations:

1/
2/
3/3a
e = [A] / [cm X C]
e = [0.4900] / [1.00 cm X 0.00500 mg/mL] ^ e = [98.0 $cm^{-1}(mg/mL)^{-1}$]
3.b1 A = elC
A = [98.0 $cm^{-1}(mg/mL)^{-1}$] X 1.00 cm X [0.00250] ^ 0.2450 3. b2 A = elC
A = [98.0 $cm^{-1}(mg/mL)^{-1}$] X 5.00 cm X [0.00250] ^ A = 1.225
4/4a
A = -log[I/I0]
-0.4900 = log[I/I0] ^ 0.3236 = [I/I0]
[0.3236 X I_0] = I ^ 32.36% of the light is transmitted 4b
50.0% ^ 0.500
-log [0.500] = A ^ 0.3010 = A = elC 0.3010 = [98.0 $cm^{-1}(mg/mL)^{-1}$] X 1.00 cm X [C]
C = 0.00307 mg/mL
5/5a
e = [A] / [cm X C] ^ e = [0.209] / [1.00 cm X 10 pg/mL] e = [0.209 $cm^{-1}(pg/mL)^{-1}$]
5b
A = elC
A = [0.209 $cm^{-1}(pg/mL)^{-1}$] X 1.00 cm X [5.00 pg/mL]
A = 0.01045 5c
A = elC = [0.209 $cm^{-1}(pg/mL)^{-1}$] X 5.00 cm X [10.00 pg/mL] = 1.045

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Chapter 2

Bacterial growth curve problems

Introduction:
Microbial growth involves the understanding of exponential growth and associated mathematical equations. These exponential growth equation is useful for population growth problems from the ecology chapter as well. It requires you to remember values of log to solve those equation.

Chapter content:

1. Bacterial Growth and division problems

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1. Bacterial Growth and division problems:

Theory behind bacterial growth

Most bacteria reproduce by binary fission, which results in doubling of the number of viable bacterial cells. Therefore, during active bacterial growth, the number of bacterial cells and, hence their population, continuously doubles at specific time intervals because each binary fission takes a specific duration of time.

This 'specific time interval' between two subsequent binary fissions is known as generation time or doubling time.

It we start with a single bacterial cell, its fission proceeds as a geometric progression (exponential growth) with one cell dividing to form two, these two to four, further to eight and so on i.e. each succeeding fission (generation), assuming no cell death, doubles the population size.

$$1 \rightarrow 2^1 \rightarrow 2^2 \rightarrow 2^3 \rightarrow 2^4 \rightarrow 2^5 \rightarrow \rightarrow 2^n$$

Geometric progression. *n* = the number of generations

Generation time or doubling time varies considerably among different bacteria. A bacterium such as *E. coli* enjoys generation time as short as 20 minutes under optimal conditions, although in nature many bacteria have generation times of several hours. One cell of E. coli with a 20-minute generation time, for convenience, will multiply to 512 cells in 3 hours, to 4096 cells in 4 hours, and to 32768 cells in 5 hours, and so on.

How to solve problems?

To calculate the generation time of individual microorganisms the following experimental data are required:

1. The number of organisms present at the beginning.
2. The number of organisms present at the end of a given time interval.
3. The time interval.

The relationship of cell numbers and generations can be expressed in a series of equations. Starting with a single cell, the total population B at the end of a given time period would be expressed as

$B = 1 \times 2^n$

Where 2^n is the bacterial population after n generations. However, under practical conditions several thousands of bacteria are introduced into the medium at zero time and not one, so the formula now becomes.

$B_n = B_0 \times 2^n$

Where B_0 = number of organisms at zero time.

B_n = number of organisms after n generations.

n= number of generations.

Solving the equation for n, we have

$\log B_n = \log B_0 + n \log 2$

$n = \log B_n - \log B_0 / \log 2$

Thus, we can calculate the number of generations if we know the initial population B_0 and the population B_n after time t. The generation time G is equal to t (the time which elapsed between B_0 and B_n) divided by the number of generations n, or

$G = t/n = t \log 2 / \log B_n - \log B_0$

An alternative method is used to describe bacterial growth in mathematical terms when the culture is undergoing balanced growth. The rate of increase in bacteria at any particular time is proportional to the Cell number of mass or bacteria present at that time (Fig. 19.1).

The constant of proportionality is an index of the rate of growth and is called the exponential growth rate constant (K). It is defined as number of doublings in unit time, and is usually expressed as the number of doubling in an hour.

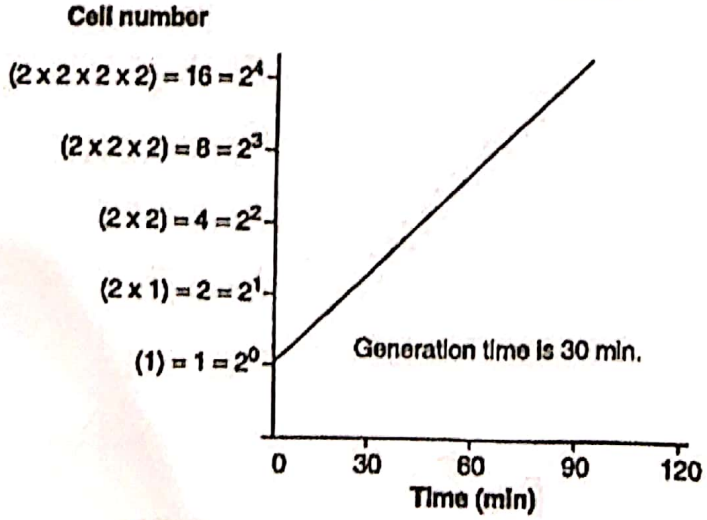

FIG. 19.1. Relationship of generation time and growth.

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It is calculated from the following equation:

$B_n = B_0 \times 2^{Kt}$

B_n = Population at time t.

B_0 = Population at time zero.

Taking the logarithms

$\log B_n = \log B_0 + Kt \log 2$, and

Solving the equation for K

$K = \log B_n - \log B_0 / t \log 2$

The exponential growth rate constant is therefore reciprocal to generation time, i.e.

$G = 1/K$

For example, generation time of E. coli is 20 minutes, i.e. 1/3 hour.

$1/3 = 1/K$

K = 3 doublings per hour.

Another Approach to solve this:

Calculation of Generation Time

When growing exponentially by binary fission, the increase in a bacterial population is by geometric progression. If we start with one cell, when it divides, there are 2 cells in the first generation, 4 cells in the second generation, 8 cells in the third generation, and so on.

The **generation time** is the time interval required for the cells (or population) to divide.

G (generation time) = (time, in minutes or hours)/n (number of generations)

$G = t/n$

t = time interval in hours or minutes

B = number of bacteria at the beginning of a time interval

b = number of bacteria at the end of the time interval

n = number of generations (number of times the cell population doubles during the time interval)

$b = B \times 2^n$ (This equation is an expression of growth by binary fission)

Solve for n:

$\log b = \log B + n \log 2$

$n = \frac{\log b - \log B}{\log 2}$

$n = \frac{\log b - \log B}{.301}$

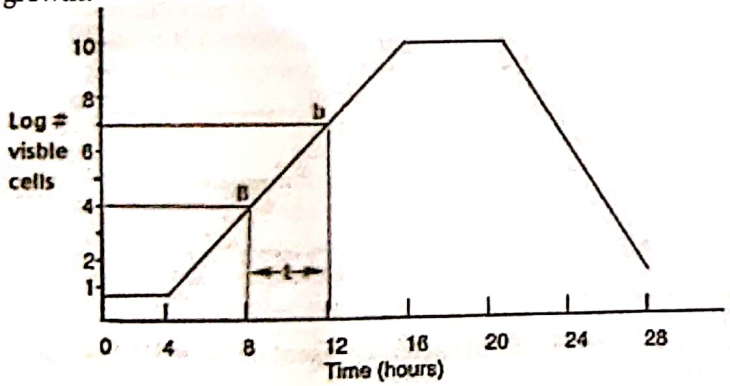
$n = 3.3 \log b/B$

$G = t/n$

Solve for G

$G = \frac{t}{3.3 \log b/B}$

Example: What is the generation time of a bacterial population that increases from 10,000 cells to 10,000,000 cells in four hours of growth?



$G = \frac{t}{3.3 \log b/B}$

$G = \frac{240 \text{ minutes}}{3.3 \log 10^7/10^4}$

$G = \frac{240 \text{ minutes}}{3.3 \times 3}$

G = 24 minutes

Example Problems

Problem 1-Calculate the number of bacteria in a culture at a given time

1. **How many bacteria are present after 51 hours if a culture is inoculated with 1 bacterium? Use the model, $N(t) = N_o e^{kt}$, and assume the population doubles every 3 hours. ($N(t)$ is the population size at time t and k is a constant.)**

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Now that we have convinced ourselves that an exponential model is appropriate, but not perfect, we will use the general model

$$N(t) = N_0 e^{kt} \tag{1}$$

where $N(t)$ represents the population size at time t (Note: you may also use the model, $N(t) = N_0 a^t$). The question asks for the population size when $t = 51$. In order to use this model, we will need to figure out the values of the constants, N_0 and k. At time $t = 0$, there is a single bacterium, therefore $N_0 = 1$. Substituting $N_0 = 1$ into (1) gives,

$$N(t) = 1 \cdot e^{kt} = e^{kt}, \tag{2}$$

We now use the fact that when $t = 3$, $N(3) = 2$ (the population has doubled). Substituting the point $(t, N(t)) = (3, 2)$ into (2),

$$N(3) = 2 = e^{3k} \tag{3}$$

We can now solve for k in (3) by "undoing" the exponential using the *natural logarithm*,

$$\ln 2 = \ln e^{3k} = 3k,$$

$$\Longrightarrow k = \frac{\ln 2}{3}.$$

Using this value of k our model in (2) becomes,

$$N(t) = e^{\frac{\ln(2)}{3} t}. \tag{4}$$

Now that we have our model, we need to find the population size after 51 hours. Substituting $t = 51$ into (4) yields,

$$N(51) = e^{\frac{\ln(2)}{3} 51} = 131,072.$$

Thus, we find that after 51 hours there are 131,072 bacteria.

Problem 2-Caluclate the number of bacteria you will need to start a new culture

2. **With how many bacteria should a culture be inoculated if there are to be 81,920 bacteria present on hour 42? Use the model, $N(t) = N_0 e^{kt}$, and assume the population doubles every 3 hours.**

We are asked to find the number of bacteria to inoculate a culture with if there are to be 81,920 bacteria present on hour 42. Since our bacteria are growing at the same rate in this problem as the last (i.e. doubling every three hours), our model is given by,

$$N(t) = N_0 e^{\frac{\ln(2)}{3} t}. \tag{5}$$

Our goal in this problem is to determine N_0, the initial number of bacteria. Using the information provided, we know that there are 81,920 bacteria on hour 42. Substituting this information into (5),

$$N(42) = 81920 = N_0 e^{\frac{\ln(2)}{3} 42} = 16384\, N_0$$

and solving for N_0,

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$$N_0 = \frac{81920}{16384} = 5.$$

Therefore, if we inoculate a culture with five bacteria, there will be 81,920 bacteria present on hour 42.

Problem 3-Calculate the time that a culture needs to grow to reach a given size
How long would it take for an initial population of 6 to reach a size of 12,288 bacteria? Use the model, $N(t) = N_0 e^{kt}$, and assume the population doubles every 3 hours.

This question asks us to find the time it takes for an initial population of six to reach 12,288. Since the rate of population growth is the same as in the previous questions, we use our previous model with $N_0 = 6$,

$$N(t) = 6e^{\frac{\ln(2)}{3}t}.$$

We would like to know at what time τ, does the population size reach 12,288. Incorporating this into the above equation,

$$N(\tau) = 12288 = 6e^{\frac{\ln(2)}{3}\tau},$$

$$\Longrightarrow \quad 2048 = e^{\frac{\ln(2)}{3}\tau}.$$

To solve for the variable τ , we need to take the natural logarithm of both sides of the above equation,

$$\ln 2048 = \ln e^{\frac{\ln(2)}{3}\tau} = \frac{\ln(2)}{3}\tau,$$

$$\Longrightarrow \quad \tau = \frac{3 \cdot \ln 2048}{\ln 2} = 33.$$

Therefore, it takes 33 hours for a culture inoculated with six bacteria to accumulate 12,288 bacteria.

Practice Problems (Apply your knowledge)

1. **A strep culture is sampled at two points in time, 1 hour and 10 minutes apart. 3.2 x 10^6 cells/ml are found in the first sample, and 1.8 x 10^7 cells/ml in the second sample. What was the doubling time?**
2. **A beginning yogurt maker decides to test how consistently his yogurt cultures grow. He starts with 1 x 10^5 cells/ml in each of two cultures. After 4 hours, he has 16 x 10^5 cells/ml in the first culture, and 20 x 10^5 cells/ml in the second culture. Are the doubling times within 10 minutes of each other?**
3. **A syphilis patient has tissue cultures taken one week apart. The first sample has 5000 CFUs per ???, while the second sample has 32000 CFUs. Four days later a third sample has 58000 CFUs. Is the population growing exponentially?**
4. **A doctor receives a pathogen sample which she believes to be M. tuberculosis (doubling time 13 hours in culture). She cultures the sample, finding that it grows from 0.2x10^5 cells/ml to 8 x 10^6 cells/ ml in 24 hours. Assuming that the culture was growing exponentially, is the sample likely to be M. tuberculosis?**

Answers:

1. 70/2.49 = 28.1 minutes.
2. Yes, the times are 60 and 55.5 minutes.
3. No -- Doubling time was faster for the first week (2.61 days) compared to the next four days (4.66 days)
4. No -- the culture grew 400-fold in 24 hours-- a doubling time of a little under 3 hours.

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Chapter 3

Plant Physiology / Water potential Problems

Introduction:
Water potential problems are the most important topics for asking questions from plant physiology. Water poteintial involves a very easy formula that involves 3 components and are easy to solve. These problems are not very common in CSIR NET or GATE exam but you may expect few questions from this chapter.

Chapter content:

1. Water Potential Math problems from Plant Physiology

Scan the following QR code to Know how to solve Plant Physiology problems

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Water Potential Math problems from Plant Physiology

Theory: What is it?

Because you will be working with potato cells in the laboratory, you need to understand the concept of water potential. Biologists use this term to describe the tendency of water to leave one place in favor of another. Water always moves from an area of higher water potential to an area of lower water potential.
Water potential is affected by two factors: pressure and the amount of solute. For example, imagine a red blood cell dropped into distilled water. Water will move into the red blood cell and cause the cell to expand, stretching the flexible membrane. At some point, the pressure of the incoming water will cause the cell to pop, just like an over-filled balloon.

How to solve problems?

Water potential is calculated using the following formula:
Water potential (Ψ) = pressure potential (Ψ_p) + solute potential (Ψ_s)
Pressure potential (Ψ_p):
In a plant cell, pressure exerted by the rigid cell wall that limits further water uptake.
Solute potential (Ψ_s):
The effect of solute concentration. Pure water at atmospheric pressure has a solute potential of zero. As solute is added, the value for solute potential becomes more negative. This causes water potential to decrease also.
In sum, as solute is added, the water potential of a solution drops, and water will tend to move into the solution.
In this laboratory we use bars as the unit of measure for water potential; 1 bar = approximately 1 atmosphere.

Important Things to Remember

- Water always moves from an area of higher water potential (higher concentration) to an area of lower water potential (lower water concentration)
- Water potential measures the tendency of water to leave one place in favor of another place (picture water . diffusing "down" a water potential or concentration gradient
- Water potential is affected by 2 factors: l. Addition of a solute - this always lowers the water potential since it decreases the concentration of water

* Solute potential can be calculated: Vj/s =-iCRT (i = ionization constant, C = molar concentration, R = Pressure constant = 0.0831, and T = temperature °K = 273 + °C of solution)

2. Pressure potential - an increase in pressure raises the water potential

- The pressure potential of pure water at atmospheric pressure is defined as zero
- Pressure potential is usually positive in living cells Water potential can be positive, negative or zero

Solute potential is always negative (since the solute potential of pure water is o, the addition of any solutes will make the solution have a lower (thus negative) water potential as the pressure potential increases, so does the water potential

Example Problems and solutions

Problem 1

Given: A potato cell is placed in pure water. The solute potential of the potato is -3. The solutes in the potato cell cannot diffuse out of the cell. What will be the pressure potential of the cell when it reaches equilibrium?

What you know;

- The initial pressure potential = 0 (since we are doing the experiment in normal conditions = atmospheric pressure).

OK you chemists and physicists...the fact that our elevation is significantly above sea level does mean that our atmospheric pressure is slightly less than atmospheric, but we ignore that when we are doing these types of calculations

- Water will diffuse into the potato because:

o The solution is hypotonic o The water concentration is higher outside the cell
o The water potential is higher outside the cell since there is a negative solute potential inside the cell o The potato is a plant, therefore it has a cell wall and will not burst.
o The potato cell will gain water, swell and become turgid. It will swell until the contents of the cell push against the cell wall and produce an increase in the pressure potential o Eventually, the pressure inside the cell will build up until it balances the negative solute potential inside the cell and the net movement of water will be zero

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Figure 1: Initial Conditions

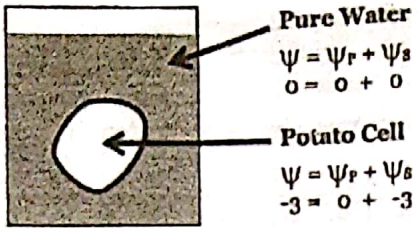
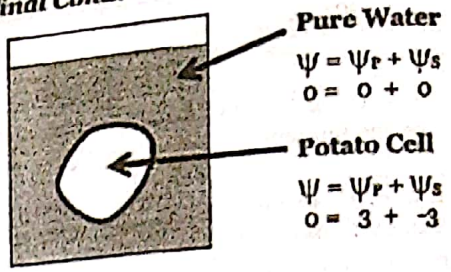


Figure 2: Final Conditions



Answer; The pressure potential of the potato cell at equilibrium is 3!

Problem 2

Given: Potato cores are placed in sucrose solutions with various molar concentrations, and the percent change in mass is calculated for each solution. The data is graphed. From this data, it is determined that the molar concentration of the potato core is 0.32

A. What is the solute potential of the sucrose solution?

B. What is the water potential of the sucrose solution?

What vou know:

- The initial pressure potential = o (since we are doing the experiment in normal conditions = atmospheric pressure).
- The molar concentration of the potato is the sucrose molarity in which the mass of the potato core does not change (where it crosses zero on the x axis)
- This point is the molar concentration of sucrose with a water potential that is equal to the potato tissue water potential - so the net movement of water is zero, and the system is isotonic

« Solute potential = -iCRT

- The ionization constant (i) for sucrose is always 1.0 because it does not ionize in water)

® The experiment is assumed to be conducted at room temperature, 22°C.

Problem 3

Given: A student calculates the water potential of the solution inside a dialysis bag to be -6.25 bar, and the solution surrounding the bag to be -3.25 bar. The system is open to the air (this can always be assumed unless otherwise stated).

A. What is the solute potential of the dialysis bag?

B. What is the solute potential of the solution?

C. In which direction will the net movement of water flow?

What you know

- The system is open to the air (this can always be assumed unless otherwise stated). Thus, the pressure potential is o (since we are doing the experiment in normal conditions = atmospheric pressure).
- The experiment is assumed to be conducted at room temperature, 22°C.

Practice problems and solutions

1. If a plant cell's Y_p = 2 bars and its Y_s = -3.5 bars, what is the resulting Y?

2. The plant cell from question #1 is placed in a beaker of sugar water with Y_s = -4.0 bars. In which direction will the net flow of water be?

3. The original cell from question # 1 is placed in a beaker of sugar water with Y_s = -0.15 MPa (megapascals). We know that 1 MPa = 10 bars. In which direction will the net flow of water be?

4. The value for Y in root tissue was found to be -3.3 bars. If you place the root tissue in a 0.1 M solution of sucrose at 20°C in an open beaker, what is the Y of the solution, and in which direction would the net flow of water be?

5. NaCl dissociates into 2 particles in water: Na^+ and Cl^-. If the solution in question 4 contained 0.1 M NaCl instead of 0.1 M sucrose, what is the Y of the solution, and in which direction would the net flow of water be?

6. A plant cell with a Y_s of -7.5 bars keeps a constant volume when immersed in an open-beaker solution that has a Y_s of -4 bars. What is the cell's Y_p?

7. At 20°C, a plant cell containing 0.6 M glucose is in equilibrium with its surrounding solution containing 0.5 M glucose in an open container. What is the cell's Y_p?

8. Surrounding solution: $Y = Y_p + Y_s$ = 0 bars + (1) (0.5 mol/L) (0.0831 L*bars/mol*K) (293 K) = -12.2 bars Cell at equilibrium: -12.2 bars = Y_p + (1) (0.6 mol/L) (0.0831 L*bars/mol*K) (293 K) = Y_p + (-14.6 bars) Y_p = -12.2 bars - (-14.6 bars) = 2.4 bars. Calculate the solute potential of a 0.15M salt solution in a beaker at 20° C. Round to nearest thousandth.

9. If the molar concentration of a sugar solution in an open beaker is 0.4M, calculate the solute potential at 27 degrees. Round to nearest hundredth.

Answers with explanations:

1. $Y = Y_p + Y_s$ = 2 bars + (-3.5 bars) = -1.5 bars

2. The pressure potential of a solution open to the air is 0. Therefore, the water potential of the sugar water is -4.0 bars [Y = 0 bars +(-4.0) bars]. Since free water always flows towards the solution with a lower water potential, the flow of water would be outside of the cell.

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3. -0.15 MPa = -1.5 bars

The water potential of the sugar water is -1.5 bars [Y = 0 bars + (-1.5 bars)]. Since the water potential of the original cell was also -1.5 bars, there would be no net flow of water. The cell and the sugar water are in equilibrium.

4. Y_s = -iCRT

Ys = -(1)(0.1 mol/L)(0.0831 L*bars/mol*K)(293 K) = -2.43 bars

Y = $Y_p + Y_s$ = 0 bars + -2.43 bars = -2.43 bars

The Y of the root tissue is -3.3 bars and the Y of the sucrose solution is -2.43 bars. Water will flow into the root tissue because free water always moves towards the lower overall water potential.

5. Ys = -iCRT

Ys = -(2) (0.1 mol/L) (0.0831 L*bars/mol*K) (293 K) = -4.87 bars

Y = $Y_p + Y_s$ = 0 bars + -4.87 bars = -4.87 bars

The Y of the root tissue is -3.3 bars and the Y of the NaCl solution is -4.87 bars. Water will flow out of the root tissue and into the salt solution because free water always moves towards the lower overall water potential.

6. The plant cell keeps a constant volume because of the buildup of turgor pressure inside the cell. The Y_p at equilibrium would be the difference between the two solute potentials, which is 3.5 bars.

7.

8. ψ_s = -iCRT

ψ_s = (-2) (.15M) (0.0831liter bars) (273+20 degrees)

ψ_s =-7.304 bars

9. ψ_s = -iCRT

ψ_s = (-1) (0.4M) (0.0831bar) (273+27)

ψ_s =-9.972 bars

Practice problems: Apply your knowledge

1. A cell is in equilibrium with its surroundings. The molarity of the surrounding solution is 0.5M.

i = ionization constant (assume that is 1)

C=molar concentration (given above)

R = pressure constant (R=0.00831-liter MPa/mole °K)

T = temperature in °K (room temp is about 293°K)

a. Calculate the solute potential of the surrounding solution.

b. Find the water potential of the surrounding solution.

What is the water potential of the cytoplasm of the cell?

2. If a cell's Ψ_P = 3 bars and its Ψ_S = -4.5 bars, what is the resulting Ψ?

The cell above is placed in a beaker of sugar water with Ψ_S = -4.0 bars. In which direction will the net flow of water be? Explain

The original cell from above is placed in a beaker of sugar water with Ψ_S = -0.15 MPa (megapascals). We know that 1 MPa = 10 bars. In which direction will the net flow of water be? Explain/support your answer.

3. Calculate the water potential of a solution of .15 M Sucrose. (assume the room temperature is 23 degrees C) Do you calculate in "bars" not Pascals.

4. At 20 C, a cell with a pressure potential of 3 bars is in equilibrium with the surrounding .4 solution of sucrose in an open beaker. What is the molar concentration of sucrose in the cell?

5. The water potential in root tissue was found to be -3.3 bars. If you place the root tissue in a .1 M solution of sucrose at 20 Celsius in an open beaker, what is the water potential of the solution, and in which direction would the net flow of water be?

6. A plant cell with a Ψs of -7.5 bars keeps a constant volume when immersed in an open-beaker solution that has a Ψs of -4 bars. What is the cell's ΨP?

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Chapter 4

Human Physiology problems

Introduction:
Human physiology is an important chapter for many life science competitive exams like CSIR NET, ICMR-NET etc. It is also required in some pre-medical entrance tests. There are several regions in human physiology to ask analytical questions. Different body systems involve neumerous formula which are not easy to remember. Although not many questions are asked from this chapter in CSIR NET exam. Here I have enlisted only few of those important formula that can be useful answering questions from this chapter.

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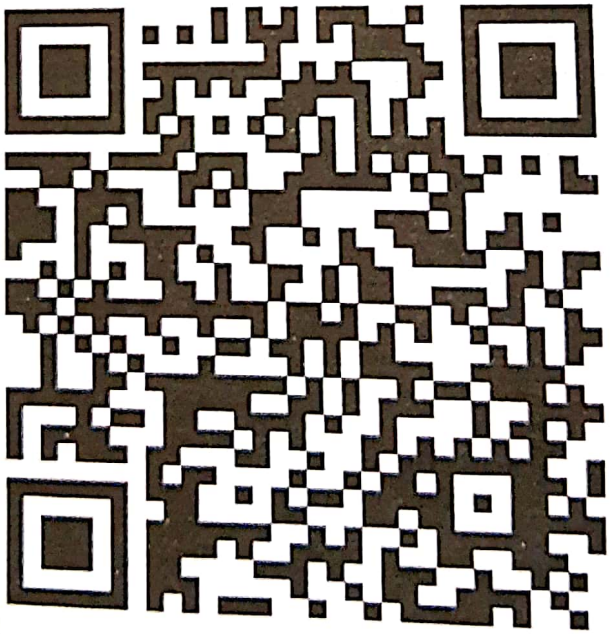

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1. Human Physiology Problems and answers

Sample problems with answers

1. Rate of urine formation

QU: Given that 250ml of urine was produced in 30 minutes. What is the rate of urine formation (in ml/minute)?

Ans: The stated rate of 250 ml/30min needs to be converted into a rate per minute

= 250 ÷ 30 ml/min = 8.3ml/min

2. Respiratory rate

QU: If the number of inhalations in 20 seconds was 5, what is the respiratory rate (in breaths/min)?

Ans: The given rate of 5 breaths per 20sec must be converted into a rate per minute.

Generally, the breaths are counted for 20 or for 30 seconds so that they may be converted into breaths per minute (60sec) easily (by multiplying by 3 or 2 respectively).

(5 breaths / 20 secs) × 3 = 15 breaths / 60sec = 15 bpm

3. Heart rate

QU: If the radial pulse is counted for 15 seconds at 14 beats, what is the heart rate (in beats per minute)?

Ans: the given rate of 14beats/15sec must be converted into a rate per minute

= (14 beats / 15s) × 4 = 56beats/60s = 56bpm

4. Mean arterial pressure

Mean arterial pressure (MAP) is the "average" of the maximum blood pressure in the aorta (i.e., during systole) and the minimum blood pressure in the aorta (i.e., during diastole). Arterial blood pressure is written with larger number (systolic pressure) first, separated from the lower value (diastolic pressure) by a slash: eg 120/40 in units of millimeters of mercury.

MAP = diastolic pressure + ⅓ × (systolic pressure – diastolic pressure)

QU: what is the MAP for a blood pressure reading of 124/78? (systolic/diastolic)?

Ans: MAP = 78 + ⅓ × (124 -78) = 78 + ⅓ × (46) = 78 + 15.3 = 93.3mmHg

5. Total magnification for a microscope

Microscopes have two sets of lenses: the eyepiece lens (which you look through) and the objective lens (which is placed just above the specimen being viewed).

The eyepiece lenses have a magnification of 10X.

The 4 objective lenses have magnifications of 4X, 10X, 40X and 100X

Total magnification = eyepiece lens magnification × objective lens magnification

QU: What is the total magnification of a microscope with eyepiece lens of 10X and objective lens of 40X?

Ans: 10×40 = 400X magnification

6. Normal saline

A solution that is infused intravenously is "normal saline". It has a concentration of dissolved particles that is the same as the concentration of dissolved particles in blood. However, saline contains only sodium chloride (Na^+Cl^- = table salt) dissolved in water.

One kilogram of normal saline solution contains 9g of sodium chloride.

%conc = mass of solute ÷ mass of solution × 100

QU: what is the percentage concentration of sodium chloride in this solution?

Ans: %conc = 9g ÷ 1000g × 100 = 0.009 × 100 = 0.9%

7. Waist to hip ratio (WHR)

In many cases persons with extra fat located around the middle ('apple' shaped) are at higher risk for diseases such as heart disease and diabetes than those who carry weight around their hips and thighs ('pear' shaped). (A waist circumference >88cm for women or >102cm for men defines abdominal obesity).

WHR = waist circumference (cm) ÷ hip circumference (cm)

(the waist circumference = circumference at or just above the umbilicus while breathing out gently)

(the hip circumference = circumference at the widest point ie at greater trochanters which are the bony bits at the side of your hips)

QU: determine your WHR using the measured circumference of your waist and your hips

8. Ankle/Brachial index (ABI)

Ankle/Brachial index (ABI) is a non-invasive test used for assessing the arterial circulation in the lower limbs. Normally the blood systolic pressure measured in the feet is greater than or equal to the blood systolic pressure measured at the wrist.

ABI = systolic pressure at radial artery ÷ systolic pressure in posterior tibial artery

QU: Determine the ABI for a male when:

1. Supine (lying on your back)
2. Standing

(use typical male values:

Brachial artery systolic pressure=119 mmHg;

Tibial artery systolic pressure (when supine) = 148 mmHg;

Tibial artery systolic pressure (when standing) = 216 mmHg)

Ans:

1. Supine ABI = 119 ÷ 148 = 0.8
2. Standing ABI = 119 ÷ 216 = 0.55

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Chapter 4

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Ans:

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2. Standing ABI = 119 ÷ 216 = 0.55

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9. Body Mass Index (BMI)

Body mass index (BMI) combines measurements of height and weight to determine whether an individual is the appropriate weight for their height.

$$BMI = \frac{mass \cdot (in \cdot kg)}{height^2 \cdot (in \cdot m)}$$

- Underweight: BMI <18.5 kg/m²
- Healthy weight: BMI between 18.5 & 24.9 kg/m²
- Overweight: BMI between 25 & 29.9 kg/m²
- Obesity: BMI > 30 kg/m²
- Bariatric: BMI > 40 kg/m²

QU: Determine the Body Mass index (corrected to 1 decimal place) using your own height and weight. Classify yourself as underweight, of healthy weight, overweight or obese.

Ans: (for example)

$$BMI = \frac{70 \cdot kg}{1.76 \cdot m \cdot \times \cdot 1.76 \cdot m} = \frac{70}{3.1} = 22.6 \cdot kg/m^2$$

6. Mosteller formula for total body surface area

Pharmacology application: anti-cancer drugs are sometimes administered at a dose depending on the individual's body surface area (BSA), e.g. 200 mg/m² (but the practice has been criticized as unscientific). BSA can be calculated from your measured height and mass.

QU: Use the Mosteller formula below to calculate body surface area (**to 2 decimal places**) for a person who is 176cm tall and has a mass of 72 kg:

$$BSA \cdot (in \cdot m^2) = \cdot \sqrt{\frac{height(cm) \times weight(kg)}{3600}}$$

[hint: Multiply height by weight, then divide the answer by 3600. Now use a calculator take the "square root" of this answer.]

Ans:

$$BSA \cdot = \cdot \sqrt{\frac{176cm \times 72kg}{3600}}$$

$$\cdot = \cdot \sqrt{\frac{12672}{3600}}$$

$$\cdot = \cdot \sqrt{3.52}$$

$$\cdot = \cdot 1.88 \ m^2$$

11. Percentage of body surface using "rule of 9s"

Total body surface area (TBSA) is an assessment of injury to or disease of the skin, such as burns or psoriasis. You can estimate the body surface area on an adult that has been burned by using multiples of 9, as follows:

Head = 9% of skin surface
Chest (front) = 9% (back = 9%)
Abdomen (front) = 9% (buttocks & back 9%)
Upper/mid/low back and buttocks = 18%
Each arm = 9% (front = 4.5%, back = 4.5%)
Groin = 1%
Each leg = 18% total (front = 9%, back = 9%)

QU: What percentage of the skin surface has been burned if both legs, the groin and the front chest and abdomen were burned?

Ans: both legs (18% x 2 = 36%), the groin (1%) and the front chest and abdomen (18%) were burned, hence 36 + 1 + 18 = 55% of the body surface.

12. ECG calculations

An ECG measures the variation with time of the voltage produced by the heart muscle as it contracts (see figure below). The graph is printed on moving graph paper marked in millimeters.

Since the paper speed was 25mm/second, each millimeter (mm) of paper corresponds to a time interval of 0.04 seconds.

Example: If an interval requires 4mm of paper its duration will be:

4mm × 0.04 sec/mm = 0. l6 seconds.

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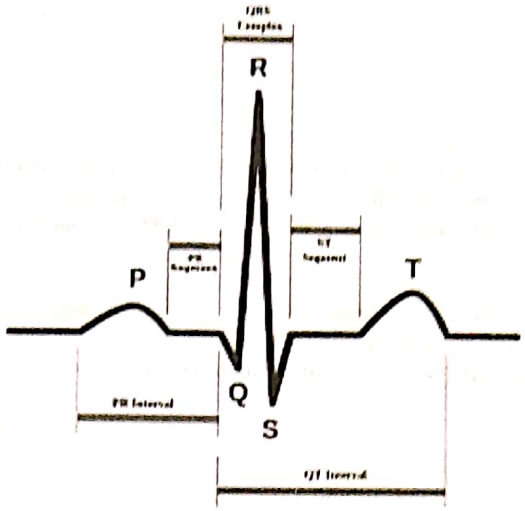

QU: Calculate the time duration of:
25mm
5mm
2.5mm
Ans:
25mm × 0.04 sec/mm = 1.0 seconds = 1000ms
5mm × 0.04 sec/mm = 0.2 seconds = 200ms;
2.5mm × 0.04 sec/mm = 0.1s = 100ms
On the ECG graph, the time from the start of the "P wave" to the start of the "R wave" is called the P-R interval (it is normally less than 0.2 second)
QU: calculate the P-R interval (in seconds) if there is 4mm between the two waves.
Ans: P-R interval duration: 4mm × 0.04 sec/mm = 0.16s
The QT interval of the ECG graph varies with heart rate - becoming shorter at faster rates. It is usually corrected (to QTc) by using Bazett's formula:
$QTc = QT \div \sqrt{(RR)}$
QU: determine QT_c given that QT = 12mm; RR = 32mm (and convert mm to time using 1mm = 0.04s)
Ans: 12mm corresponds to 12×0.04 = 0.48s

13. GFR (creatinine clearance) in ml/min estimated with Cockroft-Gault formula

Creatinine clearance rate (CCR) is the volume of blood plasma that is cleared of creatinine per unit time and is a useful measure of how well the kidneys are working (renal function). Glomerular filtration rate (GFR) describes the flow rate of filtered fluid through the kidney. CCR is used for approximating the GFR in order to indicate of the health state of the kidney. (CCR of between 90 and 150 ml/min is healthy)
Creatinine clearance= [(140 - Age) x weight (kg) x F] ÷ (Plasma Creatinine × 0.8136) ml/min
(Where F = 1 if male, and 0.85 if female, Requires creatinine measurement in umol/L)
QU: Use plasma creatinine of 110 umol/L to determine creatinine clearance for a 59yo male of mass 82kg.
QU: Use plasma creatinine of 80 umol/L to determine creatinine clearance for a 36yo female whose weight is 72kg.
Ans (for male): CCR = [(140 – 59) ×82×1] ÷ (110×0.8136) ml/min
[(81) ×82] ÷ 89.5
= 6642 ÷ 89.5
= 74.2 ml/min
Ans (for female): CCR = [(140 – 36) ×72×0.85] ÷ (80×0.8136) ml/min
= 6365 ÷ 98.3
= 64.8 ml/min

14. Lung volumes

Forced vital capacity (FVC) is the maximum volume of that can be forcefully exhaled from the lungs after a maximum inhalation. Forced expiratory volume in one second (FEV_1) is the volume of air that can be forcefully exhaled (with a maximum effort) in 1s. FEV_1 should be greater than 80% of FVC.
QU: given an FVC of 4.93L and an FEV_1 of 4.24 L, what is FEV_1%?
Ans: $FEV_1\% = FEV_1 \div FVC \times 100\% = 4.24 \div 4.93 \times 100\% = 86\%$

15. Lung residual volume calculation

Even when you are fully exhaled, your lungs contain some air. This air volume is called the "residual volume" (RV) and is difficult to measure. However, an estimate of RV can be made using the following formula:
For males: RV = (2.29 × height) – (0.012 × weight) - 1.72 litres
For females: RV = (2.63 × height) – (0.02 × weight) – 1.78 litres
(Note: height = height in metres, e.g. 1.76 m; weight = weight in kg, e.g. 72 kg)
QU: Estimate your lung RV by substituting your height (in m) and your weight (in kg) into the formula.

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Ans: if a male ht = 176cm & weight =72kg:
RV = (2.29 × 1.76) – (0.012 × 72) - 1.72 litres
= 4.03 – 0.864 – 1.72 L
= 1.45 L

16. Oxygen transport and haemoglobin

Oxygen is transported around your body by blood while attached to the protein called "haemoglobin" (Hb). Hb is located inside your red blood cells (rbc) of which you have millions. We will calculate how many you have.
(Given that 1 microlitre (1mm³) of blood contains about 5 million (5×10^6) rbc, each rbc contains about 280 million (280×10^6) haemoglobin (Hb) molecules)

QU: How many molecules of haemoglobin does the 5.5 litres of blood within an average human contain?
[*hint*: first calculate the number of microliters of blood in 5.5L.]
Ans: Humans have about 5.5 litres of blood and 280×10^6 Hb molecules per rbc.
5.5 litres = 5500 millilitres
= 5 500 000 microlitres
= 5.5×10^6 microlitres
$(5.5 \times 10^6$ ul) × (5×10^6 rbc per ul) = 27.5×10^{12} rbc per human
(280×10^6) Hb molec/rbc × (27.5×10^{12}) rbc/human = 7700×10^{18}
= 7.7×10^{21} Hb molecules/human
(= 7 700 000 000 000 000 000 000 molecules of Hb/human)

17. Anion Gap

Anion Gap (IONGP) is a characteristic of blood electrolytes that is routinely determined for diagnostic purposes. It is the difference between measured concentration of cations (positive ions) and anions (negative ions) in the serum. (healthy range = 7-17 mmol/L)
Anion Gap = (sodium + potassium) – (chloride + bicarbonate)
= $(Na^+ + K^+) - (Cl^- + HCO_3^-)$
QU: Calculate the anion gap for a patient whose electrolytes concentrations (in mmol/L) are: sodium=140, potassium=3.9, chloride=101, bicarbonate=26. Is it in the normal range?
Ans:
= (140 + 3.9) – (101 + 26)
= (143.9) – (127)
= 16.9 (within normal range)

18. Blood pH

The following calculation involves logarithms. You may wish to work through the module on Logarithms first.
Blood pH is a way of expressing the concentration of hydronium (that is hydrogen) ions in the blood which gives us a measure of its acidity. Blood pH is found by calculating the negative logarithm of the hydrogen ion concentration. Blood pH is strictly regulated by the body to be between 7.35 and 7.45 in healthy people.
$pH = -\log_{10}(H^+$ conc in mol/L)
QU: Use your calculator's logarithm function to calculate blood's pH given that its H^+ concentration is 4.0×10^{-8} mol/L
Ans:
$pH = -\log(4.0 \times 10^{-8})$
$= -(\log(4.0) + \log(10^{-8}))$
$= -(0.6026 - 8)$
$= -(-7.4)$
$= 7.4$

Practice problems: Apply your knowledge

1. Female subject weighs 60 kilograms and is composed of 60% water by weight. Assuming that one liter of water weighs one kg, how many liters of total body water does she have?
60 kg x 0.6 = 36 kg = 36 L
2. In the example above, if the extracellular fluid (ECF) makes up one third of her total body water, how many liters of ECF does she have?
36 L x (1/3) = 12 L
3. Suppose the subject in the example above drinks three liters of water and that, initially, all this water is absorbed into her extracellular fluid.
a) What is the percent change (increase or decrease) in her ECF volume?
New ECF volume = 12 L + 3 L = 15 L % change = 100 x (15 L - 12 L) / (12 L)
= 100 x (0.25) = 25% increase
b) What is the percent change in her total body water?
New TBW volume = 36 L + 3 L = 39 L % change = 100 x (39 L - 36 L) / (36 L)
= 100 x (0.083) = 8.3% increase
4. In this 60-kg woman, the left ventricle of her heart pumps 5 liters of blood per minute at rest. Her kidneys have a blood flow of 1.2 L/min and the kidneys have a combined weight of 0.3 kg.

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a) What percent of her body mass do the kidneys make up?
(0.3 kg / 60 kg) x 100 = 0.5% of body mass
b) What percent of the total blood flow do the kidneys receive?
(1.2 L/min / 5 L/min) x 100 = 24% of total blood flow
5. Suppose the 60-kg female requires 1,500 calories per day to meet her energy needs. If you assume that caloric requirem
is directly proportional to body mass*, how many calories would a 90-kg woman require?
1,500 calories / 60 kg = x / 90 kg 60 x = (1,500 x 90) x = (1,500 x 90) / 60 = 2,250 calories

Answers and explanations:
1. 60 kg x 0.6 = 36 kg = 36 L
2. 36 L x (1/3) = 12 L
3. a) What is the percent change (increase or decrease) in her ECF volume?
New ECF volume = 12 L + 3 L = 15 L % change = 100 x (15 L - 12 L) / (12 L)
= 100 x (0.25) = 25% increase
b) What is the percent change in her total body water?
New TBW volume = 36 L + 3 L = 39 L % change = 100 x (39 L - 36 L) / (36 L)
= 100 x (0.083) = 8.3% increase
4. a) What percent of her body mass do the kidneys make up?
(0.3 kg / 60 kg) x 100 = 0.5% of body mass
b) What percent of the total blood flow do the kidneys receive?
(1.2 L/min / 5 L/min) x 100 = 24% of total blood flow
5. 1,500 calories / 60 kg = x / 90 kg 60 x = (1,500 x 90) x = (1,500 x 90) / 60 = 2,250 calories

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Chapter 5

Genetics math problems

Introduction:
Genetics is the most important chapter to understand the mathematical aspects of it. Every single concept of genetics requires the understanding of the background mathematical calculations. You need to know all the mathematical sides of different genetics concepts including Classical genetics and molecular genetics and human genetics like pedigree analysis and gene mapping. This chapter will make your understanding about genetics roick solid. Practice as many problems as you can to answer all the genetics math questions in upcoming competitive exams.

Chapter content:

1. Mendelian and non-Mendelian Genetics Problems
2. Gene Mapping Problems
3. Pedigree analysis problems:
4. Additional problems from different topics
5. Hfr Mapping or Bacterial Gene Mapping problems

Scan the following QR code to Know how to solve Genetics problems

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1. Mendelian and Non-Mendelian Genetics Problems

How to solve genetics problems?

1.Read the problem carefully two or three times.
2.Determine which character is dominant and which one is recessive.
3.Use capital letter for dominant character and small letter for recessive character.
4.Plot the data carefully on the paper.
5.Determine the genotype of the parents. Genotype can be shown as A--B—for dominant character and aabb for recessive character respectively.
6.Determine all possible types of gametes by each parent.
7.Work out the cross carefully.
8.Read the problem again and answer to the exact question asked to do so.

Sample problems with solutions

MONOHYBRID CROSS

Problem No.1

A red-fruited tomato plant is crossed to a yellow-fruited one produce 173 offspring, 84 of which were yellow and 89 red. Determine the genotype of the parents.

Red fruit colour 'R' is dominant over yellow 'r'.

Answer:

Red fruit – RR or Rr

Yellow fruit –rr

Red is completely dominant over yellow.

In the given data above the offspring segregate in an approximate ratio of 1:1. So it is a case of monohybrid test cross. Test cross is a cross of a hybrid back with its recessive parent. So the genotype of the parents are - Red –Rr; Yellow – rr.

R r X r r

Gametes- (R), (r) (r)

Offspring- R r (Red 89) & r r (yellow 84) Ratio – 1:1

Problem No. 2

In man brown eye colour 'B' is dominant over blue eye colour 'b'. A brown-eyed man marries a brown-eyed woman. Their first child is a blue eyed one. Determine the genotype of the parents. Give the reason of it.

Answer :

Brown eyes- BB, Bb

Blue eyes – bb

In order to produce a blue-eyed child, the genotype of the parents should be heterozygous for the character. So the genotype of the man – Bb and the genotype of the woman is- Bb.

Bb Bb

bb (Child – blue eyed)

Problem No. 3

In garden peas tall nature is dominant over dwarf nature. A tall plant is crossed with dwarf one. Find out the genotype and phenotype of F1 generation. If the F1 plants are allowed to self-fertilize, what would be the appearance of F2 generation?

Phenotype Tall Plant X dwarf Plant

TT t t

Gametes T t

Offspring F1 T t (Selfing of F1)

Gamets – (T) ,(t)

	(T)	(t)
(T)	TT Tall	Tt Tall
(t)	Tt Tall	-tt- Dwarf

F2 Offspring ratio – 3 Tall: 1 Dwarf. 1TT : 2Tt : 1 tt

PROBLEM :4

If a red flowered four- o' clock plant is crossed with a white flowered one, what will be the flower colour of the F1 : of the F2 : of the offspring of a cross of F1 with its red parent and with its white parent ?

Answer :

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Phenotype of parent - Red X White
Genotype - RR X rr
Gametes - (R) &(r)
F1 offspring - Rr All Pink X Rr
F1 gametes - (R) (r)

	(R)	(r)
(R)	RR Red	Rr Pink
(r)	Rr – Pink	-rr- white

F2 ratio – 1 Red : 2 Pink : 1 white
Cross of F1 with red parent - R r X RR
Gametes – (R), (r) & (R)
Offspring – RR Red & R r – Pink (ratio – 1 Red : 1 Pink) Cross of F1 with white parent – R r X r r
Gametes – (R) , (r) & (r) O ffpring – 1 R r Pink : 1 r r white.

DIHYBRID CROSS:

Problem No.5

In Drosophila, ebony body colour is produced by a recessive gene 'e' and wild type grey body colour is by its dominant allele 'E'. Vestigial wings are governed by a recessive gene 'v' and normal wings by its dominant allele 'V'. If the wild type dihybrid flies are crossed to produce 256 progenies, how many of these progenies are expected in each phenotypic class?

Answer:

Grey body colour – E
Ebony body colour – e
Normal wings – V
Vestigial wings –v Cross- Wild type flies X Wild type flies
Genotype of parents E e Vv - Ee Vv

Gametes EV , Ev, eV , ev (four types of gametes

	(EV)	(Ev)	(eV)	(ev)
(EV)	EEV V Grey nornal	EEVv- Grey nornal	EeV V Grey nornal	EeVe- Grey nornal
(Ev)	EeVv- Grey nornal	Eevv- Grey Vestigial	EeVv- Grey nornal	-Eevv- Grey Vestigial
(eV)	EeV V Grey nornal	EeVv Grey nornal	-eeVV- Ebony Normal	-eeVv- Ebony Normal
(ev)	EeVv Grey nornal	Eevv Grey Vestigial	-eeVv- Ebony Normal	-eevv- Ebony vestigial

It is a case of dihybrid cross and in F2 generation the offspring are expected in the proportion of 9:3:3:1.So the offspring are distributed as (256X9 / 16)144 Grey Normal (256X3 /9) :48 Grey Vestigial : 48 Ebony Normal : 16 Ebony vestigial

Problem No.6

A ragged leaved corn plant with round pollen was crossed to a ragged leaved corn plant with angular pollen produced offspring in the following manner - 186 ragged leaved and round pollen, 174 ragged leaved and angular pollen , 57 smooth leaved and round pollen and 64 smooth leaved and angular pollen . (Ragged – S, Smooth – s, Round – A, Angular –

a) Give the genotypes of the two parents and what numbers would you have expected for each of the four types of progenies?

Answer :

Ragged- S--, Smooth –s
Round –A, Angular- a
Ragged leaved plant with round pollen X Ragged leaved with angular pollen
S—A-- X S—aa
186 Ragged round: 174 Ragged angular: 57 Smooth round: 64 Smooth angular(ssaa)

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In the offspring generation there is a group of double recessive offspring. So, the genotype of Ragged leaved corn plant with round pollen is heterozygous. So, the genotype of this parent is - SsAa. The genotype of the other parent is -Ssaa (angular is already a recessive character)

The cross of the above parents can be shown as below.

Ragged leaved plant with round pollen X Ragged leaved with angular pollen

SsAa X Ss aa

Gametes - SA,Sa,sA,ss ---- Sa,sa

	(Sa)	(sa)
(SA)	SSAa-Ragged round	SsAa -Ragged Round
(Sa)	Ssaa-Ragged angular	Ssaa- Ragged angular
(sA)	SsAa-Ragged round	-ssAa-Smooth round
(sa)	Ssaa-Ragged angular	-ssaa-Smooth angular

The expected ratio is – 3 Ragged round :3 Ragged angular: 1 Smooth round : 1 Smooth angular.

Problem No.7

Suppose two strains of Jimson Weeds are crossed and they produce a progeny in the following proportions. spiny red 360, spiny purple 724, spiny white 362, smooth red 121, smooth purple 240, smooth white 120. Explain the reason for the diverse phenotypes? What would be the genotype and phenotype of the above parental strain? Represent the cross and progeny by means of a checkerboard.

Answer :

It is a case of incomplete dominance. There are 6 classes of offspring. Spiny character is completely dominant over smooth. But Red is incompletely dominant over white. The intermediate character is purple. The data can be summarised into an approximate ratio of 3:6:3:1:2:1. This ratio is obtained in the case of a dihybrid cross where one character is completely dominant over the other and the other character shows incomplete dominance.

Phenotype - Spiny Purple X Spiny Purple

Genotype - Ss Rr X SsRr

Gametes – SR, Sr, sR, sr

	(SR)	(Sr)	(sR)	(sr
(SR)	SSR R Spiny red	SSRr Spiny Purple	SsRR Spiny red	SsRr Spiny Purple
(Sr)	SSRr Spiny Purple	SSrr Spiny white	SsRr Spiny Purple	Ssrr Spiny white
(sR)	SsRR Spiny red	SsRr Spiny Purple	-ssRr- Smooth purple	-ssRr- Smooth purple
(sr)	SsRr Spiny Purple	Ssrr Spiny white	-ssRr- Smooth purple	-ssrr- Smooth white

OffspringRatio–

3 spiny red : 6 spiny purple :3 spiny white : 1 smooth red : 2 smooth purple : 1 smooth white.

Problem No.8

White flower colour 'W' is dominant over cream 'w' and salver shaped corolla 'S' is dominant over funnel shaped corolla's'. A white funnel (x) cream salver cross gives ¼ white salver, ¼ white funnel, ¼ cream salver, ¼ cream funnel plants. Determine the genotype of the parents? What is the chance of these parents producing only two types of offspring, one white salver and the other cream salver?

Answer

White flower –W, Cream flower –c.

Salver shaped corolla—S, Funnel shaped corolla – s

Phenotype of Parents – White funnel X Cream salver Genotype of parents - W –ss X wwS—

Offspring- 1 White salver : 1 White funnel : 1 Cream salver : 1 Cream funnel (wwss)

In order to produce a double recessive offspring in the offspring generation, the parents should contain at least one recessive gene for both the characters.So the dashes contain recessive genes. So the genotype of White funnel is – Wwss. Cream salver is – wwSs.

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The cross can be shown as follows.

	(Ws)	(ws)
(wS)	WwSs White salver	-wwSs- cream salver
(Ws)	Wwss White funnel	-wwss- cream funnel

There is no chance of producing only two types of offspring such as white salver and cream salver.

Problem No.9

When two squash plants having the same phenotype of seed shape and colour were crossed, the offspring appeared in the following proportions. Find out the genotype of the parents and offspring and also the phenotype of the parents? 1/16 round yellow, 1/8 disc yellow, 1/8 elongated grey, 1/16 elongated yellow, 1/8 round grey, 1/8-disc green, ¼ disc grey, 1/16 round green, 1/16 elongated green.

Answer :

It is a case of dihybrid incomplete dominance because there are nine different phenotypic classes of offspring. As there are 4 disc grey , these characters are the intermedeate characters or heterozygous condition. Round and Yellow characters may be taken as dominant and elongated and green may be taken as recessive characters. In order to produce a double recessive offspring in the offspring generation the parents may be heterozygous for both the characters. So the genotype of the parents – RrYy. and Phenotype – Disc Grey.

The cross can be shown as follows-

Round – R, elongated – r.

Yellow-Y , Green – y.

Disc grey X Disc Grey

RrYy X RrYy

	(RY)	(Ry)	(rY)	(ry)
(RY)	RRYY Round Yellow	RRYy Round Grey	RrYY Round Green	RrYy Disc Grey
(Ry)	RRYy Round Grey	RRyy Round Green	RrYy Disc Grey	Rryy Disc Green
(rY)	RrYY Disc Yelloow	RrYy Disc Grey	-rrYY- Elongated Yellow	-rrYy- Elongated Grey
(ry)	RrYy Disc Grey	Rryy Disc Green	-rrYy- Elongated Grey	-rryy- Elongated Green

atio – 1: Round Yellow 2 Round Grey:1 Round Green:2 Disc Yelloow :4 Disc Grey:2 Disc Green:1 Elongated Yellow:2 Elongated Grey:1 Elongated Green

Problem No.10

In poultry black plumage 'B' is dominant over red 'b' and crested head 'C' is dominant over plain 'c'. A red crested male bird is mated to a black plain female.

They produced offspring half of which were black crested and half red crested. What was the genotype of the parents?

Answer:

Black Plumage – B–, Red Plumage –bb Crested head – C–, Plain head – cc

Red Crested male X Black Plain female

bbC– B– cc

50 % Black Crested -- 50% Red Crested offspring

B–C– bb C–

1.In the offspring generation there is recessive red character. So, the black female parent is heterozygous for black character. So black plain female is Bbcc.

2.Both the offspring are showing crested characters. But one parent is plain. So, the Red Crested male parent is homozygous for crested character. So, the genotype of Red Crested male parent is -bbCC

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Problem No.11

In Rice lax panicle 'S' is dominant over dense panicle's'. The clustered spike let 'C' is incompletely dominant over non-clustered 'c' and heterozygous condition is having intermediate spike let. Give the genotype and phenotype of F1 when a plant with dense panicle and clustered spike let is crossed to a plant with pure lax panicle and non- clustered spike let. Determine the genotype and phenotype of F2 generation if the F1 are allowed to self-fertilize?

Answer:

Lax panicle -S, Dense panicle – s

Clustered spikelet – C, Non-clustered spikelet – cc, Intermediate spikelet - Cc

Phenotype – Dense panicle and clustered spike let X Lax panicle and non-clustered

Genotype - ssCC X SScc

Gametes - sC Sc

F1 offspring - SsCc

Phenotype of F1 -Lax panicled and Intermediate spikelet.

Gametes produced by F1 offspring-SC,Sc,sC,sc

The formation of F2 can be shown in the checker board as follows:

	(SC)	(Sc)	(sC)	(sc)
(SC)	SSRCC	SSCc	SsRCC	SsCc
(Sc)	SSCc	SScc	SsCc	Sscc
(sC)	SsCc	SsCc	-ssCc	-ssCc-
(sc)	SsCc	Sscc	-ssCc	-sscc-

F2 Ratio – 3 Lax panicled clustered spike let :6 Lax panickled Intermediate :3 Lax panicled non-clustered:1 Dense panicled clustered :2 Dense panicled Intermediate clustered :1Dense panicled non-clustered.

Problem No.12

The shape and colour of Radishes are controlled by two independent pairs of alleles that show no dominance. The colour may be red (RR), purple (Rr), white (rr). And the shape may be long (LL), oval (Ll), round (ll).

Using a checker board (a) diagram a cross between red long and white round (b) summaries the F2 segregation giving the genotypic ratios.

Answer:

Phenotype of parents - Red Long X White Round

Genotypeo of parents - RR LL rrll

Gametes RL rl

F1 Offspring RrLl

F1 is self fertilized

F1 Gametes RL, Rl, rL, rl

	(RL)	(Rl)	(rL)	(rl)
(RL)	RRLL	RRLl	RrLl	RrLl
(Rl)	RRLl	RRll	RrLl	Rrll
(rL)	RrLL	Rrll	-rrLl-	-rrLl-
(rl)	RrLl	Rrll	-rrLl-	-rrll-

F2 Phetypic and genotypic ratios are the same –

F2 ratio – 1Red long: 2Red oval :1Red round: 2Purple long :4 Purple oval :2 Purple round: 1White long :2 White oval: 1White round

Problem No.13

In Man assume that spotted skin 'S' is dominant to non-spotted's' and woolly hair 'W' is dominant over non woolly 'w'.

a) List the genotypes and phenotypes of the children to be expected from a marriage of spotted (Ssww) man and a woolly haired (ssWw) woman.

b) Also give the genotypic and phenotypic ratio of children to be expected from a marriage between a man and a woman both are heterozygous for spotted skin and woolly hair.

Answer :

Spotted skin – S , Non spotted skin – s

Woolly hair – W, Non woolly hair – w

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a)Phenotypes of parents – Spotted non woolly man X Non spotted woolly haired woman
Genotype of parents - Ssww ssWw
Gametes - Sw, sw - sW ,sw

	(sW)	(sw)
(Sw)	SsWw Spotted woolly	-Ssww- Spotted nonwoolly
(sw)	-ssWw- Nonspotted woolly	-ssww- non spotted and non woolly

Offspring – 25% Spotted woolly : 25% Spotted nonwooly : 25%Non spotted woolly : 25% Non spotted and non wolly
b) A marriage between a man and woman both are heterozygous
Phenotypes of parents – Spotted and woolly X spotted and woolly
Genotypes of parents SsWw - SsWs
Gametes – SW, Sw, sW, sw (both parents produce same type of gametes) Draw the checker board -
Expected offspring in the marriage – 9/16 Spotted woolly: 3/16 Spotted non woolly : 3/ 16 Non spotted woolly : 1/16 non spotted and non woolly.

Problem No.14
Two rough black Guinea pigs bred together have two offspring, one of them rough white and the other smooth black. Find out the genotypes of the parents? If these same parents were again bred together what offspring in proportion could you expect from them? In guinea pig, rough coat 'R' is dominant over smooth coat 'r' and black coat 'B' is dominant over white coat 'b'.
Answer :
Rough coat – R, Smooth coat – r.
Black coat – B , White coat – b
Phenotypes of Parents – Rough black Guinea pigs X Rough black Guinea pigs
Genotype of Parents - R--B-- -- R--B—
Offspring Phenotype - Rough white(R—bb) & smooth black(rrB--)
In order to produce a white coat the parents should have one small 'b' in their genotype and also to produce a smooth coat the genotype of the parents should have atleast one 'r'. So the genotype of the parents – RrBb
If we cross the same parents again we expect the offspring in the proportion of 9/16 Rough black : 3/16 Rough white 3/16 Smooth black : 1/16 Smooth white, because it is a typical dihybrid cross. The result is shown below.

	(RB)	(Rb)	(rB)	(rb)
(RB)	RRBB	RRBb	RrBb	RrBb
(Rb)	RrBb	RRBb	RrBb	Rrbb
(rB)	RrBB	RrBb	-rrBB-	-rrBb-
(rb)	RrBb	Rrbb	-rrBb-	-rrbb-

Problem No.15
In man assume that brown eyes B are dominant over blue eyes b and right handedness R is dominant over left handedness r. A left handed , brown eyed woman , hailing from a family having brown eyed members for several generations, is married to a right handed, blue eyed man whose father was brown eyed and left handed. Can you predict the appearance of children to be expected from this couple as to the two traits mentioned?
Answer :
Brown eyes – B, Blue eyes – b
Right handedness- R , Left handedness-r.
Phenotype of parents –
Left handed brown-eyed woman X Right handed blue-eyed man
(from a family of brown eyes) - (Father brown eyed left handed)
Genotype of parents - rr BB - Rrbb
Gametes - rB & Rb, rb
Offspring - RrBb(Right handed brown eyed) & rrBb (Left handed brown eyed) There is a chance of 50% RrBb % 50% rrBb

Problem No.16

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Offspring produced by inbreeding a hybrid snapdragon plant produced the following number of phenotypes. Explain the appearance of diverse phenotypes, and account for the ratio modification.
Red flowered broad leaved- 120, Red flowered intermediate breath of leaf 240, Red flowered narrow leaved 120, White flowered broad leaved 120, White flowered narrow leaved 120, White flowered intermediate breath of leaf 240, Pink flowered broad leaved 240, Pink flowered narrow leaved 240, Pink flowered intermediate breath of leaf 480.

Answer :

It is a case of dihybrid cross showing incomplete dominance. Here both the characters show incomplete dominance. And we got a ratio of 1:2:1:2:4:2:1:2:1.Both the phenotypic and genotypic ratios are the same. The cross can be shown as follows.
Phenotype of the parents –
Pink Flowered intermediate leaf breath X Pink Flowered plant with intermediate leaf breath
Genotype RrBb - RrBb
Gametes – RB,Rb,rB,rb

	(RB)	(Rb)	(rB)	(rb)
(RB)	RRBb	RRBb	RrBb	RrBb
(Rb)	RRBb	RRbb	RrBb	Rrbb
(rB)	RrBB	Rrbb	-rrBb-	-rrBb-
(rb)	RrBb	Rrbb	-rrBb-	-rrbb-

Ratio – 1 Red broad :2 Red intermediate: 1 Red narrow: 2 Pink broad: 4 Pink intermediate: 2 Pink narrow: 1 White broad : 2 White intermediate : 1 White narrow.

Problem No.17

In cattle the polled condition 'P' is dominant over the horned 'p' and the heterozygous condition of red coat 'R' and white 'r' is roan . A polled roan bull bred to a horned white cow produces a horned roan daughter. If this daughter is bred to a heterozygous polled roan bull what offspring may be expected as to horns and coat colour? Could the same result be expected from a cross between a polled roan PpRr cow and a polled white Pprr bull?

Answer :

Polled condition – P , Horned condition –p
Red coat – RR, White – r r, Roan – R r
Phenotype of parents – Polled Roan Bull X Horned white cow
Genotype of parents - P—Rr - pp rr
Phenotype of offspring – Horned roan calf (pp R--)
Explanation –1) In order to produce a horned calf the genotype of the bull should be PpRr.2) The genotype of the daughter is – ppRr
1) If this daughter is bred to heterozygous Polled Roan bull, the result is ----
ppRr X PpRr Gametes – pR, pr & PR,Pr,pR,pr

	(PR)	(Pr)	(pR)	(pr)
(pR)	PpRR Polled red	PpRr Polled Roan	-ppRR Horned red	-ppRr Horned Roan
(pr)	PpRr Polled Roan	Pprr Polled white	-ppRr- Horned Roan	-pprr- Horned white

Result - 1 Polled Red : 2 Polled Roan :1 Polled White : 1 Horned red :2 Horned Roan : 1 Horned white.
2) If the the cow is ------- Polled Roan PpRr and bull is Polled white Pprr
Gametes - PR,Pr,pR,pr & Pr, pr

	(PR)	(Pr)	(pR)	(pr)
(Pr)	PPRr Polled roan	PPrr- Polled white	PpRr Polled Roan	Pprr- Polled white
(pr)	PpPr Polled Roan	Pprr Polled white	-ppRr- Horned Roan	-pprr- Horned white

esult – 3 Polled Roan: 3Polled white :1 Horned Roan: 1horned white. The result is different.

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Problem No.18

In the following question the appearance of the parents and the offspring are stated. Determine in each case the genotype of the parents. a) White disk crossed with yellow sphere gives half white disc and the other half white sphere. b) White disc crossed with white sphere gives ¾ white sphere and ¼ yellow sphere. Explain the result-using checkerboard.

Answer :-

A) Phenotype of parents – White Disc X Yellow Sphere Genotype of parents – WW dd & ww Dd

Gamtes - Wd & wD, wd

	(wD)	(wd)
(Wd)	WwDd -White Sphere	Wwdd -White Disc

Offspring - 1/2 White Disc & ½ White Sphere.

Explanation-

White is dominant over yellow. Sphere is dominant over disc

In order to produce a disc offspring, the sphere may be heterozygous. In order to produce only white offspring, the parent may be homozygous in this situation.

B) Phenotype of parents – White Disc x White sphere

Genotype of parents – Wwdd WwDD

Offspring - ¾ White Sphere & ¼ Yellow Sphere

Explanation – Here white and yellow segregated in to 3:1 ratio.So the white colour may be heterozygous for the character. So the white is dominant over yellow. A disc character is crossed with sphere produced all sphere. So sphere is dominant over disc. In order to produce all sphere the dominant character may be homozygous. So the genotype of parents are - White Disc (Wwdd) & White sphere (WwDD)

The cross can be shown as follows

C) Phenotype of parents – White Disc x White sphere

Genotype of parents – Wwdd WwDD

Gametes – Wd,wd & WD, wD

	(Wd)	(wd)
(WD)	WwDd White sphere	WwDd White sphere
(wD)	WwDd White sphere	-wwDd yellow sphere

Offspring - ¾ White Sphere & ¼ Yellow Sphere .

Problem No.19

In garden phlox white flower colour is due to a dominant gene W and cream to its allele w, salver shaped flower is due to a dominant gene S and funnel shaped flower to its recessive. allele' s '. A plant producing white salver shaped flower is crossed with one producing cream funnel shaped flowers. Of the 110 offspring 56 produced white salver shaped flowers and 54 produced cream salver shaped flowers. What were the genotypes of the parents? Also state the probability of obtaining four phenotypic classes of offspring in almost equal proportion from the above a white salver X cream funnel cross.

Answer –

White flower colour- W, Cream- w

Salver shaped flower- S, funnel shaped flower- s

Phenotype of parents –White Salver X Cream funnel

Genotype of Parents – W—S-- & wwss

Offspring- 56 White Salver : 54 Cream salver

Explanation – Here the characters white and cream segregates to 1:1 ratio. So with respect to this character this is a test cross. So the genotype of white is heterozygous.The salver is not segregated and it produced only salver offspring . So salver is homozygous.

So the genotype of parents – White salver (WwSS)& Cream funnel (wwss)

The cross can be shown as follows-

Phenotype of parents –White Salver X Cream funnel

Genotype of Parents – WwSS & wwss

Gametes - WS,wS& ws

	(WS)	(wS)
(ws)	WwSs - White Salver	-wwSs - Cream salver

OffspringRatio - 1 White Salver: 1 Cream salver

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There is no chance of getting four types of offspring in the above cross. If both white and salver are heterozygous, we will get four types of offspring.

Problem No.20

Offspring produced by inbreeding a hybrid snapdragon plant produced the following number of phenotypes. Explain the appearance of diverse phenotypes, and account for the ratio modification.

Red flowered broad leaved- 340, Red flowered intermediate breath of leaf 680, Red flowered narrow leaved 340, White flowered broad leaved 340, White flowered narrow leaved 340, White flowered intermediate breath of leaf 680, Pink flowered broad leaved 680, Pink flowered narrow leaved 680, Pink flowered intermediate breath of leaf 1360.

Answer : It is a case of dihybrid cross and both the characters showing incomplete dominance.The ratio is 1:2:1:2:4:2:1:2:1 (Explained earlier the same type of problem)

Problem No.21

Offspring produced by inbreeding a hybrid snapdragon plant produced the following number of phenotypes. Explain the appearance of diverse phenotypes, and account for the ratio modification.

Red flowered broad leaved- 120, Red flowered intermediate breath of leaf 240, Red flowered narrow leaved 120, White flowered broad leaved 120, White flowered narrow leaved 120, White flowered intermediate breath of leaf 240, Pink flowered broad leaved 240, Pink flowered narrow leaved 240, Pink flowered intermediate breath of leaf 480.

Answer : It is a case of dihybrid cross and both the characters showing incomplete dominance.The ratio is 1:2:1:2:4:2:1:2:1(Explained earlier the same type of problem)

Problem No.22

In Tomato, red flesh of the fruit 'R' is dominant over yellow 'r' and spherical shape 'S' is dominant over cylindrical shape's'. The two allelic pairs are inherited independently. A cross between red spherical X red cylindrical produce 122 red spherical, 120 red cylindrical, 39 yellow spherical and 41 yellow cylindrical . Determine the genotype of the parents. Represent the cross and progeny by means of a checkerboard.

Answer :

Red Fruit colour – R, Yellow- r

Spherical shape- S, Cylindrical shape – s

Phenotype of parents - Red Spherical X Red Cylindrical

Genotype of parents - R—S-- & R—ss

Offspring – 122 Red spherical R-S-) , 120 Red cylindrical (R—ss), 39 Yellow spherical (rrS--), 41 Yellow cylindrical (rrss)

Explanation – In order to produce a double recessive offspring, the parents should be at least heterozygous for the character. So the genotype of parents – RrSs & Rrss. Representation of the cross –

Parents - RrSs X Rrss

Gametes – RS,Rs,rS,rs & Rs, rs

	(RS)	(Rs)	(rS)	(rs)
(Rs)	RRSs- Red Spherical	RRss- Red cylindrical	RrSs- Red Spherical	Rrss- Red cylindrical
(rs)	RrSs- Red Spherical	Rrss- Red cylindrical	-rrSs- Yellow spherical	-rrss- Yellow cylindrical

Ratio -3 Red Spherical:3 Red cylindrical:1 Yellow spherical :1 Yellow cylindrical

Problem No.23

Give the most probable genotype of the tomato plants involved in the following crosses.

A) Purple cut tomato X green cut tomato produces 321 purple cut , 101 purple potato, 310 green cut, and 107 green potato. B) Purple potato X Green cut produce 70 purple cut, 91 purple potato, 86 green cut and 77 green potato. (Purple stem A, Green stem a, Cut leaf C, Potato leaf c).

Answer:

Purple stem A, Green stem a.

Cut leaf C, Potato leaf c).

a) Phenotype of parents – Purple cut tomato X Green cut tomato Genotype of parents - A—C-- & aa C--

Off spring – 321 purple cut(A—C--) : 101 purple potato(A—cc) : 310 Green cut(aaC--) : 107 Green potato (aacc)

Explanation -

In order to produce a double recessive offspring green potato the parents should be at least heterozygous for the characters.)

So the genotype of parents – Purple cut tomato is AaCc & Green cut tomato is - aaCc

b) Phenotype of parents - Purple potato X Green cut

Genotype of offspring - A—cc & aaC-

Offspring – 70 Purple cut(A—C--) : 91 purple potato(A—cc) : 86 Green cut(aaC--) : 77 Green potato(aacc)

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Explanation - In order to produce a double recessive offspring green potato the parents should be at least heterozygous for the characters.)
So the genotype of parents- Purple potato-Aacc &Green cut - aaCc

Problem No.24
In across involving both seed shape and seed colour, Mendel found in a total of 2156 F2 seeds of the following combinations - 1215 round yellow, 408 round green , 400 wrinkled yellow, and 133 wrinkled green. Using symbols draw a checkerboard to arrive at the result, and find out the genotypic segregation ratio.
Answer :
Here the data comes to an approximate ratio of 9:3:3:1.Whenever dihybrid parents showing complete dominance is crossed such a ratio is obtained or it can be explained in another way . In order to produce a double recessive offspring wrinkled greenin the offspring generation the parents should be at least heterozygous for the characters. So the parents are heterozygous for the characters.
Round - W, wrinkled - w.
Yellow - Y , green - y
The cross can be shown as follows -
Phenotype of parents - Round Yellow X Round Yellow Genotype of parents - WwYy & WwYy Gametes - WY,Wy,wY,wy

	(WY)	(Wy)	(wY)	(wy)
(WY)	WWYY	WWYy-	WwYY	WwYy-
(Wy)	WWYy-	Wwyy-	WwYy-	Wwyy-
(Wy)	WwYY	WwYy-	-wwYY-	-wwYy-
(wy)	WwYy-	Wwyy-	-wwYy-	-wwyy-

Genotypic ratio of offspring - 9 round yellow: 3 round green: 3 wrinkled yellow: and 1 wrinkled green.

Problem No.25
Two rough black guinea pigs bred together have two offspring, one of them rough white and the other smooth black. Find out the genotype of the parents. If these same parents were again bred together, what offspring in proportions would you expect from them.(In guinea pigs rough coat 'R' is dominant over smooth coat 'r' and black coat' B' is dominant over white 'b')
Answer :
Rough coat - R , Smooth coat - r
Black coat - B , White coat - b
Phenotype of Parents - Rough Black X Rough Black
Genotype of Parents - R—B-- & R—B--
Phenotype of Offspring - Rough white & Smooth black
Genotype of offspring - R—bb & rr B—
Explanation - There is a recessive character white is seen in one of the offspring. This gene come from both the parents. In the same way another recessive character smooth is seen in the other offspring. This will also come from both parents. So the parents will be heterozygous for both the characters. So the genotype of the parents are - RrBb If the parents are again bred together we expect a ratio of 9Rough black: 3 Rough white: 3 Smooth black: 1 Smooth white as it is a dihybrid cross.

Problem No.26
Find out the genotypes of the parents and offspring s involved in the following crosses of dogs were dark coat colour 'C' is dominant over albino 'c', short hair 'S' is dominant over long hair 's'. a) Dark short X dark short produced 88 dark short, 31 dark long, 29 albino short and 11 albino long. b) Dark short X dark long produced 29 dark short, 31 dark long, 9 albino short, and 11 albino long.
Answer -
Dark coat colour 'C' , albino 'c'
Short hair 'S', long hair 's'.
A) Phenotype of Parents- Dark Short X Dark Short
Genotype of Parents - C—S-- & C—S--
Phenotype of Offspring- 88 dark short : 31 dark long : 29 albino short :11 albino long.(ccss) Explanation - Inorder to produce a double recessive offsprig the parents may be at least heterozygous for the character . So the genotype of parents are - CcSs & CcSs
B) Phenotype of Parents- Dark short X Dark long
Phenotype of Offspring-29 Dark short: 31dark long : 9 albino short :11 albino long. (ccss) Explanation - In order to produce a double recessive offspring the parents may be at least heterozygous for the character. The long is a recessive character.So the genotype of parents are - CcSs & ccss

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Problem No.27

True breeding flies with long wings and dark bodies are mated with true breeding flies with short wings and tan bodies. All the F1 progeny have long wings and tan bodies. The F1 progeny are allowed to mate and produce- 44 tan long, 14 tan short, 16 dark long, 6 dark short. What is the mode of inheritance? Explain.

Answer :

Long wings -L, short wing – l
Tan body- D, dark body -d
Phenotype of Parents – Long dark X Short tan
F1Offspring – Long tan X Long tan
F2 offspring – 44 tan long: 14 tan short: 16 dark long: 6 dark short.
Explanation - 1) The F1 offspring are all long tan . So long wing is dominant over short wing and tan body is dominant over dark body. 2) The F2 generation is segregated to an approximate ratio of 9:3:3:1 and there is a double recessive offspring group dark short .So the genotype of P1 parents are – Long dark (Ll dd) and Short tan (ll Dd). It is Mendelian pattern of inheritance. (Show the crossing and draw the checker board)

Problem No.28

In Drosophila the wild type(normal) is grey in colour with wings that extend beyond the tip of the abdomen. Among the mutants of drosophila are two which are respectively distinguished by dark body colour (ebony) and vestigial condition of wings. (vestigial). A fruit fly with vestigial wings and ebony body colour is crossed to the wild type. The F1 flies are backcrossed to the double recessive (ebony vestigial) and the result is -32 wild type, 33 ebony normal wing, 33 vestigial grey body colour and 34 ebony vestigial. Discuss the result with the aid of a diagram commenting on the relationship between phenotypic appearance and genotypic make up. What would be the result of mating the wild type of this last experiment among themselves.

Answer :

Dominant - Wild type – (Grey body colour), & Normal wings
Recessive – Ebony (dark body colour) & vestigial wings
Phenotype of parents – Wild type X Ebony vestigial
F1 flies crossed back with ebony vestigial (It is a test cross)
Here the data comes to an approximate ratio of 1:1:1:1
1) Wild type Grey – E , Ebony –e.
Normal wings – V , Vestigial wings –v.
F1 Flies EeVv X eevv Gametes - EV,Ev,eV,ev & ev

	(EV)	(Ev)	(eV)	(ev)
(ev)	EeVv- Grey nornal	Eevv- Grey Vestigial	-eeVv- Ebony Normal	-eevv- Ebony vestigial

Ratio – 1 Grey Normal :1 Grey Vestigial : 1 Ebony Normal : 1 Ebony Vestigial

2) Crossing of F1 Flies among themselves.

	(EV)	(Ev)	(eV)	(ev)
(EV)	EEV V Grey nornal	EEVv Grey nornal	EeV V Grey nornal	EeVe Grey nornal
(Ev)	EeVv Grey nornal	Eevv Grey Vestigial	EeVv Grey nornal	Eevv- Grey Vestigial
(eV)	EeV V Grey nornal	EeVv Grey nornal	(eeVV) Ebony Normal	(eeVv) Ebony Normal
ev	EeVv Grey nornal	Eevv Grey Vestigial	(eeVV) Ebony Normal	(eevv) Ebony vestigial

Result - 9 Grey Normal :3 Grey Vestigial : 3 Ebony Normal : 1 Ebony vestigial

Problem No.29

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In poultry feathered legs 'F' are dominant over clear legs 'f' and pea comp 'P'is dominant over single comp 'p'. Two cocks A & B were bred to two hens C & D, all the four birds were feathered legs and pea combed. The crosses A x C and A x D yielded all feathered and pea combed birds, Cross B x C produced feathered and clean legged but all pea combed birds. Crosses B x D produced all feathered birds but segregation for pea combed and single combed. Find the genotype of all birds.

Answer :

Feathered legs - F, lean legs - f
Pea comp - P, Single comp - p
Crosses:
1) Feathered Pea X Feathered Pea
A X C
F--P-- F--P--
Offspring All Feathered pea
F--P--
2) Feathered Pea X Feathered Pea
A X D
F--P-- F--P--
Offspring All Feathered pea
F--P--
3) Feathered Pea X Feathered Pea
B C
Offspring - 1) Feathered Pea 2) Clean pea
F—P-- & ff P—
4) Feathered Pea X Feathered Pea
B X D
Offspring - 1) Feathered Pea 2) feathered single Offspring genotype - F—P-- & F- pp Explanation:
1)Genotype of bird B is heterozygous for feathered and pea characters because in the crosses between bird B and C and with B and D produced recessive clean and single characters.So genotype of B is- Ff Pp
2)Genotype of bird D is FFPp because in the cross between B and D there is only the segregation of single character and no clean legs.
3) The genotype of bird C is Ff PP because in the cross between birds B and C there is segregation of only clean legs and no single character.
4) The genotype of bird C is known as Ff PP and in the cross between birds A and C there is no segregation of clean or single .So the genotype of bird A may be FFPP and not FFPp. Because even in the cross between birds A and D (FFPp) there is no single offspring.

GENE INTERACTION

Problem No. 30

In poultry rose comb and pea comb patterns are produced by the dominant effect of two different genes. The recessive nature of both these alleles produces single comb. But in a cross between rose combed and pea combed fowls, a new type known as walnut was also produced along with rose, pea, and single in 9:3:3:1 ratio. Suggest the possible genotypes of F 1 and F2 offspring and explain the type of gene interaction involved in this cross. Answer:
Rose comb - R—pp, Pea comb - rrP—
R—P— =Walnut comb, rrpp =single comb.
Explanation- 1) R gene in homozygous or heterozygous condition and in the absence of 'P' produces Rose comb in fowls.2) P gene in homozygous or heterozygous condition and in the absence of' R' produces Pea comb.3) When both R & P comes together they interact and produces a novel character Walnut. in F1 generation. 3) Both the recessive genes interact and produces single comb.4) Even though the F2 ratio is numerically same as a dihybrid ratio it is not at all related to the pattern of inheritance. Because the F1 is a different character not seen among the parents. The F2 single is also not seen anywhere in the parental lines. The cross can be shown as follows.
Phenotype of parents - Rose X Pea
Genotype of parents - RRpp & rrPP
Gametes - Rp & rP
F1 Offspring - RrPp (Walnut)
The F1 is self-fertilised- RrPp X RrPp F1 gametes- RP,Rp,rP,rp

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	(RP)	(Rp)	(rP)	(rp)
(RP)	RRPP	RRPp	RrPp	RrPp
(Rp)	RRPp	RRpp	RrPp	Rrpp
(rP)	RrPP	RrPp	-rrPp-	-rrPp-
(rp)	RrPp	Rrpp	-rrPp-	-rrpp-

The F2 ratio - 9 walnuts :3rose:3pea:1 single.

Problem No.31

Nilsson - Ehle made between two types of oats. One with white hulled and the other with black hulled and F2 had black 418, grey 106, white 36. Give the type of interaction and explain it.

Answer : Firstly find out the possible ratio and find out the type of interaction.

Total number of offspring - 418+106+36= 560

Find out the proportion of the first group of offspring among the sum total of 560.We are considering it , as a multiple of 16 . (dihybrid cross)

So the first group is - 418 X 16 / 560 = 11.9428=12 Black.

Middle group - 106 X16 / 560 =3.02 = 3 Grey

Last group - 36 X16 / 560=1.02 = 1 white

Here we will get an approximate ratio of 12:3:1

It is a case of dominant epistasis. Explain the theory of dominant epistasis.

Problem No.32

Two white flowered strains of sweet peas were crossed, producing an F1 with only purple flowers. Random crossing among the F1 produced 96 progeny plants of which 53 are purple, 43 with white flowers. A) What phenotypic ratio is shown by the F2 ?. B) What type of interaction is involved? C) What were the probable genotypes of the parental strains? Answer - Firstly find out the possible ratio and find out the type of interaction.

Total number of offspring -96.

Find out the proportion of the first group of offspring among the sum total of 560.We are considering it , as a multiple of 16 . (dihybrid cross)

So the first group is- 53 X16 / 96 =8.83 = 9 purple

The second group is - 43 X16 / 96 =7.16= 7 white.

It is a case of complementary factor interaction. Here we will get a ratio of 9:7. Explain the theory of complementary factor interaction. The genotype of parental strains -ccPp (white) X Ccpp (white). C -- in single dose or double dose produce white flower colour. Like wise P-- in single dose or double dose produce white flower colour. But when they come together they interact and produce purple flower colour. The double recessives will also produce white. The last three groups of Mendelian ratio of 9:3:3:1 is indistinguishable and modified to 9:7 ratio.

Problem No.33

Four walnut hens are crossed to single and each produces a large number of chicks. The first produces only walnuts, the second produces only walnuts and peas, the third produces only walnuts and rose and the fourth produces among other types single also. Give the genotypes of each of the four walnut hens used as parents.

Answer -

1) Walnut X Single (rrpp)
   Offspring - all walnuts

The walnut parent is homozygous for R. & P

RRPPX rrpp = RrPp All walnuts

2) Walnut X single

Offspring - Walnuts & Peas Genotype of parents - RrPP X rrpp

Gametes - RP,rP & rp Offspring- RrPp (walnut) rrPp (Pea)

3) Walnut X single Offspring - Walnut & Rose

Genotype of parents- RRPp X rrpp Gametes - RP,Rp & rp

Offspring - RrPp (walnuts) &Rrpp (rose)

4) Walnut X Single

Offspring - Walnuts , Rose , Pea & single .

Genotype of parents- RrPp X rrpp

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Offspring – RrPp (walnuts) ,Rrpp (rose) ,rrPp (pea) & rrpp (single)

Problem No.34

In summer squash spherical fruit is recessive to disc. Spherical races from different geographical regions were crossed. The F1 were disc and the F2 were segregated to 35 discs, 25 spherical, 4 elongate. Explain the result?

Answer:

Firstly find out the possible ratio and find out the type of interaction.

Total number of offspring = 35+25+4 =64.

Find out the proportion of the first group of offspring among the sum total of 560.We are considering it , as a multiple of 16 . (dihybrid cross).

The first group is – 35 X16 /64 = 8.75 = 9 Discs

The second group is – 25 X16 / 64 =6.25 = 6 spherical

The third group is – 4 X 16 / 64 =1 elongate.

So the ratio is 9:6:1

It is a case of dominant and recessive interaction. There are two pairs of genes, which produces the same character separately. When they come together they interact and produce a novel character. When both the recessive genes come they interact and produce another character. For example, if the genes are A & B interact and produce Disc. The genes A alone produces spherical and gene B alone produces spherical. Both the recessive genes interact and produce elongate. In the Mundelian ratio 9:3:3:1, the two middle groups are indistinguishable and so modified to 9:6:1

Problem No.35

A variety pepper having brown fruit was crossed with a variety having yellow fruit. The F1 plants had red fruits and the F2 consisted of 182 plants with red fruits 59 with brown fruits, 61 with yellow fruits and 20 with green fruits. What is the genetic basis of these fruit colours in pepper?

Answer :

Firstly find out the possible ratio and find out the type of interaction.

Total number of offspring = 182+59+61+20= 322

Find out the proportion of the first group of offspring among the sum total of 322.We are considering it , as a multiple of 16 . (Dihybrid cross)

The first group is –182X16 / 322 =9.04

The second group =59X16/322 =2.93

The third group – 61X16/322=3.03

The fourth group – 20 X16 /322= 0.993

The approximate ratio is 9:3:3:1. It is a case of factor interaction like comb shape in fowls.

Problem No.36

Red colour in wheat kernels produced by genotype R-B- , white by double recessives. The genotype R- bb, rrB- produces brown kernels. A homozygous red variety is crossed to white. What phenotypic results are expected in F1 and F2?

Explanation -

It is a case of dominant and recessive interaction. There are two pairs of genes, which produces the same character separately. When they come together they interact and produce a novel character. When both the recessive genes come they interact and produce another character. For example, if the genes are R-- & B-- interact and produce Red colour. The genes R alone produces brown and gene B alone produces brown. Both the recessive genes interact and produce white. In the Mendelein ratio 9:3:3:1, the two middle groups are indistinguishable and so modified to 9:6:1. The F1 are all Red and F2 are segregated to

9 red : 6 brown : 1 white.

Problem No.37

White fruit colour in summer squash is governed by the dominant gene 'W and coloured fruit by its recessive allele 'w'. Yellow fruit is governed by an independently assorting gene 'G' and green by its recessive gene 'g'.When dihybrid plants were crossed 50 offspring were produced of which 36 plants were with white fruits, 10 with yellow fruits and 4 with green fruits. a) What phenotypic ratio can be deduced from the above data? B) What type of interaction is operative?

Answer : Firstly find out the possible ratio and find out the type of interaction.

Total number of offspring – 50

Find out the proportion of the first group of offspring among the sum total of 50. We are considering it , as a multiple of 16 . (Dihybrid cross)

So the first group is – 36 X 16 / 50 = 11.52 White fruits

Middle group – 10 X16 /50 =3.2 yellow

Last group – 4 X16/ 50 =1.28 green.

Here we will get an approximate ratio of 12:3:1. It is a case of dominant epistasis. Explain the theory of dominant epistasis.

Problem No.38

Crosses were made between two types of Oats. One with white hulled and the other with black hulled and the F2 had 178 black, 126 grey and 26 white. Give the type of interaction and explain it.

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Answer :
Firstly find out the possible ratio and find out the type of interaction.
Total number of offspring – 178+126+26 = 330
Find out the proportion of the first group of offspring among the sum total of 330.We are considering it , as a multiple of 16 . (Dihybrid cross)
So the first group is –178X16 /330 =8.63 = 9 black
The second group is –126 X16 / 330 =6.01 = 6 grey.
The third group is – 26 X16 / 330 =1.26 = 1 white.
Explanation -
It is a case of dominant and recessive interaction. There are two pairs of genes, which produces the same character separately. When they come together they interact and produce a novel character. When both the recessive genes come they interact and produce another character. For example if the genes are R-- & B-- interact and produce black colour. The genes R alone produce grey and gene B alone produces grey. Both the recessive genes interact and produce white. In the Mendelein ratio 9:3:3:1, the two middle groups are indistinguishable and so modified to 9:6:1. The F1 are all black and F2 are segregated to -9 black: 6 grey: 1 white.

Problem No.39
In capsella, the capsule shape is due to the interaction of two separate genes. A cross between triangular and ovoid-fruited plants produced all triangular fruited plants. This F1 upon selfing yielded 152 triangular fruited and 10 ovoid-fruited plants in the F2. What are the possible genotypes of the parents and the offspring? Explain the pattern of inheritance in this cross.
Answer:
Firstly find out the possible ratio and find out the type of interaction.
Total number of offspring – 152 +10 = 162
Find out the proportion of the first group of offspring among the sum total of 330.We are considering it, as a multiple of 16. (Dihybrid cross)
So the first group is –152 X16 / 162 =15 triangular
The second group is- 10 X16 / 162=. 0.99 =1 ovoid fruit.
The F2 ratio – 15:1. It is a case of duplicate genes. Two pairs of non – allelic genes (TTDD) produce the same character known as triangular seeds in Capsella bursa –pastoris. The double recessives produce ovoid seeds(ttdd). The two pairs of dominant genes have the effect on the phenotype. They together or individually produce the same effect. So they are called duplicate genes. One gene is the duplicate of the other. So in the Mendelian ratio the first three classes are indistinguishable and modified to 15:1 .

Problem No.40
Two white flowered plants of Pea varieties when crossed produced purple flowered F1 plants. Selfing of F1 plant produced 112 progeny plants, 62 with purple flowers and 50 with white flowers. A) What type of interaction is involved? B) Give the phenotypic ratio approximated by the F2 progeny. C) Give the genotypes of the parents.
Answer - Firstly find out the possible ratio and find out the type of interaction.
Total number of offspring –112
Find out the proportion of the first group of offspring among the sum total of 560.We are considering it, as a multiple of 16 . (dihybrid cross)
So the first group is- 62 X16 / 112 =8.85 = 9 purple
The second group is – 50 X16 /112 =7.14= 7 white.
It is a case of complementary factor interaction. Here we will get a ratio of 9:7. Explain the theory of complementary factor interaction. The genotype of parental strains –ccPp (white) X Ccpp (white). C -- in single dose or double dose produce white flower colour. Like wise P-- in single dose or double dose produce white flower colour. But when they come together they interact and produce purple flower colour. The double recessives will also produce white. The last three groups of Mendelian ratio of 9:3:3:1 is indistinguishable and modified to 9:7 ratio.

Problem No.41
Two non- allelic, dominant genes when together in a genotype produce coloured aleurone in maize seeds. Alone each produces colourless aleurone. Two colourless strains when crossed produced F1 progeny, all with coloured aleurone in seeds. 1) Give the genotypes of the parents. 2) Give the genotypes of the F2 progeny. 3) How many phenotypes, in what proportion are expected in F2 population?
Explanation -
It is a case of complementary factor interaction. Here we will get a ratio of 9:7. Explain the theory of complementary factor interaction. The genotype of parental strains – ccPP(colourless aleurone) X CCpp (colourless aleurone). C -- in single dose or double dose produce colourless aleurone. Like wise P-- in single dose or double dose produce colourless aleurone. But when they come together they interact and produce coloured aleurone in maize seeds. The double recessives will also produce colourless aleurone. The last three groups of Mendelian ratio of 9:3:3:1 is indistinguishable and modified to 9:7 ratio. The cross can be shown as follows –(ccPP X CCpp)gives an F1 Coloured – CcPp . The F1 is self-fertilized and we got 9 coloured: 7 colourless offspring.(Draw the checker board)

Problem No.42

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The colour of Ginger rhizome is governed by two pairs of alleles. A pure red strain crossed to a pure white strain produces yellow F1. The F2 progeny is represented by 49 white, 37 red, and 109 yellow rhizomes. 1) Calculate the expected phenotypic ratio of the progeny. 2) What is the basis of this type of inheritance?

Answer –

Firstly find out the possible ratio and find out the type of interaction.

Total number of offspring –49+37+109=195

Find out the proportion of the first group of offspring among the sum total of 195 .We are considering it , as a multiple of 16 . (Dihybrid cross)

So the first group is- 49 X16 / 195 = 4.02 white

The second group is – 37 X16 /195 =3.03 red

The third group is –109 X16 / 195 =8.94 yellow.

It is a modified Mendelian ratio 9:3:4 Supplementary factor inheritance. Here the last two classes are indistinguishable. There are two pairs of non-allelic genes- C and A. The genes C and A together interact and produce yellow ginger. The gene C-- alone in the absence of A produces red ginger. In the absence of C the colour of ginger is white. No matter even if A-- is present is present in the absence of C—the colour will also be white. This means that 'A' or 'a' have no effect on the phenotype.

Problem No.43

Mating between black rodents of the same genotype produced offspring in the ratio of 15 cream coloured, 46 black, and 19 albinos. 1) Give the type of interaction between the non- allelic genes responsible for the ratio.2) Give the approximate phenotypic ratio of the offspring. 3) Give the genotypes of the parents and the offspring.

Answer –

Firstly, find out the possible ratio and find out the type of interaction.

Total number of offspring –15+46+19=80

Find out the proportion of the first group of offspring among the sum total of 80 .We are considering it , as a multiple of 16 . (Dihybrid cross)

So the first group is- 15X16 / 80 = 3 Cream coloured

The second group is – 46 X16 / 80 = 9.2 black

The third group is-19X16 /80 = 3.8 albinos

Approximate ratio – 9 black: 3 cream: 4 albinos.

It is a modified Mendelian ratio 9:3:4 Supplementary factor inheritance. Here the last two classes are indistinguishable. There are two pairs of non-allelic genes- C and A. The genes C and A together interact and produce black colour. The gene C-- alone in the absence of A produces cream coloured. In the absence of C the colour is white or albino. No matter even if A-- is present is present in the absence of C—the colour will also be white or albino. This means that 'A' or 'a' have no effect on the phenotype. Genotype of parents = CcAa (black) X CcAc (black)

Problem No.44

The coat colour of mice depends upon the action of at least two genes. A dominant gene I - inhibits the expression of another non-allelic colour gene B and white colour is produced. The gene is expressed only when the recessive condition exists at the inhibitor locus (ii), thus iiB-genotype produces black coat and iibb genotype produce brown colour. If dihybrid white mice are mated together, determine 1) the ratio of the three phenotypes in the F1 progeny 2) the chance of selecting a genotype homozygous at both the loci: from among the white and black progeny.

Answer: I – = white colour , ii B—black, ii bb –brown

Phenotype of parents – White X white

Genotype of parents – Ii Bb X Ii Bb Gametes – IB,Ib, iB,ib

	(IB)	(Ib)	(iB)	(ib)
(IB)	IIBB White	IIBb- White	IiBb- White	IiBb- White
(Ib)	IIBb- White	Iibb- White	IiBb- White	Iibb- White
(iB)	IiBB- White	Iibb- White	(iiBb- black)	(iiBb- black)
(ib)	IiBb White	Iibb- White	(iiBb- black)	(iibb – brown)

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1)The F1 ratio – 12 white: 3 black :1 brown.
2) Only one coloum – IIBB (1/16 of offspring

Problem No.45
When dogs from a true breeding brown coat line were mated to dogs from a true breeding white coat line, al the F1 progeny were with white coat colour. Mating among the F1 progeny produced F2 phenotypes in the ratio of 130 white, 35 black, 11 brown. Explain this result.
Answer –
Firstly, find out the possible ratio and find out the type of interaction.
Total number of offspring –130+35+11= 176
Find out the proportion of the first group of offspring among the sum total of 176. We are considering it, as a multiple of 16. (Dihybrid cross)
So, the first group is- 130X16 / 176 = 11.818 = 12 white
The second group is – 35 X16 / 176 = 3.18 = 3 black
The third group is-11X16 /176 = 1 brown
Approximate ratio – 12 white: 3 black: 1 brown
Here we will get an approximate ratio of 12:3:1. It is a case of dominant epistasis. Explain the theory of dominant epistasis.

Problem No.46
Two pairs of non-allelic genes control the coat color in mice. A homozygous yellow mouse when mated with a homozygous black produces F1, all grey. Brother- Sister mating of F1 produces an F2 progeny in the phenotypic ratio of 27 grey, 9 yellow, 8 black, 3 cream colored. Explain this result.
Answer - Firstly find out the possible ratio and find out the type of interaction.
Total number of offspring –27+9+8+3=47
Find out the proportion of the first group of offspring among the sum total of 47. We are considering it, as a multiple of 16. (Dihybrid cross) The approximate ratio = 9:3:3:1
Explanation- 1) R gene in homozygous or heterozygous condition and in the absence of 'P' produces Yellow mice. 2) P gene in homozygous or heterozygous condition and in the absence of' R' produces black mice .3) When both R & P comes together they interact and produces a novel character Grey colour in F1 generation. 3) Both the recessive genes interact and produces cream colour.4) Even though the F2 ratio is numerically same as a dihybrid ratio it is not at all related to the pattern of inheritance. Because the F1 is a different character not seen among the parents. The F2 cream colour is also not seen anywhere in the parental lines. The cross can be shown as follows.
Phenotype of parents – Yellow X Black
Genotype of parents - RRpp & rrPP
Gametes - Rp & rP
F1 Offspring - RrPp (Grey)
The F1 is self-fertilised- RrPp X RrPp F1 gametes- RP,Rp,rP,rp

	(RP)	(Rp)	(rP)	(rp)
(RP)	RRPP-Grey	RRPp Grey	RrPp Grey	RrPp Grey
(Rp)	RRPp Grey	(RRpp-yellow)	RrPp Grey	(Rrpp-yellow)
(rP)	RrPP Grey	RrPp Grey	(rrPp-black)	(rrPp-black)
(rp)	RrPp Grey	(Rrpp-yellow)	(rrPp-black)	(rrpp- cream)

The F2 ratio – 9Grey:3Yellow :3 black :1 cream

Practice problems and solutions

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Find out the proportion of the first group of offspring among the sum total of 47. We are considering it, as a multiple of 16. (Dihybrid cross) The approximate ratio = 9:3:3:1
Explanation- 1) R gene in homozygous or heterozygous condition and in the absence of 'P' produces Yellow mice. 2) P gene in homozygous or heterozygous condition and in the absence of' R' produces black mice .3) When both R & P comes together they interact and produces a novel character Grey colour in F1 generation. 3) Both the recessive genes interact and produces cream colour.4) Even though the F2 ratio is numerically same as a dihybrid ratio it is not at all related to the pattern of inheritance. Because the F1 is a different character not seen among the parents. The F2 cream colour is also not seen anywhere in the parental lines. The cross can be shown as follows.
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Gametes - Rp & rP
F1 Offspring - RrPp (Grey)
The F1 is self-fertilised- RrPp X RrPp F1 gametes- RP,Rp,rP,rp

	(RP)	(Rp)	(rP)	(rp)
(RP)	RRPP-Grey	RRPp Grey	RrPp Grey	RrPp Grey
(Rp)	RRPp Grey	(RRpp-yellow)	RrPp Grey	(Rrpp-yellow)
(rP)	RrPP Grey	RrPp Grey	(rrPp-black)	(rrPp-black)
(rp)	RrPp Grey	(Rrpp-yellow)	(rrPp-black)	(rrpp- cream)

The F2 ratio – 9Grey:3Yellow :3 black :1 cream

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These problems are divided into subdivisions composed of problems that require application of a specific genetic principle. These problems are intended to complement the lecture portion of this course; specifically, the material described in lecture is to be applied to solve these problems. The answers are provided in Appendix A. You are strongly advised against 'consulting this appendix before you have made a serious attempt to answer a problem.

A. Monohybrid Crosses

1. In *Coleus*, some plants have shallowly crenated edges and others have deeply incised leaves. A cross is made between homozygous deep and shallow individuals. The shallow trait is dominant.

a. Using S and s to symbolize the genes for this trait, give the phenotypic and genotypic ratios for the Fi generation.

b. If self-pollination is allowed, what is the phenotypic ratio for the F2 generation?

In a pea plant that breeds true for tall, what possible gametes can be produced? Use the symbol *D* for tall, *d* for dwarf.

In a pea plant that breeds true for dwarf, what possible gametes will be produced? What will be the genotype of Fi offspring from a cross between these two types?

Assuming that the allele for tall is dominant, what will be the phenotype of F_1 offspring from a cross between these two types?

What will be the probable distribution of traits in the F2 generation? (Illustrate with a Punnett square).

3. The ability to taste a bitter chemical, phenylthiocarbamide (PTC), is due to a dominant gene. Use *T* and *t* to symbolize the two alleles of this gene.

a. What is the genotype of a nontaster? What are the possible genotypes of a taster?

b. Could a person with two tasters as parents be a non-taster? How?

4. A woman heterozygous for polydactyly (extra fingers and toes), a dominant trait, is married to a normal man. What is the probability of producing an offspring that has extra fingers or toes?

5. Parents who do not have Tay Sachs disease produce a child who has this terrible affliction. What are the chances that each child born of this union will be affected?

6. In human beings, ability to curl the tongue into a U-shaped trough is a heritable trait. "Curlers" always have at least one curler parent, but "noncurlers" may occur in families where one or both parents are curlers. Using C and *c* to symbolize this trait, what is the genotype of a noncurler?

7. Albinism, the total lack of pigment, is due to a recessive gene. A man and woman plan to marry and wish to know the probability of their having any albino children. What are the probabilities if:

a. both are normally pigmented, but each has one albino parent.

b. the man is an albino, the girl is normal, but her father is an albino.

c. the man is an albino and the girl's family include no albinos for at least three generations.

8. In a certain plant, both purple x purple and purple x blue yield purple and blue colored progeny, but blue x blue gives rise only to blue.

a. What does this tell you about the genotypes of blue- and purple-flowered plants?

b. Which gene is dominant?

9. Two short-haired female cats are mated to the same long-haired male. Several litters are produced. Female No. 1 produced eight short-haired and six long-haired kittens. Female No. 2 produced 24 short-haired ones and no long-haired. From these observations, what deductions can be made concerning hair-length inheritance in these animals? Assuming the allelic pair S and s, give the likely genotypes of the two female cats and the male.

10. In human beings, a downward pointed frontal hairline ("widow's peak") is a heritable trait. A person with a widow's peak always has at least one parent who also has this trait, whereas persons with a straight frontal hairline may occur in families in which one or even both parents have widow's peak. When both parents have a straight frontal hairline, all children also have a straight hairline. Using W and w to symbolize genes for this trait, what is the genotype of an individual without widow's peak?

11. Rh negative children (those not producing rhesus antigen D) may be born to either Rh positive or Rh-negative parents, but Rh-positive children always have at least one Rh positive parent. Which phenotype is due to a dominant gene?

Answers with explanation:

A. <u>Monohybrid Crosses</u>

1. a. *SS* x
Shallow deep
genotypic = all *Ss* phenotypic = all shallow
b. F_2 Genotypic = 1 : 2 : 1 or 1/4 *SS:* 1/2 *Ss:* 1/4 ss
Phenotypic = 3 : 1 or 3/4 shallow; 1/4 deep

2. a. D b. d c. Dd d. All tall e. F_2

	D	d
D	DD	Dd
d	Dd	dd

3. a. tt; TT or Tt

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b. Yes, if both his or her parents were heterozygous.
4. 50%
5. 25%
6. cc
7. a. 25% b. 50%
c. Probably zero: the girl's family history suggests that she is homozygous normal.
8. a. Purple is heterozygous; blue is homozygous b. Purple
9. Short hair is the dominant trait. Genotypes: Female No. 1 is Ss; Female No. 2 is SS; the male is ss.
10. ww - homozygous recessive
11. Rh+ is dominant.

Practice problems and solutions

B. Modifications of Complete Dominance

Incomplete Dominance

1. The so-called "blue" (really gray) Andalusian variety of chicken is produced by a cross between the black and white varieties, both of which breed true (i.e., both are homozygous). What color chickens (and in what proportions) would you expect if you crossed two blues? a blue and a black?
2. In four o'clock, red color exhibits incomplete dominance over white; when both exist together, the flowers are pink.
a. In a cross between a red flower and a white one, what is the genotype of the offspring?
b. What is the genotypic ratio of the F2 generation if two of the F1 from (a) are crossed?
c. List the genotypes of offspring produced by a cross between the F_1 generation and red parent.
3. It has long been known in the field of human genetics that wavy hair is the expression of a heterozygous genotype in which the allele for straight hair is paired with the allele for curly hair. Lucinda Lovelee married Larry Legg. Both of these charmers have wavy hair. What is the probability that their offspring, the littlest Legg, will have:
a. wavy hair?
b. curly hair?
c. straight hair?
4. If pale colored horses are crossed with chestnut-colored horses to produce "palomino", an intermediate coat color:
a. What type of expression is suggested?
b. A number of matings between palominos produced 19 pale, 21 chestnut, and 44 palominos. Does this evidence support or contradict your answer to (a)? Why?

Codominance

1.For each of the following pairs of parental genotypes, calculate the phenotypic ratios for the F_1 generation.
a. I^AI^A x ii b. I^AI^A x I^AI^B c. I^AI^A x I^Bi
d. I^AI^A x I^Ai e. I^Ai x I^Ai f. I^Ai x I^AI^B g. I^Ai x ii
2. If Mr. and Mrs. Fecundity, both having blood type B, have 12 children, 3/4 of whom are type B and 1/4 of whom are O, what are the genotypes of the parents?
3. A family of six includes four children, each of whom has a different blood type: A, B, AB and O. What are the genotypes of parents for this trait?
4. A man with blood type B, with one parent of blood type O, marries a woman with blood type AB. What will be the theoretical percentage of their children with blood type B?
5. Mrs. Smith and Mrs. Doe were roommates at Harris Hospital and both had daughters at about the same time. After Mrs. Smith took Susie home, she became convinced that the babies had been switched. Blood tests were performed with the following results:
Mr. and Mrs. Smith were both type AB;
Mr. and Mrs. Doe were both type A;
Susie Smith was type A and Debbie Doe was type O.
Had a switch occurred?
6. Mortimer has type B blood. His wife Murgatroyd is unsure of her blood type. If their first child, Magnifica, is type B, their second offspring, Maximum, is AB and the twins, Maud and Lyn, are A, can you determine the genotypes of Mort and Murg?
7. In a well-publicized paternity case, the following facts were unearthed; the mother, a strikingly beautiful, twice-convicted axe murderess, is blood type A, her child, Lizzie, is type O, and the alleged father, a mild-mannered felon, is type B. Could he be the father? Explain. Is there any chance that little Lizzie, the "Bad Seed", will grow up to be a missionary lady?

C. Lethal Alleles

1. Cystic fibrosis is caused by a recessive lethal gene and can be detected by an excess concentration of chloride in sweat. If a sweat test reveals a man to be heterozygous and his wife to be homozygous normal, what are the chances that their children will have the disease? Could any of their grandchildren have the disease?
2. In humans, sickle-cell anemia is caused by a recessive lethal allele Hb^s; individuals who are Hb^aHb^s have sickle-cell trait, but are healthy.

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a. What is the probability of two heterozygous individuals giving birth to a child with sickle-cell trait?
b. What is the probability of two heterozygous individuals giving birth to a child with sickle-cell anemia?
c. If a normal Hb^AHb^a individual receives a blood transfusion from a Hb^AHb^a individual (heterozygous for sickle cell) what are the chances that the Hb^AHb^a man and his Hb^AHb^a wife will have Hb^AHb^a children?

3. Albinism in corn plants is caused by a recessive lethal gene that results in death before maturity. What will the adult phenotypic ratio be for the F1 generation of heterozygous parents?

4. Huntington's chorea is a dominant lethal in humans. The disease does not appear until later in life, so that afflicted individuals may already have produced children. What are the F_1 genotypic and phenotypic ratios of parents who are homozygous dominant and heterozygous?

D. Sex-linked Genes

1. Red-green color blindness is inherited as a sex-linked recessive. If a color-blind woman marries a man who has normal vision, what would be the expected phenotypes of their children with reference to this character?

2. Suppose that gene b is sex-linked, recessive, and embryonic lethal. A man marries a woman who is heterozygous for this gene. If this couple had many normal children, what would be the predicted sex ratio of these children?

3. A man and his wife both have normal color vision, but a daughter has red-green color blindness, a sex-linked recessive trait. The man sues his wife for divorce on grounds of infidelity. Can genetics provide evidence supporting his case?

4. In the mouse, the dominant sex-linked gene B results in a short, crooked tail. Its recessive allele b produces a normal tail. If a normal-tailed female is mated with a benttailed male, what phenotypic ratio should occur in the F_1 generation?

5. In cats, a gene for coat color is sex-linked. Cats homozygous for allele A have yellow coats; those homozygous for allele a have black coats; and heterozygotes have tortoiseshell coats. What type(s) of offspring would result from a mating of a black male and a tortoise-shell female? Is it possible to obtain a tortoise-shell male?

6. On the X chromosome of *Drosophila* there may occur a recessive gene l, which is lethal in the larval stage. A heterozygous female is crossed to a normal male; what F_1 adult sex phenotypic ratio results?

Answers with explanations:

B. Modifications of Complete Dominance

Incomplete Dominance

1. The genotypes of the black and white may be represented as BB and bb, respectively. A "blue" chicken has the genotype Bb. Crossing two blues would produce 1/4 black, 1/2 blue, and 1/4 white. Crossing a blue and a black would produce 1/2 blue and 1/2 black.

2. a. If the parental genotypes are RR and rr, the offspring would be Rr.
b. 1 RR : 2 Rr : 1 rr
c. 1 RR : 1 Rr

3. The genotypes may be symbolized as follows: AA = straight hair, aa = curly hair, Aa = wavy hair.
a. 50%
b. 25%
c. 25%

4. a. incomplete dominance
b. We suspect palominos to be heterozygous. If they are crossed, a 1:2:1 ratio should result. It does, which supports our answer.

Codominance

1. a. 100% A
b. 50% A, 50% AB
c. 50% AB, 50% A
d. 100% A
e. 75% A, 25% O
f. 50% A, 25% AB, 25% B
g. 50% A, 50% O

2. I^Bi x I^Bi
AB

3. Parents must be I^Ai and I^Bi

4. 50%. The parental genotypes are: I^Bi x I^AI^B
AB

5. Mr. and Mrs. Smith must both have genotypes I^AI^B. Mr. and Mrs. Smith could not have given birth to Debbie Doe - Type O. A switch had not occurred.
B A A B

6. Mortimer is I^Bi. Murgatroyd is either I^Ai or I^AI^B.
A B

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7. The mother may be I^Ai, while the father is I^Bi. Consequently, there would be a 25% chance of the child being type O. Criminal behavior has never been found to be correlated with chromosomal inheritance. With the proper environment, Little Lizzie could turn out to be an angel.

C. Lethal Alleles

1. None of the their children will have cystic fibrosis. Their grandchildren could have the disease if the heterozygous children marry other heterozygotes.
2. a) Hb^sHb^s 50%
   b) Hb^sHb^s 25%
   c) 0: the genotype is not affected by the blood transfusion.
3. All adult F1 will be green, albinos die.
4. Genotypic ratio: 1 homozygous dominant: 1 heterozygote Phenotypic ratio: All F1 will have Huntington's chorea.

D. Sex-Linked Genes

1. All girls are normal, but carriers.
   All boys are color-blind.
2. Two-thirds female: one-third male.
3. A color-blind girl could only be the result of a union of (a) a mother with normal vision but a carrier (X^oX), with a colorblind male (X^oY); or (b) a spontaneous mutation of the X chromosome handed down by the father. The smart money goes for (a).
4. One normal male: one bent female.
5. 1/4 tortoise female; 1/4 black female; 1/4 yellow male; 1/4 black male. No.
6. Two normal females: 1 normal male (male with lethal dies).

Practice problems and solutions

E. Dihybrid Crosses

1. In the fruit fly *Drosophila melanogaster,* vestigial wings and hairy body are produced by two recessive genes located on different chromosomes. The normal alleles, long wings and hairless body, are dominant. Give the genotype and phenotype of F_1 progeny obtained from a cross between a vestigial-winged, hairy male and a normal, homozygous female. If the F_1 from this cross are permitted to mate randomly among themselves, what phenotypic ratio would be expected in the F_2 generation?
2. In peas, a gene for tall plants *(T)* is dominant over its allele for short plants (t). The gene for smooth peas (S) is dominant over its allele for wrinkled peas (s). The genes are not linked. Calculate both phenotypic and genotypic ratios for the results of each of the following crosses:
   a. *TtSs* x *TtSs* b. *Ttss* x *ttss* c. *ttSs* x *Ttss* d. *TtSS* x *ss*
3. In a particular species of flower, tall is dominant to short, and orange petals are dominant to the recessive white color. Use *T* and *t* to symbolize the alleles for height, and *F* and *f* to symbolize the alleles for flower color. A homozygous tall white flower is crossed with a flower heterozygous for both traits. List the genotypes of the parents. What are the F1 genotypic and phenotypic ratios?
4. How many phenotypic classes are produced by a dihybrid test-cross where one parent is heterozygous for both pairs of genes?
5. In hogs, an allele that produces a white belt around the animal's body *(W)* is dominant over its allele for a uniformly colored body (w). The dominant allele of another gene (F) produces a fusion of the two hoofs on each foot. Suppose a uniformly-colored hog homozygous for fused hoofs is mated with a normal-footed hog homozygous for the belted character.
   a. What are the genotypes of the parents?
   b. What are the genotypic and phenotypic ratios of the F1?
   c. If the F_1 were allowed to interbreed, what are the genotypic and phenotypic ratios of the F_2?
6. In watermelons, the genes for green color and for short length are dominant over their alleles for striped color and for long length. Suppose a plant with long striped fruit is crossed with a plant heterozygous for both of these characters. What phenotypes would this cross produce and in what ratios?

Answers and explanations: -

E. Dihybrid Crosses

1. The genes may be symbolized as:
   v = vestigial h = hairy body V = long wingH = hairless body
   The genotypes of the parents are: Male vvhh, Female VVHH.
   One hundred percent of the F1 will be VvHh (normal long wing, hairless body).
   The F2 phenotypic ratio is 9 normal long wings, hairless body; 3 vestigial wings, hairless body; 3 normal long wing, hairy body; 1 vestigial wing, hairy body. This can be obtained by multiplying the two phenotypic ratios times one another.

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The phenotypic ratio for wings is 3 normal long: 1 vestigial. The phenotypic ratio for hair is 3 hairless: 1 hairy. Since the two traits segregate independently -

3 :	1
3	1
9 : 3 : 3	1

2. The two ratios can again be multiplied, since the genes (S+T) are not linked.

Genotypic Ratio

a) 1TT : 2Tt : 1tt 1SS : 2Ss : 1ss 1TTSS : 2TtSS : 1ttSS 2TTSs : 4TtSs : 2 ttSs short, smooth 1TTss : 2Ttss : 1 ttss

b) 1Tt : 1tt

1ss

1Ttss : 1ttss

Phenotypic Ratio

3 tall : 1 short 3 smooth : 1 wrinkled 9 tall, smooth : 3 short, smooth: 3 tall, wrinkled : 1 short, wrinkled

1 tall : 1 short all wrinkled

1 tall, wrinkled : 1 short, wrinkled

1 tall : 1 short

1 smooth : 1 wrinkled

1 tall, smooth : 1 short,

smooth : 1 tall, wrinkled:1 short, wrinkled

All tall All smooth All tall, smooth

c) 1tt : 1Tt 1Ss : 1ss 1ttSs : 1TtSs 1ttss : 1Ttss

d) 1Tt 1Ss 1TtSs

3. The parental genotypes are: TTff x TtFf

The F_1 genotypic ratio is: 1 : 1 : 1 : 1

The F_1 phenotypic ratio is: 1 : 1

4. Four. For example: CcBb x ccbb —Cc Bb

Cc bb cc Bb cc bb

5. a) The parental genotypes are: ww FF x WW ff

b) All the Fi's are WwFf (white belted with fused hoofs).

c) The genotypic ratio is: 1:2:1: 2:4:2: 1:2:1 The phenotypic ratio is: 9:3:3:1

6. 1 green short: 1 green long: 1 striped short: 1 striped long

Practice problems: Apply your knowledge

More Example questions

The following problems are provided to develop your skill and test your understanding of solving problems in the patterns of inheritance. They will be most helpful if you solve them on your own. However, you should seek help if you find you cannot answer a problem. Most of these problems are fairly simple, yet mastering their solutions will provide the background to solve many genetic puzzles and will strengthen your understanding fundamental principles of genetics.

The completion of this set of problems constitutes an assignment worth 10 points. To receive credit, you must work each of the problems to its correct solution. I have provided the answers, I want you to show how to arrive at them. You must show how you solved each problem, even if you can do it in your head. Number the problems and make your work neat enough for me to read. You must turn in your worked problems by October 10 (the last class before the exam) to receive any credit. NOTE: The points for this assignment will be awarded on an ALL-OR-NONE basis. That means you must work ALL problems to their correct solution in order to receive ANY credit. You may ask me or your TAs for help on any of these problems. I recommend that you make a copy of your work before you hand it in so that you can use it while studying for the exam.

A. PROBABILITY

1. You and your spouse have no children. You stand to inherit a sizeable fortune from your crazy Uncle Irving if you can produce three daughters in your family of three children. What is the probability of doing just that?

2. If you could convince Uncle Irving that simply having three children all of the same sex would do, then what would be the probability of your receiving the inheritance?

3. In quest of the family stipulated in #2 above, you produce a boy as your first child. Now what is the probability of inheriting the fortune?

4. Why are the answers to #2 and #3 the same?

5. Finally, you have convinced Uncle Irving that you will agree to try for at least three girls out of four children. How likely are you to become wealthy given those conditions?

B. MONO-, DI-, AND POLYHYBRID CROSSES; DOMINANCE AND RECESSIVENESS

In all of the following problems, capital letters will be used to denote a dominant trait, and lower-case letters will be used for the recessive trait.

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6. In peas, seeds may be round (R) or wrinkled (r). What proportion of the offspring in the following crosses would be expected to be wrinkled?
a. RR x rr b. Rr x Rr c. Rr x rr

7. In peas, seeds may be yellow (Y) or green (y). What proportion of the offspring in the following crosses would be expected to be yellow?
a. YY x Yy b. Yy x Yy c. yy x yy

8. In peas (again), the stem length may result in a tall (T) or dwarf (t) plant. What proportion of the offspring in the following crosses would be expected to be tall, and what proportion dwarf?
a. TT x tt b. TT x Tt c. Tt x Tt d. tt x Tt

9. What proportion of the plants from the following crosses would be tall with yellow, wrinkled seeds?
a. TtYYRr x ttYYrr b. TTYyRr x TtYyRr
c ttYyrr x ttyyRr d.TtYyRr x TtYyRr

10. From the crosses TTYyRr x TtYyrr, what proportion of the offspring would be expected to be
a. tall plants with round, yellow seeds
b. tall plants with round, green seeds
c. dwarf plants with round, green seeds
d. tall plants with yellow, wrinkled seeds
e. tall plants with green, wrinkled seeds

11. For the purpose of this problem assume that in humans the gene for brown eyes is dominant to that for blue eyes.
a. A brown-eyed man marries a blue-eyed woman, and they have eight brown-eyed children. What are the genotypes of all the individuals in the family?
b. What is the probability that the first child produced in parents who are both heterozygous for brown eyes will be blueeyed?
c. If the first child is a brown-eyed girl (same parents as in b), what is the probability that the second child will be a blueeyed boy?
d. Again, referring to the marriage in b, what is the probability that the first three children will be blue-eyed girls and the fourth a brown-eyed boy?

12. Eye color in certain species of flies is controlled by a single pair of genes. A white-eyed fly, both of whose parents had white eyes, was crossed with a red-eyed fly, and all of their offspring (both male and female) were red-eyed.
a. Is the gene for red eyes or that for white eyes dominant? Proof?
b. What was the genotype of the white-eyed parents?
c. What was (were) the genotype(s) of the red-eyed offspring?
d. If one of the red-eyed offspring was mated with the white-eyed parent, what would be the expected ratio of offspring, with respect to eye color?
e. If two of the red-eyed offspring are mated, how many genetically different kinds of zygotes, with respect to eye color, will be formed, and what will the proportions be?

13. In cattle, the gene for hornless (H) is dominant to the gene for horned (h), the gene for black (B) is dominant to that of red (b), and the gene for white face (or Hereford spotting) (S) is dominant to that for solid color (s). A cow with the genotype BbHhSs is inseminated by a bull of the genotype bbhhSs. What is the probability of the calf's being:
a. a black, hornless cow with Hereford spotting
b. a red, horned bull with solid color
c. a red, hornless bull with Hereford spotting

14. Assume that D, E, F, G, H, and I are autosomal genes on different chromosomes. From the mating DdeeFfGGHhIi x DdEEFFGgHhii:
a. What is the probability that one of the offspring will have the genotype DdEeFFGghhIi?
b. What is the probability that one of the offspring will be heterozygous for each allele?
c. What is the probability that one of the offspring will have the genotype DDEEFfGGhhii?

C. INCOMPLETE DOMINANCE

15. In cattle, RR = red, Rr = roan, and rr = white. What are the predicted color phenotypes and their frequencies for the offspring from crosses between:
a. a red bull and a white cow
b. a red bull and a roan cow
c. a roan bull and a roan cow

16. Given the following information about the inheritance of characteristics in pea plants, answer the questions below:
Y (yellow) is dominant to y (green)
R (round) is dominant to r (wrinkled)
B (bitter) is dominant to b (sweet)
S (smooth) is dominant to s (hairy)
L (long pod) shows incomplete dominance to l (short pod) (Ll is medium in length)
Given this cross: (P1) Yy Rr Bb SS Ll (male) x yy RR Bb Ss Ll (female)
a. How many different gametes can be formed by the female plant?
b. How many different genotypes are possible in the F1 offspring?

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c. How many different phenotypes are possible in the Fi offspring?
d. What percent of the F_1 individuals will be
green, bitter, and smooth
hairy, medium, and sweet
round, bitter, and long

D. MULTIPLE ALLELES

17. In humans, the ABO blood groups are controlled by three alleles (only two of which occur in any one individual): the alleles for A and B type blood are co-dominant toward each other, and both are dominant to the allele for O type blood.
a. If a person with type AB blood marries someone with type O blood, what are the possible phenotypes of their offspring? In the following, determine the genotypes of the parents:
b. One parent has type A and the other has type B, but all four blood groups are represented in the children.
c. Both parents have type A, but 3/4 of the children are A and 1/4 are O.
d. One parent has type AB and the other has type B, but of the children 1/4 have type A, 1/4 have type AB, and 1/2 have type B,
18. In the following cases of disputed paternity, determine the probable parent.
a. Mother is type B, child is type O. Father #1 is A; father #2 is AB.
b. Mother is type B, child is type AB. Father #1 is A; father #2 is B.
c. Mother is type O and bears non-identical twins, one type A and one type B. Father #1 is type A; father #2 is type B.
19. Two babies in a maternity ward have lost their identity bands, and there is some confusion about their footprint records. Baby #1 is type A; baby #2 is type B. If you are one of the mothers and your blood type is O, which one of the following statements applies.
a. Neither baby could be yours.
b. The type A baby is yours.
c. The type B baby is yours.
d. Either baby could be yours.
20. A woman with type A blood has parents who are both type AB and a husband who is a type B. What is the probability that their first child will be a son with type O blood?
21. In a local court, a woman is suing a male acquaintance for financial support of her recently born child. If the woman is blood type B, Rh+, and the baby is type O, Rh-, and the man is blood type AB, Rh-, what are her chances of success in the lawsuit?
22. In the organism under consideration, r* acts like r allele, except when homozygous (r*r*). From the information given below, work out the phenotypic and genotypic ratios for each of the crosses.
Given: RR (red) x rr (white) ® Rr (pink); and
rr* x rr* ® 3/4 white, 1/4 dead zygotes
a. Rr x Rr b. Rr x rr c. Rr* x Rr* d. Rr* x rr*
23. In rabbits, fur color is determined by a set of multiple alleles at one locus (gene) that have the following relationship:
C^+ (agouti) is dominant to all other alleles c^h (himalayan) is dominant to c^a (albino)
c^u (chinchilla) shows incomplete dominance with regard to c^h and c^a
The genotypes c^uc^h and c^uc^a are light-grey phenotypes
a. What breeding stock (parents) would you select if you wished all of the offspring to be chinchilla?
b. In one of the matings of rabbits, the litter contained 4 grey bunnies, 2 albino bunnies, and 2 himalayan bunnies. What were the genotypes of the parents?
c. In another mating, the litter contained 3 agouti bunnies and 3 light-grey bunnies. What were the genotypes of the parents of this litter?

E. MULTIPLE GENES

24. In cocker spaniels, the following genotypes and phenotypes are found:
AABB = white A-bb = red aabb = lemon AaB- = black aaB- = liver AABb = grey
a. A red female is mated with a liver-colored male, and one of the pups produced is lemon-colored. What are the genotypes of the parents?
b. What proportion of these offspring would be expected to be black?
c. A black male is mated with a liver-colored female, and they produce the following pups:
3/8 black 1/8 red
3/8 liver-colored 1/8 lemon-colored
What are the genotypes of the two parents?
25. If two cocker spaniels of the genotypes below are mated, and eight pups are born, what is the most likely distribution of coat colors in that litter?
P1 AaBb x AABb
white red lemon black liver grey
26. A dominant gene, A, causes yellow color in rats. The dominant allele of another independent gene, R, produces black coat color. When the two dominant genes occur together (A-R-), they interact to produce grey coat color. Rats of the double recessive genotype are cream-colored. If a grey male and a yellow female are mated and produce approximately 3/8 yellow, 3/8 grey, 1/8 cream, and 1/8 black, what were the genotypes of the parents?

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F. SEX LINKAGE

27. In humans, dark hair (B) is dominant over blondness (b), and color blindness (c) is a sex-linked recessive trait. A woman has a blond brother, a blond mother, and a dark-haired father. Her brother and her parents have normal vision. She bears the following three children by her blond, normal-versioned husband:
a dark-haired son with normal vision a dark-haired daughter with normal vision, and a dark-haired color-blind son
a. Make a pedigree of the entire family showing the probable genotypes of all individuals.
b. What is the probability that her next (fourth) child will be a color-blind boy?
c. If her fourth child is a boy, what is the probability that he will have dark hair?
d. What is the chance that her next four children will all be girls?

28. Your research in hematology has led you to the discovery of a new type of inherited anemia, which exists in two forms--mild and severe. The information you have gathered on the familial patterns for the disease is summarized below. From these data determine a mechanism for the inheritance of these traits that is consistent with all of the information.
a. normal female x normal male = all normal children
b. normal female x anemic male = 1/2 mild females
= 1/2 normal males
c. mild female x normal male = 1/4 normal males
= 1/4 anemic males = 1/4 normal females = 1/4 mild females
d. mild female x anemic male = 1/4 normal males
= 1/4 anemic males = 1/4 mild females = 1/4 anemic females
e. anemic female x normal male = 1/2 mild females
= 1/2 anemic males f. anemic female x anemic male = all anemic children

29. In chickens and other birds, the chromosomal basis of inheritance is the opposites of that in man; i.e., in birds, XX individuals are males, and XY individuals are females.
In chickens, barred plumage is dominant to nonbarred plumage; the gene is sex-linked.
Suppose that you were a poultry breeder and that you needed large numbers of barred males and nonbarred females. Describe a breeding stock that you could assemble for this purpose, which would produce only barred males and nonbarred females.
Be certain that you show the genotypes of both the roosters and the hens in your breeding stock and also the genotypes of all the offspring that this stock will yield.

30. The inheritance of color blindness in humans is due to a recessive gene located on the X chromosome (X linked).
X^+ (normal) > X^c (color blind)
a. If a color-blind boy is born to parents both of whom have normal vision, what are the genotypes of the three individuals?
b. What is the probability that the second child born to that couple will be a color-blind daughter?
c. If this couple has four children, what is the probability that the first two children will be color-blind boys and the last two children will be girls with normal vision?

31. A particular inherited abnormality in humans has been shown to be an X-linked recessive trait. In one family seeking professional help from a genetic counselor, the following information is known:
1. The mother shows the abnormality.
2. Their only child is a daughter who is normal.
a. If their next child is a son, what is the probability that he will show the abnormality?
b. What is the probability that the next child born will be a daughter showing the abnormality?

G. HUMAN GENETICS

32. Research has shown that a particular eye defect is represented in a family pedigree as follows:

Answers to the "Apply your knowledge" questions

1. 1/8
2. 1/4
3. 1/4
4. Because the sex of the first child is irrelevant.
5. 5/16
6. a. none
   b. 1/4
   c. 1/2
7. a. all
   b. 3/4
   c. none
8. a. all tall
   b. all tall
   c. 3/4 tall; 1/4 dwarf
   d. 1/2 tall; 1/2 dwarf
9. a. 1/4

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b. 3/16
c. 0
d. 9/64

10. a. 3/8
b. 1/8
c. 0
d. 3/8
e. 1/8

11. a. female bb; male BB; children all Bb
b. 1/4
c. 1/8
d. 3/4096

12. a. red
b. homozygous recessive
c. heterozygous
d. 50% red-eyed; 50% white-eyed
e. 3; 1/4 homozygous, 1/2 heterozygous, 1/4 homozygous recessive

13. a. 3/32
b. 1/32
c. 3/32

14. a. 1/64
b. 1/32
c. 0

15. a. all roan
b. 1/2 red, 1/2 roan
c. 1/4 red, 1/2 roan, 1/4 white

16. a. 8
b. 72
c. 12
d. 37.5%
0
18.75%

17. a. 1/2 A; 1/2 B
b. AO and BO
c. both AO
d. AB and BO

18. a. type A father (1#)
b. type A father (#1)
c. either the real father must be AB or both men are the fathers, one of each twin. (There is documented evidence for such an occurrence.)

19. d; either baby could be yours
20. zero
21. zero

22. a. 1/4 red (RR), 1/2 pink (Rr), 1/4 white (rr)
b. 1/2 pink (Rr), 1/2 white (rr)
c. 1/3 red (RR), 2/3 pink (Rr*); (r*r* dead)
d. 2/3 pink (Rr and Rr*), 1/3 white (rr*); (r*r* dead)

23. a. cucu x cucu
b. cuca x chca
c. various correct answers
e.g. c+ca x cucu

24. a. female = Aabb
male = aaBb
b. 1/4
c. female = aaBb
male = AaBb

25. 1 white, 3 black, 2 red, 0 liver, 0 lemon, 2 grey

26.

27.
a. BbXCy BbXCX? BbXcY
b. 1/4

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c. 1/2
d. 1/16

28. Normal (A) is incompletely dominant to anemic (a); allele is sex-linked

AA = normal
Aa = mild anemia
aa = severe anemia

Genotype of crosses are:

a. AA x Ay
b. AA x aY
c. Aa x AY
d. Aa x aY
e. aa x AY
f. aa x aY

29. P1 (male) XbXb x XBY (female)
F1 XBXb -- all barred
Xby female -- all non-barred

2. Gene Mapping Problems

How to solve Gene mapping problems?

The dihybrid crosses that we've considered up to this point are those segregating for genes on *different chromosomes*. What genotypes might we expect to see if the genes are located on the same chromosome?

Genes on the same chromosome assort together more often than not

In dihybrid crosses, departures from a 1:1:1:1 ratio of F1 gametes indicates that the two genes are on the same chromosome (linked). Among the F2 progeny, there are more **parental types** than **recombinant types**.

Consider two *Drosophila* genes linked to the X-chromosome, the eye color gene *white* and a body color gene *yellow*. The wild-type *white* allele is *w+* (red eyes); the recessive mutant allele is *w* (white eyes). The wild-type allele is y+ (brown bodies); the recessive mutant allele is *y* (yellow bodies).

Cross:

P: white, brown female x red, yellow male
w y+ /*w* y+ x *w+y* / Y

F1: red, brown female x white brown male
w+ y / *w* y+ x *w* y+ / Y

F2 males:

white, brown males	*w* y+ / Y	Parental type	4484
red, yellow males	*w+ y* / Y	Parental type	4413
red, brown males	*w+*y+ / Y	Recombinant type	76
white, yellow males	*w y* / Y	Recombinant type	53

If the white and yellow genes sort independently of one another, than the F1 female would produce 4 kinds of games with equal probability (w y+, w+ y, w+ y+, w y). We can reveal the ratio of her gametes by observing only the male progeny, as sons receive their only X from their mother. As you can see from the data, the four kinds of gametes do not occur with equal frequency. Instead **parental types** vastly outnumber **recombinant types** (99% to 1%).

Bottom line: When two genes assort independently, the numbers of parental and recombinant F2 progeny are equal. Two genes are considered **linked** when the number of F2 progeny with parental genotypes exceeds the number of F2 progeny with recombinant genotypes.

The **recombination frequency** (**RF**, the percentage of total progeny that are recombinant) depends upon the gene pair under consideration. Linked genes have a recombination frequency of less than 50%. The example we use above indicates tight linkage (the genes are close together), whereas other gene pairs give different percentages. A RF of 1% indicates tight linkage, whereas a RF closer to 50% would indicate that the genes lie farther apart.

Autosomal traits also exhibit linkage; using the testcross

Transmission of sex-linked traits is easier to follow, since the genotypes of the F1 female chromosomes can be determined by examining her sons (they receive an X from their mother and a Y from their father). Autosomal traits are a bit harder to follow because all progeny receive two copies of autosomal genes. This is where the **test cross** comes in handy!

In this cross, we look at an autosomal gene for body color (*b+* is brown and recessive *b* is black) and a gene for wing shape (*c+* is straight and recessive *c* is curved).

P: black, straight female x brown, curved male

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b c+ / b c+ x b+ c / b+ c
F1: brown, straight progeny
All b c+ / b+ c
Test cross: brown, straight F1 female x black, curved male

b c+ / b+ c x b c / b c

Test cross progeny:			
black, straight	b c+ / b c	Parental type	2934
brown, curved	b+ c / b c	Parental type	2768
black, curved	b c / b c	Recombinant type	871
brown, straight	b+ c+ / b c	Recombinant type	846

The test cross shows that these autosomal genes exhibit linkage; they do not assort independently. Instead the parental types are transmitted together > 50% of the time.

Experimental recombination frequencies between two genes are never greater than 50%.

Recombinants among the F2 progeny are never in the majority. Genes on different chromosomes yield 50% recombination frequency because of independent assortment. Genes that lie far apart on the same chromosome also show 50%. The only way to tell for sure whether the two genes are on the same chromosome is to show definite linkage with other genes that lie in between them. How do we do that? By mapping.

Simple Gene Mapping problems and Answers:

Q. 1. How many different phenotypes can appear in the progeny, if an individual of genotype AaBbCcDd is test crossed?

a. 4
b. 8
c. 16
d. 64

Ans. 1: c. 16

Number of F2 phenotypes, assuming complete dominance = 2n, n = number of heterozygotes.

2. When a cross involving mutant genes b and e is done 3% recombinants are observed. When a cross involving mutant genes t and e is done 22% recombinants are observed. While a cross involving mutant genes b and t is done 19% recombinants are observed. What is the correct order of the genes on the chromosome?

a. t, b, e
b. b, t, e
c. t, e, b
d. b, e, t

Ans. 2: a. t, b, e

Since the frequency between t and e is largest they must lie far apart in the linkage map and frequency of b-t, b-e adds up to form t-e. Therefore, the correct order is t, b, e.

Q. 3. In a cross involving polygenic inheritance where three gene pairs control plant height, the shortest and tallest plants are 6 cm and 24 cm, respectively. What height should all FI display if homozygous 12- cm and 24-cm plants are crossed, assuming their environments are the same?

a. 20 cm
b. 43 cm
c. 6 cm
d. 18 cm

Ans. 3: d. 18 cm

FI offspring will all have the same genotype (1/2 of genome from each parent). Therefore, the genes coding for the extreme phenotypes are present in equal numbers. This results in a phenotype which is approximately halfway between the two extremes.

Q. 4. Two different traits affecting pod characteristics in garden pea plants are encoded by genes found on chromosome 5. Narrow pod is recessive to normal pod; yellow pod is recessive to green pod. A true breeding plant with narrow, green pods was crossed to a true breeding plant with normal, yellow pods. The FI offspring were then test crossed to plants with narrow, yellow pods. The following results were obtained- 144 narrow green pods, 150 normal yellow pods, 11 narrow yellow pods, 9 normal green pods. How far apart are these two genes?

a. 3.2 mu
b. 4.6 mu
c. 6.4 mu
d. None of these

4.Ans. 108: c. 6.4 mu

11 + 9/144+150+11+9 X 100 = 6.4 mu.

Two-Point cross mapping of gene

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Two-point crosses: comparisons help establish relative gene positions

Let's look at Sturtevant's undergraduate thesis data. Consider three X-linked genes y, m and w. By looking at two-point crosses (crosses tracing two genes at a time), he used the recombinant data to order the genes.

Recomb. Frequency	
y-w	1.1
w-m	32.8
m-y	34.3

Map

y---------- w ----------m

1.1 m.u. 32.8 m.u.

More data (work through the numbers to add v and r to the map and fill in the distances):

y-v	33		
y-r	42.9 (farthest apart)	v-m	4.0
w-v	32.1	v-r	24.1
w-r	42.1	m-r	17.8

MAP

y --- w ---------- v ------- m ---------- r

There are limitations to two-point crosses. With crosses involving only two genes at a time, it may be difficult to determine gene order if some of the gene pairs lie close together. See the smaller map above. In addition, the actual distances often do not match up. The distance between y-r in a two-point cross is 42.9 m.u., but the distance calculated by adding up all the intervening distances (y-w + w-v + v-m + m-r) is 55 m.u.

Three-point crosses: a faster, more accurate way to map genes

Example:

P: vg b pr/ vg b pr female x vg+ b+ pr+ / vg+ b+ pr+ male

F1: vg b pr / vg+ b+ pr+

Test cross: vg b pr/ vg+ b+ pr+ F1 female x vg b pr / vg b pr male

Test Cross Progeny:

1779	vg b pr	Parental type
1654	vg+ b+ pr+	
252	vg+ b pr	recombinant for vg relative to b and pr
241	vg b+ pr+	
131	vg+ b pr+	recombinant for b relative to vg and pr
118	vg b+ pr	
13	vg b pr+	recombinant for pr relative to vg and b
9	vg+ b+ pr	
4197		

distance between vg and b is: (252 + 241 + 131 + 118) / 4197 x 100 = 17.7 m.u. **(*** this is not correct - see below!)**

distance between vg and pr is (252 + 241 + 13 + 9) / 4197 x 100 = 12.3 m.u. distance between b and pr is (131 + 118 + 13 +9) / 4197 x 100 = 6.4 m.u.

The recombination frequencies show that vg and b are separated by the largest distance and therefore are the outside genes. The two smallest classes represent the double crossover events, in which pr alleles are recombined relative to vg and b. The arrangement of alleles in double recombinants indicates the relative order of the three genes, because in gametes formed by double crossovers, the gene whose alleles have recombined relative to the parental configuration must be the one in the middle.

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Map:

vg----------pr---------- b (stopped about here -- will continue next time)

12.3 6.4

But, again, the distance separating the outside genes vg and b (17.7 map units) is smaller than the sum of the two intervening distances (12.3 + 6.4 = 18.7). Why? Because the distance between vg and b that we calculated does not account for all the recombination events (double crossovers). Our three-point cross data lets us adjust the recombination frequency. How? We add the recombination frequency of the double crossovers twice, since each individual in the double crossover groups is a result of two exchanges between vg and b. DOUBLE CROSSOVERS are needed to generate gametes in which the middle gene is recombined relative to the two flanking genes. Thus when we calculate the genetic distance between the two outside markers, vg and b, we must add in the double crossovers twice.

***** CORRECTION TO ABOVE TO ACCOUNT FOR DOUBLE CROSSOVERS:**

distance between vg and b is (252 + 241 + 131 + 118 + 13 +13 + 9 + 9) / 4197 x 100 = 18.7 m.u.

A few additional points about mapping:

- Mapping reveals the relative order of genes, not the actual physical distance.
- The most accurate maps are made by summing the genetic distances of genes lying close together (small intervals). One needs to connect genes that lie far apart through the genes that lie between them.
- INTERFERENCE: A measure of the independence of crossovers from each other. (That is, does a crossover in one region affect the likelihood of a crossover in an adjacent region?)

Calculating interference: First of all, what is the probability of double crossovers occuring? Consider our example of vg, pr, and b linkage. We can calculate the probability of a double crossover using the Law of the Product rule. As long as a crossover in one region does not affect the probability of a crossover in another region, the probability of a double crossover is simply the product of their separate probabilities.

Probability of single crossover between vg and pr = 0.123 (corresponds to 12.3 map units) Probability of single crossover between pr and b = 0.064 (corresponds to 6.4 map units) Probability of both is 0.123 x 0.064 = 0.0079 or 0.79%

But the number actually observed was different: ([13 + 9]/4197) x 100 = 0.52%, meaning that the two crossovers are not occurring independently, but instead a crossover in one region reduces the likelihood of a crossover in an adjacent region.

Coefficient of coincidence = frequency observed / frequency expected = 0.52 / 0.79 = 0.66 (thus, in our example, only 66% of the expected DCOs took place).

Interference = 1 - coefficient of coincidence = 1- 0.66 = 0.34.

If interference is zero, this means that the frequency of double crossovers is as expected and that a crossover in one region occurs independently of a crossover in an adjacent region. If interference is 1, this means that interference is complete and that no double crossovers are observed because a crossover in one region completely blocks a crossover in an adjacent region.

Sample problems on 2-point cross

Problem 1

In the absence of tetrad analysis, some double crossovers can be detected by using threefactor crosses. Drosophila X-linked genes: yellow body = y; cut wings = ct; echinus eyes = ec Female heterozygous at all three loci y/+ ct/+ ec/+ X + + +. This is test cross if look at male progeny because male contributes Y with no genes to male offspring

phenotypes	gametes	# males
wild type	+ + +	1080
yellow cut echinus	y ct ec	1071
cut	+ ct +	293
yellow echinus	y + ec	282
yellow	y + +	78
echinus cut	+ ct ec	66
echinus	+ + ec	6
yellow cut	y ct +	4
		2880

Note that the genes are linked; if they weren't, we would have 8 phenotypes and 8 gamete genotypes in approximately equal numbers. Arranged in pairs of equal numbers, in order of magnitude. Which are parental genotypes? Which are double crossover genotypes?

Answer with Explanation:

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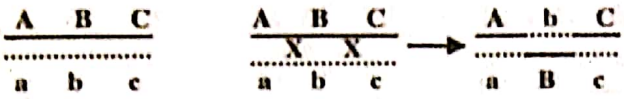
phenotypes	gametes	# males	
wild type	+ + +	1080	parental
yellow cut echinus	y ct ec	1071	parental
cut	+ ct +	293	
yellow echinus	y + ec	282	
yellow	y + +	78	
echinus cut	+ ct ec	66	
echinus	+ + ec	6	double x-over
yellow cut	y ct +	4	double x-over
		2880	

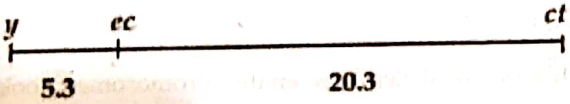
Find parental (most common) and double-crossover (least common) types. Middle gene is the one that is switched relative to the other two in doubles vs. parentals.



Which gene is in the middle: y, ct, or ec?
Analyze data as three two-factor crosses:

y - ct	ct - ec	y – ec
293	293	78
282	282	66
78	6	6
66	4	4
719	585	154
0.250	0.203	0.053

Longest distance is y - ct, so these are the outside markers on the map. Agrees with previous conclusion that ec is middle marker.



distance in map units

What is best estimate of distance between y and ct: 25.0 or 20.3 + 5.3 = 25.6?
25.0 includes only the single crossovers and omits the doubles. 6 + 4 = 10 doubles × 2 = 20 crossovers occurring in pairs.

Interference

Morgan's group first assumed x-overs occurred independently. Then found out that was wrong:
expected doubles if independent: P(double) = P (single y-ec) P (single ec-ct) = (0.053) (0.203) = 0.0108. Observed doubles = 10/2880 = 0.0035.
Or: Expected number of doubles = (0.0108) (2880) = 31;

observed number of doubles = 6 + 4 = 10. Can do statistics with numbers, not with frequencies. Observed < expected, therefore one crossover interferes with occurrence of another, but not completely.

This works in our favor, means that problem of double crossovers isn't quite as bad as it might be.

Also means that we can't predict the number of double crossovers exactly from the number of singles, without correcting for interference.

Interference doesn't always happen; mapping very large number of markers in rice showed negative interference over short distances. Maybe because some sites are hotspots for recombination? Recombination frequencies do vary along chromosomes and between chromosomes. Drosophila: No crossing-over in males.

Problem 2

If a multiply heterozygous individual is crossed to an individual homozygous at all loci, the relative positions of 3 loci can be mapped. (Cross to homozygote so gametic genotypes can be recognized):
Maize:

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l l—lazy or prostrate growth
g g--glossy leaves
s s--sugary endosperm
Order Unknown, SO L G S IS ARBITRARY ORDER
Triple Heterozygote: L G S / l g s X Recessive homozygote: l g s / l g s

Progeny Phenotype	Genotypes of offspring	Number
wildtype	L G S / l g s	286
lazy	l G S / l g s	33
glossy	L g S / l g s	59
sugary	L G s / l g s	4
lazy,glossy	l g S / l g s	2
lazy,sugary	l G s / l g s	44
glossy,sugary	L g s / l g s	40
lazy,glossy,sugary	l g s / l g s	272
Total		740

Q: Calculate the frequency of recombination between L and S, given that the parental classes are L G S and l g.
Q: Calculate the frequency of recombination between S and G, given that the parental classes are L G S and l g s.

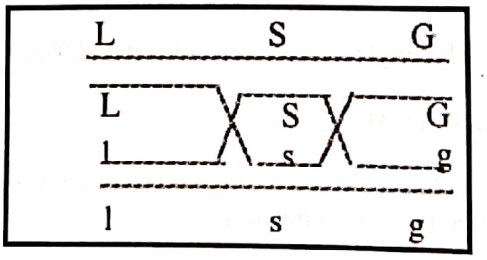
Answer with explanation:

Rule 1: The two most-frequent types of gametes are nonrecombinant. These provide the linkage phase (cis vs. trans) of the alleles in the heterozygous parent. This is because recombinant gametes can never account for more than 50% of the gametes, even if genes are unlinked.
Therefore: L G S and l g s are the nonrecombinant (parental) classes

Rule 2: The double-recombinant gametes will be the least frequent type.
Therefore: L G s and l g S are the double crossover classes

Rule 3: The effect of double crossovers is to interchange the members of the middle pair of alleles between the chromosomes. (look at figure above)
Therefore: S occurs between L and G, and the arrangement of the genes on the chromosome is:

Why? Because this is what double-crossovers do:



The gene in the middle gets shuffled
Single crossover between L and S loci===> L s g and l S G
Single crossover between S and G loci===> L S g and l s G

Can now label the genotypes with the processes that produced them in parents, L goes with S and l goes with s. So, any time these combinations get broken up --> recombination between L and S loci. In parents, S goes with G and s goes with g. So, any time these combinations get broken up --> recombination between S and G loci.

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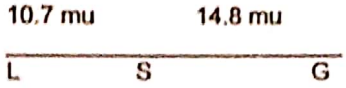
Progeny Phenotype	Genotypes of offspring	Number	Crossover or Non-X over?
wildtype	L G S / l g s	286	Parental
lazy	l G S / l g s	33	single Xover between L and S
glossy	L g S / l g s	59	single Xover between S and G
sugary	L G s / l g s	4	double Xover
lazy,glossy	l g S / l g s	2	double Xover
lazy,sugary	l G s / l g s	44	single Xover between S and G
glossy,sugary	L g s / l g s	40	single Xover between L and S
lazy,glossy,sugary	l g s / l g s	272	Parental
Total		740	

Rule 4: Reciprocal products expected to occur in approximately equal numbers, so LGS ≈ lgs, (286 ≈ 272) LgS ≈ lGs (59 ≈ 44), Lgs ≈ lGS (40 ≈ 33), and LGs ≈ lgS (4 ≈ 2).

Q: Calculate the frequency of recombination between L and S, given that the parental classes are L G S and l g s:

79/740 or 10.7% of gametes recombinant between L & S. So, map distance between L & S = 10.7 map units.

Q: Calculate the frequency of recombination between S and G, given that the parental classes are L G S and l g s:
109/740 or 14.8 % of gametes recombinant between S & G. So, map distance between S & G=14.8 map units
So, the data are given in the following map:



Rule 5: Don't forget to include the double recombinants when calculating recombination frequency!

Interference
Using information from 3-point mapping, we can determine if crossing over in two different regions of the same chromosome occur independently. If they are independent, crossing over in one region has no effect on the probability of crossing over in another region of the same chromosome. Assuming independence, the percentage of expected double crossovers is calculated by multiplying the probability of exchange in one region by probability of exchange in other (Product Rule).

MAIZE EXAMPLE:
Probability of recombination between L and S is 10.7% = 0.107, and that between S and G is 14.8% = 0.148.
Using the product rule (see Lecture 2): The probability of a detectable double crossover should then be 0.107 * 0.148 = 0.0158
So, in 740 events, the double crossover classes of gametes should occur 0.0158 * 740 = 12 times
So, the expected double crossovers between L and G should be about 12, but we observed only 6. Such deficiencies are common, and indicate that crossing over in one region of a chromosome reduces the probability of crossing over in closely adjacent regions. This phenomenon is called INTERFERENCE.
The coefficient of coincidence is the observed number of double recombinants divided by the expected number. It provides measure of the degree of interference.
Coefficient of coincidence = 6/12. Interference = 1 - (Coefficient of coincidence) = 1 - 0.5 = 0.5 in this case. = 1 - (6/12) = 1/2
Observed number of double crossovers is just 1/2 of that expected. At distances less than about 10 map units (in Drosophila) no double crossovers are observed (interference is complete, I=1). Conversely, at distances greater than about 45 map units, Interference disappears, and the observed number of double crossovers is usually nearly equal to the expected (interference is absent, I=0).

Problem 3

Assignment: Construct a genetic map using the following data:
A line of drosophila has been constructed with three recessive mutations:
sc = loss of thorax bristles ("scute")
ec = roughened eye surface ("echinus")
vg = vestigial wings
The dominant, wild-type version of each allele could be written as vg+ or ec+, but we will write them just as "+" for simplicity.

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we cross homozygous recessive flies *(sc ec vg / sc ec vg)* with homozygous dominant (wild-type) flies (+++ / +++).
All of the flies produced by the parental cross have the dominant (wild-type) phenotype. Genetically, these flies are: *(sc ec vg / +++)*
we produce an F2 generation by mating the F1 flies with homozygous recessive flies
(sc ec vg / sc ec vg) . This cross produces an F2 generation with 6 different phenotypes:

sc	ec	vg	417	*(parental combination of alleles)*
+	+	+	430	*(parental combination of alleles)*
sc	+	+	25	
+	ec	vg	29	
sc	ec	+	44	
+	+	vg	37	
sc	+	vg	4	
+	ec	+	5	
total flies in F2:			991	

From this genetic data determine the recombination frequencies between each pair of genes (there are 3 pairs: *sc/ec ec/vg sc/vg*). After you have calculated the recombination frequencies for each gene pair you should be able to construct a genetic map:

Answer With explanation

Data:					
	sc	ec	vg	417	*(parental combination of alleles)*
	+	+	+	430	*(parental combination of alleles)*
	sc	+	+	25	
	+	ec	vg	29	
	sc	ec	+	44	
	+	+	vg	37	
	sc	+	vg	4	
	+	ec	+	5	
	total flies in F2:			991	

• First, we figure out the degree of recombination between each of the 3 possible gene pairs. The key to getting this right is a correct identification of which phenotypes are "parental" and which are "recombinant." In this cross, this is pretty easy to do. Any combinations that include only sc, ec, or vg are parental. So are those with only + alleles.
All others are recombinant:

Recombination between sc & vg:
"recombinants" = 25+29+44+37 = 135
% recombination = $\frac{135}{991}$ = 13.6%

between sc & ec
"recombinants" = 25+29+4+5 = 63
% recombination = $\frac{63}{991}$ = 6.4%

between ec & vg
"recombinants" = 44+37+4+5 = 90
%recombination = $\frac{90}{991}$ = 9.1%

• The three recombination frequencies give us enough data to construct the map:

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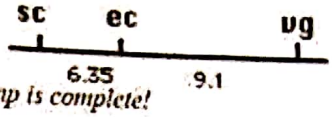
1) Most distant?? sc & vg
(total map distance = 13.6)

2) The intermediate gene is ec. It is closer to sc. So we locate it like this:



The genetic map is complete!

NOTE: The two rarest recombinant phenotypes are shown here:

sc	+	vg	4
+	ec	+	5

Practice problems

Problem No.1

You have a homozygous Drosophila line carrying the autosomal recessive genes a, b, and c linked in that order. You cross females of this line with males of a homozygous wild type line. You then cross the F1 males with their homozygous recessive sisters and you obtain the following F 2 phenotypes.

+ + + 1364, a + + 47, abc 1365, + b c 44, a b + 87, a + c 5, + + c 84, + b + 4.

What is the recombinant frequent between a, b & c? What is the coefficient of coincidence, interference and construct a linkage map of genes a b c .

Problem No.2

In corn white endosperm (p) is recessive to purple (P) and shrunken (f) is recessive to full (F) . A pure purple shrunken is crossed to a pure white full. The
F 1 is then test crossed and the offspring are as follows.
Purple shrunken – 1575, White shrunken – 58, Purple full – 60, White full – 1667. Calculate the distance between white and shrunken. Draw a chromosome map.

Problem No.3

A kidney bean shaped eye is produced by a recessive gene 'k' on the 3d chromosome of Drosophila. Orange eye colour called cardinal is produced by the recessive gene ' cd' on the same chromosome. Between these loci a third locus with recessive allele 'e' producing ebony body colour. Homozygous kidney cardinal females are mated to homozygous ebony males. The trihybrid females are then test crossed to produce the F 2.
Among the 4000 F 2 progeny are the following: Kidney cardinal- 1761 , Ebony – 1773, Kidney – 97, Ebony cardinal – 89, Kidney ebony- 128, Cardinal – 138, Kidney ebony cardinal -6, Wild type – 8.
Determine the map distance between the genes k, cd & e . Draw a chromosome map showing these genes. Which gene is in the centre and on which chromosome? Find out the coefficient of co-incidence and interference.

Problem No.4

In a three point test cross (C sh Wx / c Sh wx) x (c sh wx / c sh wx) the following data was obtained. C sh Wx = 2777, cSh wx = 2708, C Sh wx = 116, c shWx = 123, C sh wx = 643, cSh Wx = 626, C Sh Wx =4, cshwx = 3. Find the correct gene order. Calculate the recombination values and prepare a linkage map showing the relative distance and the linear order between the genes. Calculate the coefficient of coincidence and interference

Problem No.5

In Drosophilla , white eyes (w) , miniature wings (m) and forked bristles (f) are sex linked and recessive to the wild type character , red eyes, long wings, straight bristles. In a cross with wfm/wfm X + + + , the females are crossed with wfm males gave the following offspring.
White, forked, miniature- 25.8%
Red , straight, long – 25.8%
White straight long- 14.2 %
Red, forked , miniature- 14.2 %
White , straight, miniature- 7.7%
Red, forked, long – 7.7 %
White, forked , long- 2.3%
Red, straight, miniature 2.3%
Designate non-crossovers, single cross over and non- crossover classes.
Determine the percentage of crossing over between white ,forked and miniature. Calculate the coefficient of coincidence and interference.

Problem No.6

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In maize F1 plants from the cross of coloured, shrunken starchy X colourless, full, waxy were crossed with colourless shrunken waxy plant and the following progeny were obtained. Colourless, full, waxy - 2808: Coloured, shrunken, starchy- 2638: Coloured, full, waxy - 131`: Colourless, shrunken, starchy- 128: Coloured, shrunken, waxy- 631: Colourless, full, starchy - 656: Coloured, full, starchy- 5 : colourless, shrunken waxy - 3.
Construct a chromosome map of these genes.
Calculate the coefficient of coincidence interference.

Problem No.7
n Drosophila three genes f (forked), od (out stretched and g (garnet) are present in linkage group. Wild type females' heterozygous at all three loci were crossed to wild type males. The F1 reveals the following data. Females - All wild type
Males - 114 garnet outstretched; 838 garnet forked; 120 forked; 2 out stretched forked; 4 garnet ; 878 out stretched; 26 wild type; 18 out- stretched garnet forked. Total = 2000.
a) Which gene is in the middle? . b) Calculate the map distances. c) Calculate the strength of interference. & coincidence.

Problem No.8
Maize plants homozygous for the recessive gene variable sterile (va) exhibit irregular distribution of chromosomes during meiosis. Yellowish green seedlings are the result of another recessive gene called virescent (v). A third recessive called glossy (gl) produces shiny leaves. All three of these genes are linked. Two homozygous plants were crossed and produced an all-normal F1. When the F1 were test crossed, progeny phenotypes appeared as follows.
Virescent - 60 ; Virescent glossy -48; glossy -7; variable sterile, virescent, glossy- 270; Variable sterile, virescent - 4; Variable sterile- 40; Variable sterile glossy- 62; Wild type - 235.
a) What were the genotypes and phenotypes of the original parent?
b) Diagram the linkage relationship in the F1.
c) Determine the gene order.
d) Calculate the amount of recombination observed.
e) How much interference is operative? (Page 131 -Stansfield)

Answers with explanations:

Problem No.1
Answer :
Parentals - + + + =1364 & a b c =365 Gene order is same - + + +
a b c
Cross over between a & b = a + + = 47 & + b c =44
Cross over between b & c = a b + =87 & + + c =84,.
DCO = a + c = 5 & + b + 4.
% of crossing over between a & b =47 +44 + 9 X 100 / 3000 = 3.33%
% of crossing over between b & c =87 + 84 + 9 X 100 / 3000 = 6 %
% of expected DCO = 3.33 / 100 X 6 / 100 X 100 = 0.03 X 0.06 X 100 =0.18
Observed DCO = 9 X 100 / 3000 =0.03
Coincidence =% of observed crossing over / % of expected C.O =0.03/ 0.18 =0.17= 17 %
Interference = 1- 0.17 = 83%
We observed only 17% of DCOs that were expected on the basis of combining independent probabilities. (map distance). Thus 83 % of the expected DCOs did not form due to interference
Linkage map = a 3.33 % b 6% c

Problem No.2
Answer :
It is a two point test cross .
Parents of the cross = P f
p F
Parental - Purple shrunken =1575, White full - 1667 Recombinants - White Shrunken - 58, Purple full - 60 Total offspring - 3360
% of crossing over between p & f = 58 + 60 X 100 / 3360 = 3.5 %
Chromosome map = p 3.5 % f

Problem No.3
Answer :
Parents - Kidney cardinal- 1761 & Ebony - 1773
DCO = Kidney ebony cardinal -6, & Wild type - 8. In order to get the parentals and DCOs as mentioned above the gene order will be - Gene order - k + cd / + e +
Cross over between genes k & e = Kidney ebony- 128, &Cardinal - 138 Cross over between genes e & cd = Kidney - 97& Ebony cardinal - 89. DCO = Kidney ebony cardinal -6, & Wild type - 8 .

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% of crossing over between k & e = 128 +138 + 14 X 100 /4000 =7 %
% of crossing over between e & c d = 97 +89+14X 100 / 4000 = 5 %
% of Observed DCO + 14 X100 / 4000 = 0.35 %
Expected DCO = 7 / 100 X 5 / 100 X 100 =0.07 X 0.05 X 100 =0.35
Coincidence =% of observed crossing over / % of expected C.O =0.35 / 0.35 =1 Interference = 1- 1 = 0
All the expected DCOs are observed.
Linkage map = k 7% e 5% cd

Problem No.4
Answer ;
Parentals - C sh Wx = 2777 & c Sh wx = 2708 DCOs - C Sh Wx =4 & c sh wx = 3.
In majority of parental the gene order is seen as shown below and we will get the gene order of DCOs as shown in the question.
Gene order = C sh Wx
c Sh wx
Cross overs between c & sh = C Sh wx = 116 & c shWx = 123.
% of crossover between c & sh = 116 + 123 +7 X 100 /7000 =3.5 %
Cross overs between sh & wx = C sh wx = 643 & cSh Wx = 626
% of cross overs between sh & wx = 643 +626+ 7 X 100 / 7000 =18.2 %
% of observed DCO =7 X 100 / 7000 = 0.1
% of expected DCO = 3.5 /100 X 18.3 / 100 X 100 = 0.035 X 0.183 X100 =0.64 Coincidence =% of observed crossing over / % of expected C.O = 0.1/ 0.64 =0.156 = 16 % Interference = 1- 0.156 =0.84= 84 %
We observed only 16% of DCOs that were expected on the basis of combining independent probabilities. (map distance). Thus 84 % of the expected DCOs did not form due to interference
Linkage map = c 3.5% sh 18.2% wx

Problem No.5
Answer:
Gene order - W F M / w f m
Parental - White, forked, miniature- 25.8%
Red , straight, long - 25.8%
DCOs-
White, forked , long- 2.3%
Red, straight, miniature 2.3%
Single cross over between white & forked ,
White straight long- 14.2 %
Red, forked , miniature- 14.2 %
% of single cross over between white & forked = 33 %
Single cross over between forked& miniature .
White , straight, miniature- 7.7%
Red, forked, long - 7.7 %
% of single cross over between forked & miniature = 20 %
Observed DCO = 4.6 %
Expected DCO = 33 / 100 X 20 / 100 X 100 = 0.33 X 0.2 X 100 =6.6 %
Coincidence = % of observed DCO / % of expected DCO=4.6 / 6.6 X 100 = 70 % Coincidence + Interference = 1
Interference = 1- .7 = 0.3 = 30 %
We observed only 70% of DCOs that were expected on the basis of combining independent probabilities. (map distance). Thus 30 % of the expected DCOs did not form due to interference.

Problem No.6
Answer ;
Dominant characters -Coloured full starchy
Recessive characters - colourless shrunken waxy
Total offspring - 7000
Parentals - Colourless, full, waxy - 2808: Coloured, shrunken, starchy- 2638
c Sh wx = 2808 & C sh Wx = 2638
DCOs - Coloured, full, starchy- 5 & colourless, shrunken waxy - 3.
- C Sh Wx =5 & c sh wx = 3.
In majority of parental the gene order is seen as shown below and we will get the gene order of DCOs as shown in the question.
Gene order = C sh Wx c Sh wx
Cross overs between c & sh(Coloured, full, waxy - 131`: Colourless, shrunken, starchy- 128) C Sh wx = 131 & c shWx = 128.
% of crossover between c & sh = 131 + 128 +8 X 100 /7000 =3.81 %
Coloured, shrunken ,waxy- 631: Colourless, full, starchy - 656:
Cross overs between sh & wx = C sh wx = 631 & cSh Wx = 656

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% of cross overs between sh & wx = 631 +656+ 8 X 100 / 7000 =18.5 %
% of observed DCO =8 X 100 / 7000 = 0.114 %
% of expected DCO = 3.81/100 X 18.5 / 100 X 100 = 0.038 X 0.185 X100 =0.703 Coincidence =% of observed crossing over / % of expected C.O = 0.114/ 0.703 =0.16.
Coincidence + Interference = 1
Interference = 1- 0.16 = 0.84
We observed only 16 % of DCOs that were expected on the basis of combining independent probabilities. (map distance). Thus 84 % of the expected DCOs did not form due to interference

Linkage map = c 3.5% sh 18.2% wx

Problem No.7
Answer :
Parental : 838 garnet forked & 878 out stretched.
DCOs -2 out stretched forked & 4 garnet
Gene order - + + os / g f +
Single cross over
% of single cross over g & f = 114 + 120 + 6 X 100 / 2000 = 12 %
Single cross over between f & os -26 wild type, 18 out- stretched garnet forked. % of single cross over between f & os = 26 + 18 + 6 X 100 / 2000 =2.5 %
% of observed DCO = 6 X 100 / 2000 =0.3 = 30 %
Expected DCO = 12/100 X 2.5 / 100 X 100 =0.12 X 0.025 X100 = 0 .3 = 30% Coincidence = % of observed DCO / % of expected DCO = 0.3 / 0.3 = 1
Coincidence + Interference = 1
Interference = 1-1 = 0 (nil)
Linkage map = g 12 % f 2.5 % os

Problem No.8
Answer :
F1 are test crossed - $\frac{+\ +\ +}{va\ v\ gl}$ X $\frac{va\ v\ gl}{va\ v\ gl}$

Off spring - Virescent – 60 ; Virescent glossy -48;;; Variable sterile- 40; Variable sterile glossy- 62;.
Parentals - variable sterile, virescent, glossy- 270 & Wild type – 235
DCOs- glossy –7 & Variable sterile, virescent – 4 . In order to produce glossy Dco the gene order of the original parent may be + + + / va gl v . So the gene gl or glossy is in the centre.
Single cross over between va & gl
Variable sterile = 40
Virescent glossy –48
% of single cross over between va & gl = 40 + 48 + 11 X 100 / 726 =13.63 %
Single cross over between gl & v. Virescent – 60
Variable sterile glossy- 62;.
% of single cross over between gl & v.= 60 +62+11 X 100 / 726 =18.31 %
% of Observed DCO = 11 X 100 / 726 =1.5 %
Expected DCO = 13.63 /100 X 18.31 / 100 X 100 = 0.1363 X 0.1831 X100 =2.5 % Coincidence = % of observed DCO / % of expected DCO = 1.5 / 2.5 = 0.6 = 60 %
Coincidence + Interference = 1
Interference = 1- 0.6 = 0.40 = 40 %
We observed only 60% of DCOs that were expected on the basis of combining independent probabilities. (map distance). Thus 40 % of the expected DCOs did not form due to interference.

3. Pedigree analysis problems:

Pedigree analysis: How to solve problems?

Theory:

There are a number of different types of human pedigrees that you may encounter, however, there are only a few different modes of inheritance that you will need to be familiar. The following pedigrees show you different examples of human traits that can be traced through generations.

Inheritance of single gene defects.
A. Dominant and recessive as used in medical genetics.

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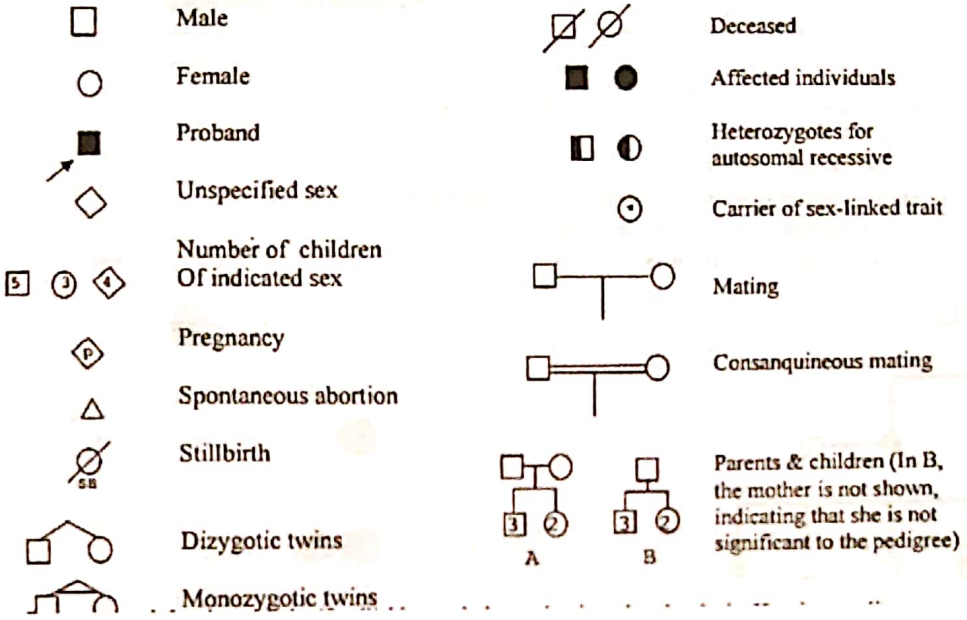
1. A trait is considered to be dominant when there is a clinical phenotype in the heterozygote. The phenotype is usually more severe in homozygotes. Example: aa is normal and Aa is affected.
2. A trait is considered to be recessive when there is a clinical phenotype only in individuals that are homozygous for the mutant allele. Example: AA and Aa are normal, aa is affected.
3. The presence and/or severity of "single gene disorders" may be modified by other genes and non-genetic factors.
4. Upper case letters are used for the dominant allele.
a. In dominant disorders this is the variant allele (the aa genotype is normal).
b. In recessive disorders this is the normal allele (the AA genotype is normal).

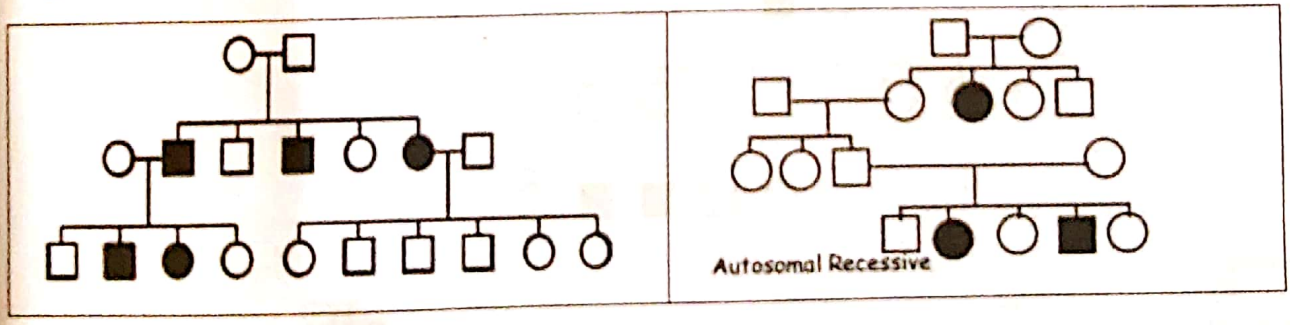
Symbols Of Pedigree chart



Rules of Inheritance

Autosomal Recessive

- Appears in both sexes with equal frequency
- Trait tend to skip generations
- Affected offspring are usually born to unaffected parents
- When both parents are heterozygote, approx. 1/4 of the progeny will be affected
- Appears more frequently among the children of consanguine marriages



Autosomal Dominant

- Appears in both sexes with equal frequency
- Both sexes transmit the trait to their offspring
- Does not skip generations
- Affected offspring must have an affected parent unless they possess a new mutation
- When one parent is affected (het.) and the other parent is unaffected, approx. 1/2 of the offspring will be affected
- Unaffected parents do not transmit the trait

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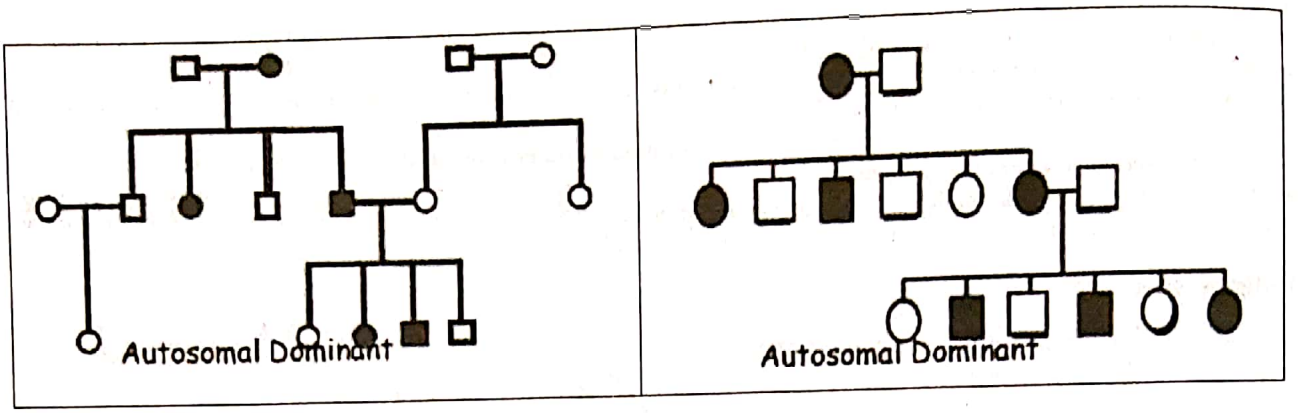
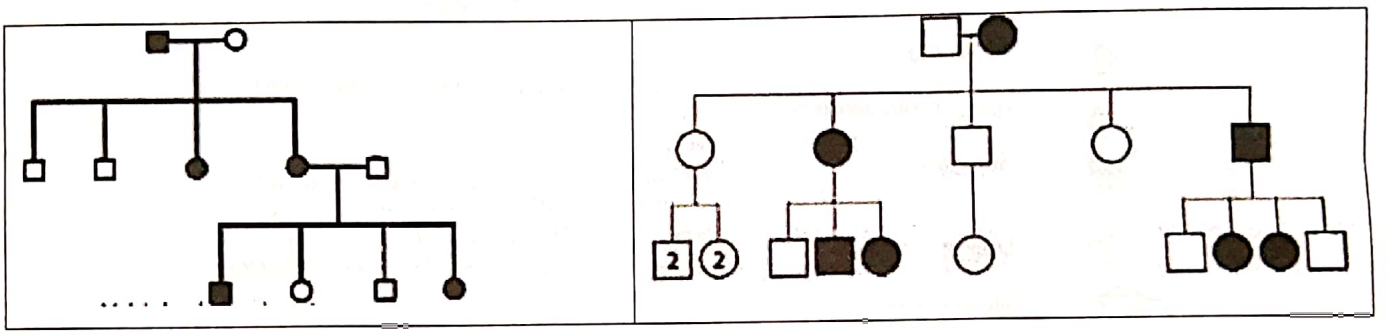
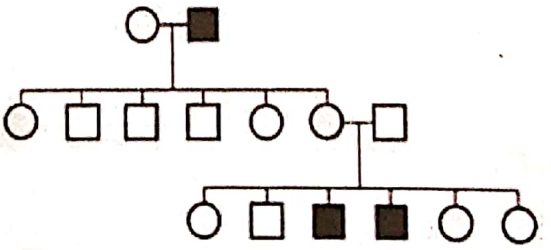


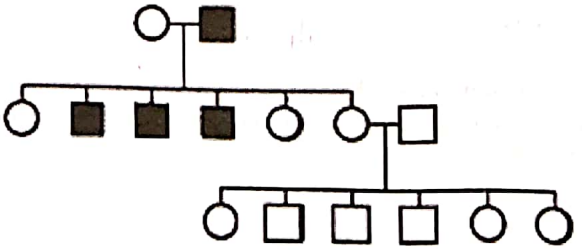
X-Linked Dominant
- Both males and females are affected; often more females than males are affected
- Does not skip generations.
- Affected sons must have an affected mother;
- affected daughters must have either an affected mother or an affected father
- Affected fathers will pass the trait on to all their daughters
- Affected mothers if heterozygous will pass the trait on to 1/2 of their sons and 1/2 of their daughters



X-Linked Recessive
- More males than females are affected
- Affected sons are usually born to unaffected mothers, thus the trait skips generations
- Approximately 1/2 of carrier mothers' sons are affected
- It is never passed from father to son
- All daughters of affected fathers are carriers



Y-Linked Dominant
- Only males are affected
- It is passed from father to all sons
- It does not skip generations



Mitochondrial
- Trait is inherited from mother only
- All children of a mother are at risk to be affected or carriers

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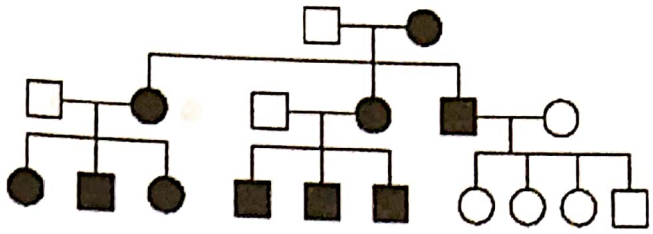


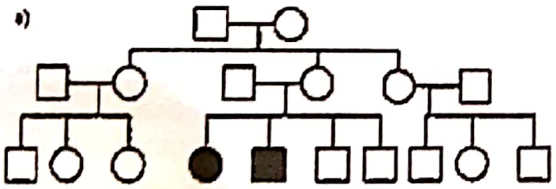
Summary of clues for solving Pedigree problems

Clues for Autosomal Inheritance	
Recessive • individual expressing trait has 2 normal parents • two affected parents can not have an unaffected child	Dominant • every affected person has at least one affected parent • each generation will have affected individuals
Clues for Sex-linked Inheritance	
Recessive • no father-to-son transmission • predominantly males affected • trait my skip generations	

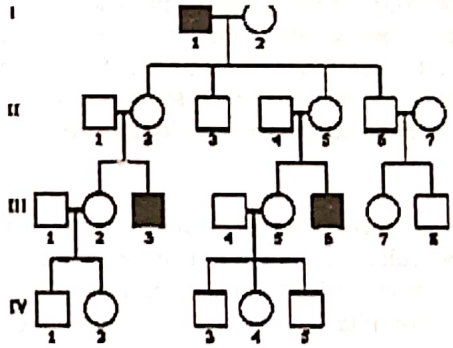
Practice problems

1. For each of the pedigrees below, identify the mode of inheritance and provide at least 2 reasons for your choice

a)



c)



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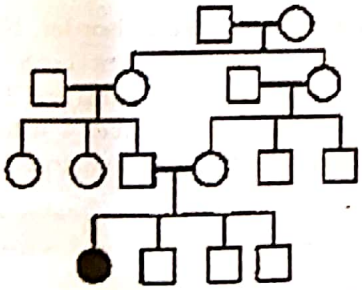
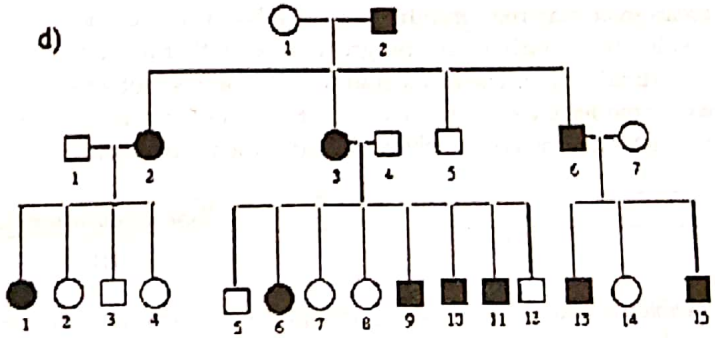


d)



2. Below is a pedigree for an inherited lung disease. Provide the genotypes of each of the individuals marked with lower case letters.

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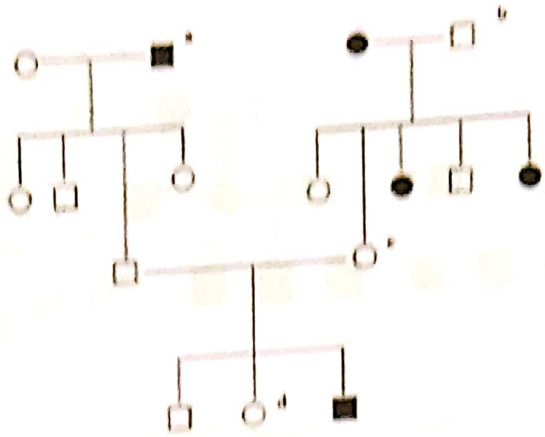
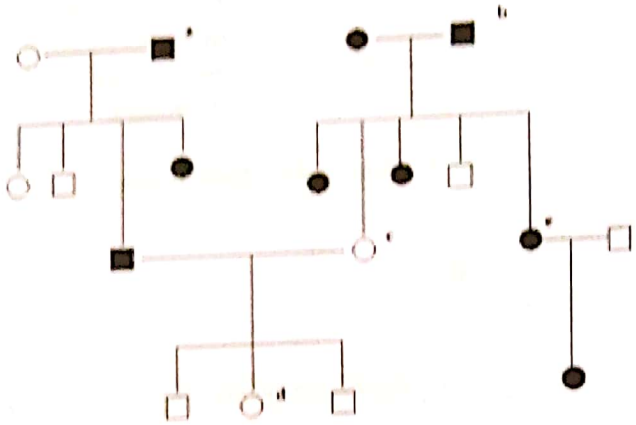


3. Below is a pedigree for an inherited brain disease. Provide the genotypes of each of the individuals marked with lower case letters.



4. Use the information provided below to create a pedigree. Then answer the question at the end of each description.

a. The ability to roll your tongue is dominant to not being able to roll your tongue.
Draw a pedigree to show the inheritance of this trait, given the following family history:
Grandpa Snow is a tongue roller but Grandma Snow is not. They have four children (2 sons and 2 daughters) who are all rollers. Their last daughter, Judy, married John Flake. John's parents are both rollers, but John's two sisters are non-rollers. John is a roller. John and Judy Snow-Flake have three children named Crystal Snow-Flake (a non-roller), Pretty Snow-Flake (a roller) and Jake Snow-Flake (a roller).

b. A man and woman marry. They have five children, 2 girls and 3 boys. The mother is a carrier of hemophilia, an X-linked disorder? She passes the gene on to two of the boys who died in childhood and one of the daughters is also a carrier. Both daughters marry men without hemophilia and have 3 children (2 boys and a girl). The carrier daughter has one son with hemophilia. One of the non-carrier daughter's sons marries a woman who is a carrier and they have twin daughters. What is the percent chance that each daughter will also be a carrier?

c. The great-great maternal grandmother of a boy was a carrier for colour-blindness, an X-linked disorder. His great uncle on his mother's side was colourblind but his great uncle's father was unaffected. The boy's mother has 2 brothers (1 colourblind, 1 unaffected) and 1 sister (unaffected). The boy's grandmother on his mother's side had 1 brother who was colourblind and 3 sisters. Two of these sisters were unaffected and one was a carrier. The boy's great grandmother on his mother's side had 4 sisters. The boy has one unaffected sister and he is colourblind. What is the probability of the boy's sons being colourblind if he marries a non-carrier?

Risk calculation in pedigree

Theory:

In these problems you examine family pedigrees displaying a rare disease trait and calculate the probability that various unaffected family members are "carriers," (that is, they carry a single disease allele).
This kind of analysis is important in genetic counseling. For example, if a male and female are both unaffected by a genetic disease trait, but are both are carriers, then for each of their children there is 1 chance in 4 that the child will be affected by the disease.
This figure displays a typical pedigree, in which a single individual is affected by a genetic disease. In each problem, the first task is to decide if the genetic trait is:

- *dominant or recessive*
- *autosomal or X-linked.*

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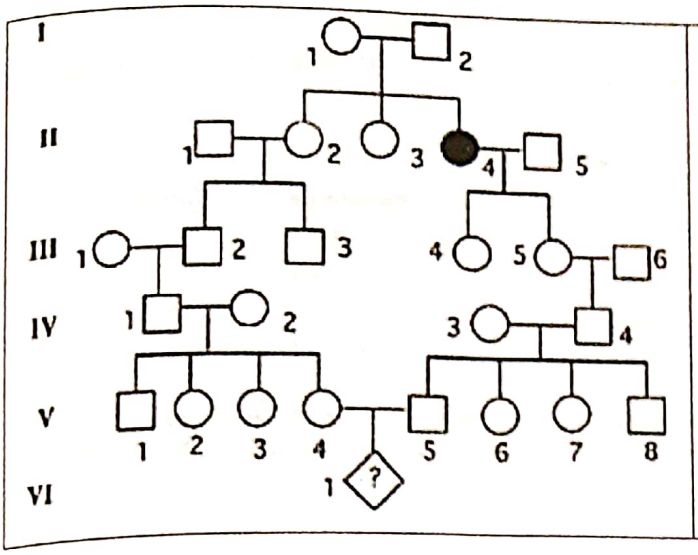

The second task is to decide the probabilities that the founding parents, I-1 and I-2, are carriers, and the probabilities that various unaffected descendants are unaffected, including V-4 and V-5.
The third task is to determine the probability that an individual VI-1, whose phenotype has not been observed, will be affected by the disease trait.
(The diamond means the gender is not specified and the question mark "?" indicates the phenotype with respect to the disease trait is not known.

In these Cognitive Tutor problems, we make three simplifying assumptions:

1. **Complete Penetrance.** If an individual represented in the pedigree carries one dominant allele or two recessive alleles for a trait, then that individual will be affected (will express the phenotype associated with the trait).
2. **Rare-in-Population.** In each problem, the trait in question is rare in the general population. **Assume for the purposes of these problems that individuals who marry into the pedigree in the 2nd, 3rd and 4th generations are not carriers.** This does not apply to the founders - the individuals at the top of the pedigree - either or both of whom could be carriers.
3. **Not-Y-Linked.** The causative genes in these problems may be autosomal or X-Linked, but are not Y-linked. **Determining Linkage**

The first important step in analyzing carrier probabilities is to determine if the pedigree is autosomal recessive or x- linked recessive, since the two different modes of transmission afford different inferences.

• The disease in the pedigree on the left must be autosomal recessive trait, since the affected daughter has two unaffected parents. (If the trait were dominant, one or the other parent would have to be affected. If the trait were X-linked recessive, her father would have to be affected.)

Autosomal Recessive Traits

A few basic facts about autosomal recessive traits will allow us to calculate the probability that unaffected individuals are carriers:

- An affected individual must be homozygous recessive and must pass one disease allele to each one of its children.
- An affected child must inherit one disease allele from its mother and one from its father.
- A parent with a single disease allele has a probability of % of passing the allele to any one of its children.

Calculating Probabilities

- The probability that an individual is a carrier is a number between 0 and 1.
- If we know an unaffected individual must be a carrier, then the probability is 1 that the individual is a carrier.
- If we know an unaffected individual cannot be a carrier, then the probability is 0 that the individual is a carrier.
- If we know that one parent is a carrier (p=1) and the other parent is not (p=0), we know there are two equally likely possibilities:
  - If the child inherits the first parent's single disease allele, the child will be a carrier.
  - If the child inherits the first parent's other allele, the child will not be a carrier.

Since there are a total of two equally likely outcomes, and in one of those two cases the child is a carrier, the probability that the child is a carrier is % or 0.5

Calculating the Probability an Unaffected Child is a Carrier When One Parent is Not a Carrier.

If one parent is not a carrier, then a child can only inherit a disease allele from the other parent. In these problems, we can assume that any individual marrying into the family is not a carrier.

As we have just discussed, if the other parent is definitely a carrier, then the probability of passing that allele to a child is % and so the probability the child is a carrier is %.

But suppose that the probability that the other parent is a carrier is only % instead of 1. The probability the child will be a carrier is % * % = %. Here's why:

- The probability the parent is a carrier is %.
- But if the parent is a carrier, the probability of passing the allele to a child is still just %.
- The probability that the parent is a carrier AND that the parent passes the allele is % times % = %.

In general, if one parent is not a carrier, the probability that a child will be carrier is:

% times (the probability the other parent is a carrier).

That is, we multiply the probability of passing a disease allele, %, times the probability that the parent does, in fact, carry the disease allele.

Calculating the Probability an Unaffected Child is a Carrier When Both Parents are Carriers.

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Finally, let's consider one more situation. Suppose both parents are carriers. What is the probability that a child who is unaffected is a carrier?
If a carrier mother and a carrier father have a child, there are four possible outcomes which are equally likely. (The probability of each outcome is %).

Inherit trait From mother	Inherit trait From father	Phenotype	Carrier?
NO	NO	Unaffected	NO
YES	NO	Unaffected	YES
NO	YES	Unaffected	YES
YES	YES	AFFECTED	NO

But if the child is **unaffected**, we can rule out the fourth outcome, in which the child inherits an allele from both parents. Given the three equally likely remaining events, what is the probability that if both parents are carriers, an UNAFFECTED child will be a carrier?

X-Linked Traits

Reasoning about carriers of x-linked recessive traits is a little more complicated than reasoning about autosomal recessive traits, but again a few basic facts will allow us to calculate probabilities.

- An affected female must be homozygous recessive and must pass one disease allele to each one of her children.
- Males are hemizygous for an X-Linked trait. An affected male must have a single disease allele.
- An affected male must have inherited the disease allele from his mother.
- An affected male must pass the disease allele to his daughter.
- A male never passes an x-linked trait to a son.
- An unaffected male cannot be a carrier and cannot transmit a disease allele to any child.

Now Try to solve probability problems from pedigree charts

Pedigree problems with probability problems and solutions

1. A man with red-green colour blindness is married to a girl with normal colour vision. What is the probability that his children will be colour blind?

Answer: Red-green colour blindness is an X-linked recessive condition. All the offspring of affected males would have normal colour vision, but all their daughters would be carriers in the very rare event of his partner being a carrier of red-green colour blindness, half their offspring, whether male or female, would be colour blind, half their daughters would be carriers and half their sons would be normal.
These results are summarized in the two following Punnett squares:
Xb - Gene for X-linked color blindness; X = normal gene
1. Affected male - genotypically normal female

	X	X	
X^b	X^b X	X^b X	All daughters are carriers
Y	XY	XY	All sons normal

2. Affected male - carrier female (Very rare)

	X^b	X	
X^b	X^b X^b	X^b X	Half daughters colour blind; half carriers
Y	X^b Y	XY	Half sons colour blind; half normal

2. A boy has Duchenne muscular dystrophy. His maternal uncle had died of muscular dystrophy at the age of 21 years. What is the probability that this child's sibs will be affected?

Answer: This boy has an X-lined recessive condition. The family history on the maternal side indicates that his mother is an obligate carrier. Consequently, half his brothers would be similarly affected and half would be normal. All his sisters would be unaffected, but half of them would be carriers. These results can be observed in the Punnett square below.
X^d - gene for Duchenne Muscular dystrophy; X - normal allele

1. Carrier female - Normal male

	X^d	X	
X	X^d X	X X	All sisters normal; half of them carriers
Y	X^d Y	XY	Half his brothers affected; half normal

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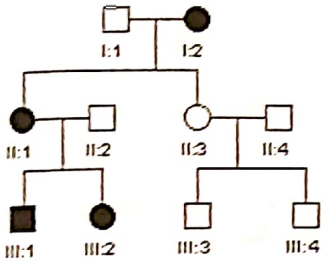
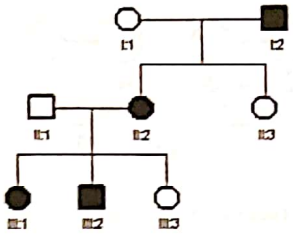
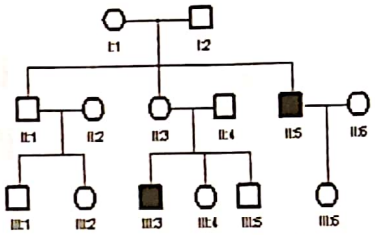
3. Explain, giving reasons, whether the following pedigrees are compatible with autosomal dominant, autosomal recessive or X-linked dominant and X-linked recessive inheritance. (Note that a pedigree may be compatible with more than one type of inheritance.







Answers with explanations:

a. This pedigree shows dominant inheritance because of linear transmission from parent to offspring. It could be either autosomal dominant or X-linked dominant

b. This is very similar to the above - autosomal dominant or X-linked dominant inheritance

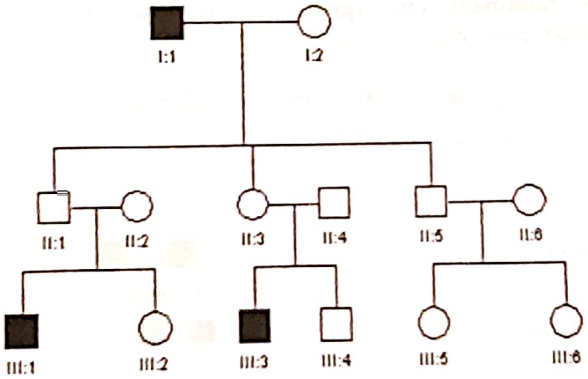
c. This pedigree is consistent with X-linked recessive inheritance because:

i. only males are affected,

ii. transmission is always through a carrier female, and there is no male to male transmission

Another possibility is autosomal dominant inheritance with decreased penetrance, thus appearing to skip generations.

4. Examine the pedigree from a family with a genetic disease and answer the questions below:



a. Does this pedigree indicate autosomal dominant, recessive or sex-linked type of inheritance? Give reasons for your choice.

b. Assuming that B and b are the normal and mutant alleles respectively, what would be the genotypes of the individuals: II.1, II.2 and III.3?

c. Individual II.3 requested genetic counselling. What is the probability that her child would be affected? Explain why.

a. Does this pedigree indicate autosomal dominant, recessive or sex-linked type of inheritance? Give reasons for your choice.

Answer: This pedigree is consistent with autosomal dominant inheritance with reduced penetrance. X-linked inheritance is excluded because of male to male transmission (II:1 to III:1).

b. Assuming that B and b are the normal and mutant alleles respectively, what would be the genotypes of the individuals: II.1, II.2 and III.3?

Answer:

II:1 - Bb. This individual, although unaffected by the disease is obviously transmitting the disease

II:2 - bb. This individual is normal and unrelated to the affected family

III:3 - Bb. Affected individual.

c. Individual II.3 requested genetic counselling. What is the probability that her child would be affected. Explain why.

Answer:

II:3 is obviously transmitting the disease although she is unaffected. The chance that her children would be heterozygotes is 50%. However, only some of these would be affected, depending on the penetrance. In the above pedigree, three out of five heterozygotes were affected, equal to 60%. This is only a rough measure of the penetrance because of the small number of individuals involved. Thus II:3 has a risk of 60% of 50% (=30%) that her offspring would be affected, and 20% risk that her normal offspring are in fact carriers of the abnormal gene.

5. Risk Calculation problem

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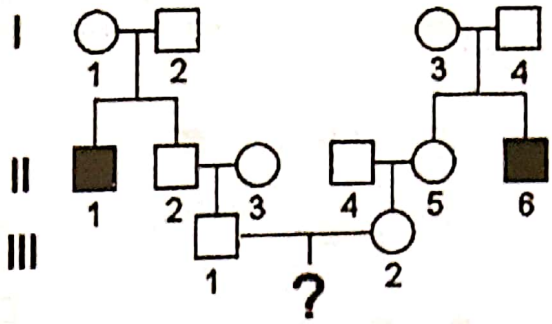


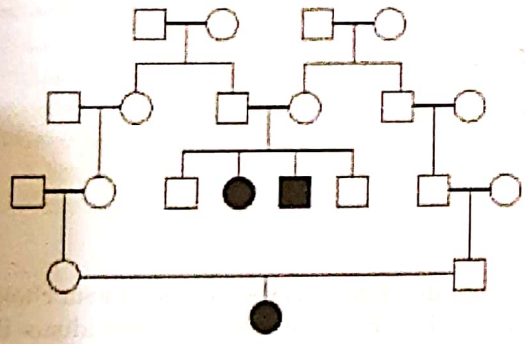
The pedigree shows the occurrence of an **autosomal recessive** trait, where the black squares have genotype **aa**. We wish to calculate the probability that **IV-1** (shown as?) will be either affected (**aa**), or a carrier heterozygote (**Aa**).

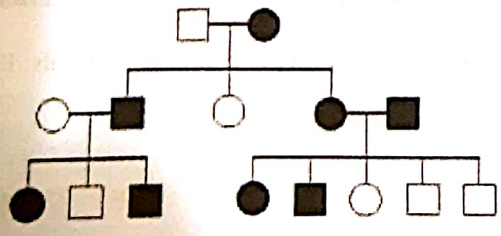
(1) For **IV-1** to be an affected **recessive homozygote**, s/he must inherit an **a** allele from the father **and the** mother. Given that **II-1** must be **aa**, both great-grandparents (**I-1** and **I-2**) must be **Aa**. **II-2** shows the dominant phenotype, and therefore has at least one **A** allele: the probability that the other is **a** is 1/2. **II-3** is from outside the affected pedigree and can be assumed to be **AA**. Like his father, **III-1** shows the dominant phenotype, and therefore has at least one **A**. Then, the probability that **III-1** is **Aa** is the probability that **II-2** is heterozygous **and passed** the **a** allele to **III-1**: (1/2) x (1/2) = 1/4. The same reasoning leads to the conclusion that **III-2** is heterozygous with a probability of 1/4. Thus, for **IV-1** to be **aa**, both parents must be **Aa, and they** must both pass the **a** allele to their offspring: 1/4 x 1/4 x 1/4 = **1/64**

(2) Alternatively, for **IV-1** to be a **heterozygous**, carrier, either s/he most inherit an **a** allele from the father, or from the mother. The chance of either parent being a heterozygote is 1/4, as calculated above. Then, the probability that both parents are heterozygotes, and the probability that two heterozygotes will have a heterozygous child, is 1/4 x 1/4 x 1/2 = **1/32**

(3) Finally, the probability that **IV-1** is a **dominant homozygote** is 1 - **1/64 - 1/32** = (64 - 1 - 2)/64 = **61/64.** This can also be calculated more tediously by summing the alternative probabilities at each of the steps above.

Practice problems and solutions


a.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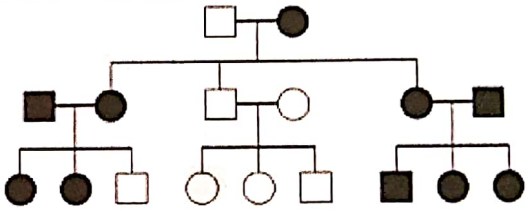
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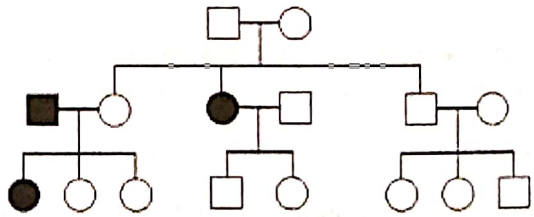
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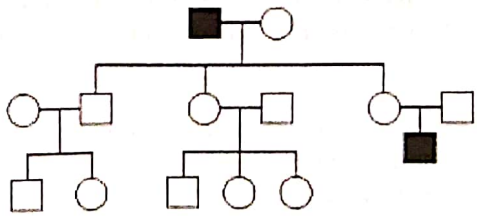
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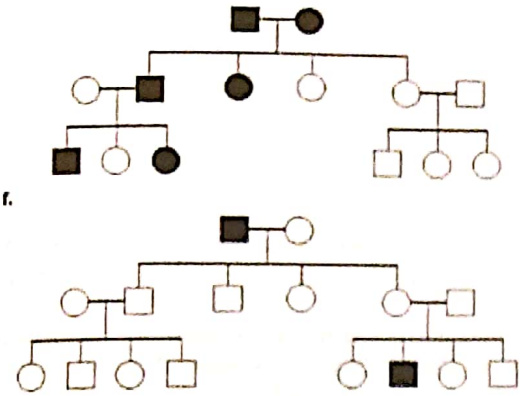

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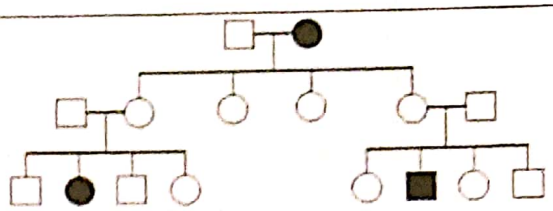


h.

Pedigree Practice Problems: Identify each pedigree as autosomal recessive, autosomal dominant, X-linked, or Y-linked

Answers:

a. autosomal recessive
b. autosomal dominant
c. autosomal dominant
d. autosomal recessive
e. x-linked recessive
f. autosomal dominant
g. autosomal recessive

4. Hfr Mapping or Bacterial Gene Mapping problems

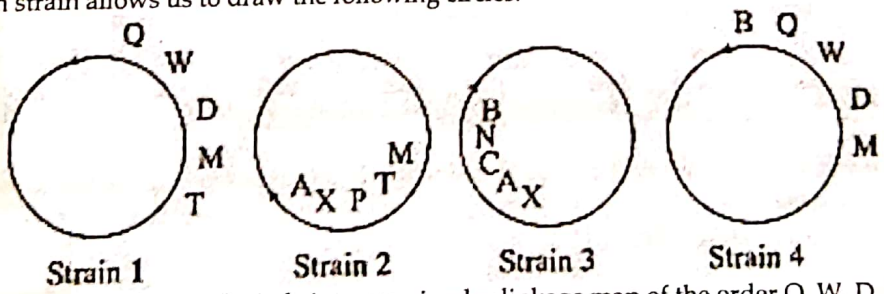
Sample problems and solution

1. In *E. coli*, four Hfr strains donate the following genetic markers shown in the order donated:

Strain 1:	Q	W	D	M	T
Strain 2:	A	X	P	T	M
Strain 3:	B	N	C	A	X
Strain 4:	B	Q	W	D	M

All of these Hfr strains are derived from the same F^+ strain. What is the order of these markers on the circular chromosome of the original F^+?

Answer: Recall the two-step approach that works well: (1) deter-mine the underlying principle, and (2) draw a diagram. Here the principle is clearly that each Hfr strain donates genetic markers from a fixed point on the circular chromosome and that the earliest markers are donated with the highest frequency. Because not all markers are donated by each Hfr, only the early markers must be donated for each Hfr. Each strain allows us to draw the following circles:



From this information, we can consolidate each circle into one circular linkage map of the order Q, W, D, M, T, P, X, A, C, N, B, Q.

2. In an Hfr × F^- cross, *leu*$^+$ enters as the first marker, but the order of the other markers is unknown. If the Hfr is wild type and the F^- is auxotrophic for each marker in question, what is the order of the markers in a cross where *leu*$^+$recombinants are selected if 27 percent are *ile*$^+$, 13 percent are *mal*$^+$, 82 percent are *thr*$^+$, and 1 percent are *trp*$^+$?

Answer: Recall that spontaneous breakage creates a natural gradient of transfer, which makes it less and less likely for a recipient to receive later and later markers. Because we have selected for the earliest marker in this cross, the frequency of recombinants is a function of the order of entry for each marker. Therefore, we can immediately determine the order of the genetic markers simply by looking at the percentage of recombinants for any marker among the *leu*$^+$ recombinants. Because the inheritance of *thr*$^+$ is the highest, this must be the first marker to enter after *leu*. The complete order is *leu, thr, ile, mal, trp*.

3. A cross is made between an Hfr that is *met*$^+$ *thi*$^+$ *pur*$^+$ and an F^- that is *met*$^-$ *thi*$^-$ *pur*$^-$. Interrupted-mating studies show that *met*$^+$ enters the recipient last, so *met*$^+$ recombinants are selected on a medium containing supplement that satisfy only the *pur* and *thi* requirements. These recombinants are tested for the presence of the *thi*$^+$ and *pur*$^+$ alleles. The following numbers of individuals are found for each genotype:

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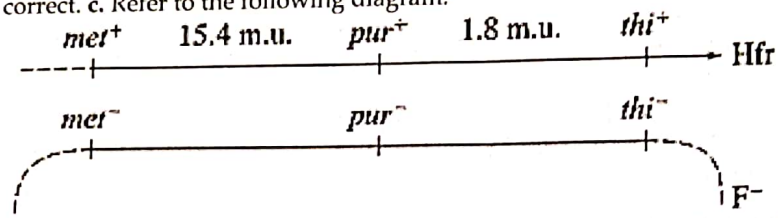
met^+ thi^+ pur^+	280
met^+ thi^+ pur^-	0
met^+ thi^- pur^+	6
met^+ thi^- pur^-	52

a. Why was methionine (Met) left out of the selection medium?
b. What is the gene order?
c. What are the map distances in recombination units?

Answer:

a. Methionine was left out of the medium to allow selection for *met*$^+$ recombinants, because *met*$^+$ is the last marker to enter the recipient. The selection for *met*$^+$ ensures that all the loci that we are considering in the cross will have already entered each recombinant that we analyze.

b. Here it is helpful to diagram the possible gene orders. Because we know that *met* enters the recipient last, there are only two possible gene orders if the first marker enters on the right: *met, thi, pur* or *met, pur, thi*. How can we distinguish between these two orders? Fortunately, one of the four possible classes of recombinants require two additional crossovers. Each possible order predicts a different class that arises by four crossovers rather than two. For instance, if the order were *met, thi, pur*, then *met*$^+$ *thi*$^-$ *pur*$^+$ recombinants would be very rare. On the other hand, if the order were *met, pur, thi*, then the four-crossover class would be *met*$^+$ *pur*$^-$ *thi*$^+$. From the information given in the table, it is clear that the *met*$^+$ *pur*$^-$ *thi*$^+$ class is the four-crossover class and therefore that the gene order *met, pur, thi* is correct. c. Refer to the following diagram:



To compute the distance between *met* and *pur*, we compute the percentage of *met*$^+$ *pur*$^-$ *thi*$^-$, which is 52/388 = 15.4 m.u. The distance between *pur* and *thi* is, similarly, 6/338 = 1.8 m.u.

4. Compare the mechanism of transfer and inheritance of the *lac*$^+$ genes in crosses with Hfr, F$^+$, and F'-*lac*$^+$ strains. How would an F$^-$ cell that cannot undergo normal homologous recombination (*rec*$^-$) behave in crosses with each of these three strains? Would the cell be able to inherit the *lac*$^+$ genes?

Answer: Each of these three strains donates genes by conjugation. In the Hfr and F$^+$ strains, the *lac*$^+$ genes on the host chromosome are donated. In the Hfr strain, the F factor is integrated into the chromosome in every cell, so efficient donation of chromosomal markers can occur, particularly if the marker is near the integration site of F and is donated early. The F$^+$ cell population contains a small percentage of Hfr cells, in which F is integrated into the chromosome. These cells are responsible for the gene transfer displayed by cultures of F$^+$ cells. In the Hfr- and F$^+$-mediated gene transfer, inheritance requires the incorporation of a transferred fragment by recombination (recall that two crossovers are needed) into the F$^-$ chromosome. Therefore, an F$^-$ strain that cannot undergo recombination cannot inherit donor chromosomal markers even though they are transferred by Hfr strains or Hfr cells in F$^+$ strains. The fragment cannot be incorporated into the chromosome by recombination. Because these fragments do not possess the ability to replicate within the F$^-$ cell, they are rapidly diluted out during cell division.

Unlike Hfr cells, F' cells transfer genes carried on the F' factor, a process that does not require chromosome transfer. In this case, the *lac*$^+$ genes are linked to the F' and are transferred with the F' at a high efficiency. In the F$^-$cell, no recombination is required, because the F' *lac*$^+$ strain can replicate and be maintained in the dividing F$^-$ cell population. Therefore, the *lac*$^+$ genes are inherited even in a *rec*$^-$ strain.

Practice problems and solutions

1. A Hfr strain possessing the markers *his*$^+$ *met*$^+$ *tyr*$^+$ *str*^{s} was mated with a F$^-$ strain possessing the markers *his*$^-$ *met*$^-$ *tyr*$^-$ *str*^{r}. The time of entry for each marker is shown below.

Donor Marker	Time (min)
his$^+$	21
met$^+$	12
tyr$^+$	46

a. Based on the results draw a map of the Hfr chromosome. Indicate the position of the F element, and the first marker to be transferred.

b. On what type of plates were cells grown to determine the above information?

2. In three different interrupted mating experiments, you obtain the following data for the time of entry of the following genes: *arg, met, leu, ser, gal, val,* and *phe*. All of these experiments were conducted using Hfr strains (Hfr 1, Hfr 2, and Hfr3) derived from the same original strain of *E. coli*.

Experiment 1 (Hfr 1)		Experiment 2 (Hfr 2)		Experiment 3 (Hfr 3)	
Met	10 min	Gal	5 min	Leu	5 min
Val	25 min	Ser	10 min	Val	10 min
Leu	30 min	Leu	35 min	Met	25 min

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met$^+$ *thi*$^+$ *pur*$^+$	280
met$^+$ *thi*$^+$ *pur*$^-$	0
met$^+$ *thi*$^-$ *pur*$^+$	6
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a. Why was methionine (Met) left out of the selection medium?
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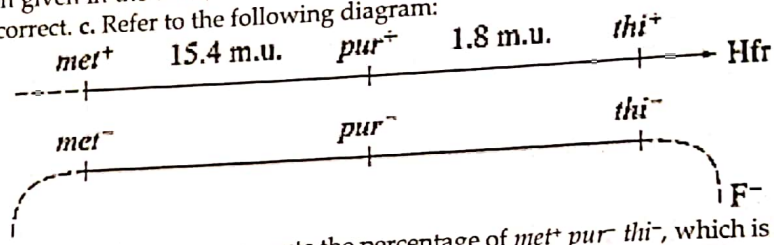

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4. Compare the mechanism of transfer and inheritance of the *lac*$^+$ genes in crosses with Hfr, F$^+$, and F'-*lac*$^+$ strains. How would an F$^-$ cell that cannot undergo normal homologous recombination (*rec*$^-$) behave in crosses with each of these three strains? Would the cell be able to inherit the *lac*$^+$ genes?
Answer: Each of these three strains donates genes by conjugation. In the Hfr and F$^+$ strains, the *lac*$^+$ genes on the host chromosome are donated. In the Hfr strain, the F factor is integrated into the chromosome in every cell, so efficient donation of chromosomal markers can occur, particularly if the marker is near the integration site of F and is donated early. The F$^+$ cell population contains a small percentage of Hfr cells, in which F is integrated into the chromosome. These cells are responsible for the gene transfer displayed by cultures of F$^+$ cells. In the Hfr- and F$^+$-mediated gene transfer, inheritance requires the incorporation of a transferred fragment by recombination (recall that two crossovers are needed) into the F$^-$ chromosome. Therefore, an F$^-$ strain that cannot undergo recombination cannot inherit donor chromosomal markers even though they are transferred by Hfr strains or Hfr cells in F$^+$ strains. The fragment cannot be incorporated into the chromosome by recombination. Because these fragments do not possess the ability to replicate within the F$^-$ cell, they are rapidly diluted out during cell division.
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Practice problems and solutions

1. A Hfr strain possessing the markers *his*$^+$ *met*$^+$ *tyr*$^+$ *str*^{s} was mated with a F- strain possessing the markers *his*$^-$ *met*$^-$ *tyr*$^-$ *str*^{r}. The time of entry for each marker is shown below.

Donor Marker	Time (min)
his$^+$	21
met$^+$	12
tyr$^+$	46

a. Based on the results draw a map of the Hfr chromosome. Indicate the position of the F element, and the first marker to be transferred.
b. On what type of plates were cells grown to determine the above information?
2. In three different interrupted mating experiments, you obtain the following data for the time of entry of the following genes: *arg, met, leu, ser, gal, val,* and *phe*. All of these experiments were conducted using Hfr strains (Hfr 1, Hfr 2, and Hfr3) derived from the same original strain of *E. coli*.

Experiment 1 (Hfr 1)		Experiment 2 (Hfr 2)		Experiment 3 (Hfr 3)	
Met	10 min	Gal	5 min	Leu	5 min
Val	25 min	Ser	10 min	Val	10 min
Leu	30 min	Leu	35 min	Met	25 min

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Arg	40 min	Val	40 min	Phe	55 min
Ser	55 min	Met	55 min	Gal	75 min
Gal	60 min	Phe	85 min	Ser	80 min

a. Determine the order of these genes on the bacterial chromosome and draw a single map showing the position of each of these genes on the bacterial chromosome. Based on the time intervals above, indicate the map distances between each gene, expressed in minutes. Also show the origin of transfer and direction of transfer for each Hfr.
b. If the crosses were set up as described in class, what are the genotypes of the donor and recipient cells?
3. In three different interrupted mating experiments, you obtain the data shown below for the time of entry. All experiments use Hfr strains derived from the same original parent strain.

Hfr1		Hfr2		Hfr3	
miy	10 min	sow	5 min	mic	5 min
sow	20 min	miy	15 min	tod	20 min
raj	35 min	chr	30 min	nub	30 min
nub	40 min	sha	40 min	raj	35 min
tod	50 min	mic	50 min	sow	50 min
mic	65 min	tod	65 min	miy	60 min
sha	75 min	nub	75 min	chr	75 min

a. Show the genetic map for each of these genes on the circular chromosome. Indicate the location of each origin and the direction of transfer. Doesn't need to be drawn to scale.
b. What are the genotype of the donor and recipient strains?
4. The exotic tropical tree frog *J. hainae* normally has 13 pairs of chromosomes (i.e. 2n = 26). You find a number of unusual individuals with the chromosome content shown below. Please fill in the blanks for each of the variants in the table below:
5. In an interrupted mating mapping experiment, you obtain the results shown in the graph below. The experiment was conducted as described in class.
a. Show the genetic map for each of these genes on the circular chromosome. Indicate the location of the origin and the direction of transfer. Doesn't need to be drawn to scale.
b. What are the genotypes of the donor and recipient strains?
c. For each gene, the number of recombinants increases with time until they reach a maximum. Why does the number of recombinants for a given gene increase with time?
d. Why does the number of recombinants reach a maximum?
6. Compare and contrast the three modes of bacterial DNA transfer.
7. In a transduction mapping experiment, you get the following results.

experiment	selected marker	unselected markers
1	a^+	15% b^+, 30% c^+
2	b^+	30% c^+, 15% a^+

a. Draw the relative position of *a*, *b* and *c*.
b. *a*, *b* and *c* are enzymes required for the synthesis of A, B and C, respectively. A mutant can't grow without A in the media, etc. How are the above recombinants identified?
8. In studying a particular animal, you find that the fourth chromosome is unusual. The two copies of chromosome 4 are shown. The normal chromosome is shown on the top, and the unusual one is shown on the bottom. The position of seven genes is shown.
a. What is the specific name of the chromosomal alteration that generated the abnormal chromosome?
b. Show the two chromosomes synapsed in meiosis. Please include the genes.
c. What effect will the rearrangement in the unusual chromosome have on recombination? Why?
d. You find an individual homozygous for the abnormal chromosome (i.e. the bottom chromosome). Show its chromosomes synapsed in meiosis. Please include the genes.

Answers to the practice problems:

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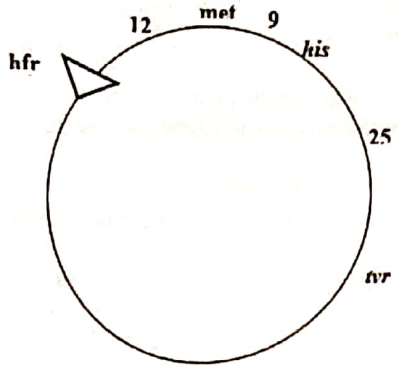
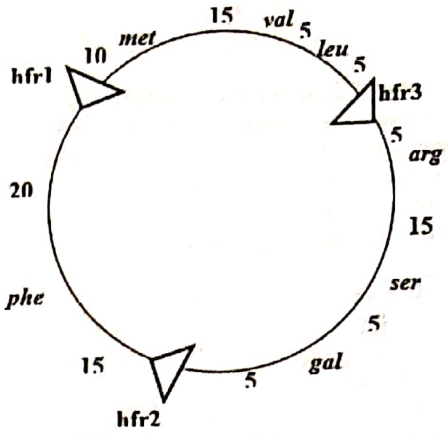


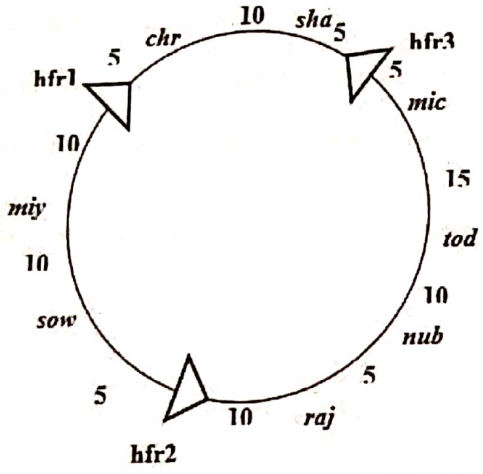
1.a.

1.b. -His + Strep
-Met + Strep
-Tyr + Strep



2.a.

2. b. Donor: Hfr arg+ ser+ gal+ phe+ met+ val+ leu+ strpS
Recipient: F- arg- ser- gal- phe- met- val-1 leu- StrpR

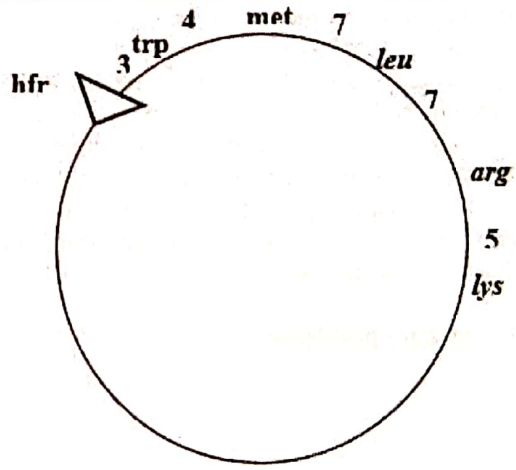


3.a.

3.b. Hfr miy+ sow+ raj+ nub+ tod+ mic+ sha+ chr+ StrS
F– miy- sow- raj- nub- tod- mic- sha- chr- StrR

4.

# chromosom	ploidy of individual
25	monosomy
13	monoploid
28	double trisomy or tetrasomy
52	tetraploid
27	trisomic



5.a

5.b. donor: Hfr trp+ met+ leu+ arg+ lys+ StrS
recipient: F- trp- met- leu- arg- lys- StrR

c. The matings to not all begin at the same time.

d. The probability of separation is balanced by the probability of new matings.

6. Conjugation – involves direct cell-cell contact through a cytoplasmic bridge. Involves transfer of fertility factor (F+), part of bacterial chromosomes (Hfr) or F'

transformation – involves take up of free DNA from the environment, no direct cell-cell contact

transduction – transfer of DNA between bacteria by virus, no direct cell-cell contact

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7. a.

a c b

b. In experiment 1, cells were plated on plates that lacked A and then replica plated to places that lacked B and plates that lacked C. In experiment 2, cells were plated on plates that lacked B and then replica plated to plates that lacked A and plates that lacked C.

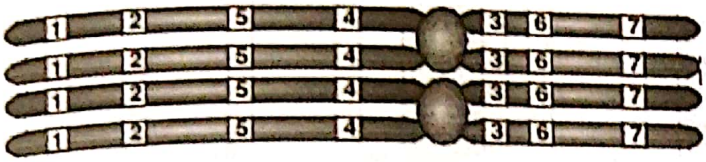
8.

a. pericentric inversion

b. too hard to draw in the computer, it involves formation of an inversion loop to bring homology together.

c. It will appear to suppress recombination because gametes produced by crossing over in the inverted region will be inviable.

d.



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